OMNICOMMERCE ENTERPRISE PLATFORM

Implementor Blueprint & Commercial Sales Manual

Comprehensive Architecture, Functional Workflows & ROI Playbook

IMMUTABLE BUILD & VERSION STAMP

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For official implementors, enterprise administrators, and authorized commercial sales personnel only. Contains detailed operational formulas, fiscal tax settings, and internal control policies.

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OVERVIEW

QSell Enterprise: Master Implementor & Sales Enablement Guide

Category: Foundations & Roadmap | Reading Time: ~11 min | Words: 2,141

*Welcome to the comprehensive operational manual for **QSell**, an enterprise-grade omnichannel commerce, inventory, route logistics, procurement, core double-entry accounting, and regulatory compliance platform.*

Welcome to the comprehensive operational manual for QSell, an enterprise-grade omnichannel commerce, inventory, route logistics, procurement, core double-entry accounting, and regulatory compliance platform.

This documentation suite is engineered specifically for System Implementors deploying and configuring customer organizations, and Sales Team Members & Solutions Consultants communicating the platform's value proposition, commercial return on investment (ROI), and competitive advantages to prospective businesses.

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1. System Architecture from a Business Perspective

QSell replaces fragmented point solutions (separate e-commerce plugins, manual POS registers, disconnected spreadsheets, third-party courier dispatch apps, and standalone accounting software) with one unified, real-time operating system.

```
graph TD
    subgraph FrontOffice["Front-Office & Customer Channels"]
        WEB["Branded Web Storefront\n(Desktop & Mobile)"]
        POS["In-Store Point of Sale\n& Counter Orders"]
        QUOTE["Proforma Sales Quotes\n& Direct B2B Portal"]
    end

    subgraph Operations["Operations & Logistics Engine"]
        ORD["Order Stage Engine\n(Timestamped Milestones & SLAs)"]
        INV["Multi-Location WMS\n(Warehouses, Stores & Van Stock)"]
        DISP["Dispatch Board &\nRoute Run Sequencing"]
        RIDER["Rider Field Mobile App\n(Dropoffs, PIN & COD Collections)"]
    end

    subgraph Governance["Governance & Supply Chain"]
        APP["Runtime Approval Engine\n(Tiered Limits & Substitute Chains)"]
        PROC["Vendor Procurement\n(RFQs, POs, RGNs & 3-Way Match)"]
        VEND["Vendor Self-Service Portal\n(Bids, Acknowledgements & Invoices)"]
    end

    subgraph BackOffice["Back-Office Finance & Compliance"]
        AR["Accounts Receivable\n(Receipts, Wallets & Ageing)"]
        GL["General Ledger\n(Immutable Double-Entry Postings)"]
        BANK["Bank, Cash & Till\nReconciliation"]
        KRA["KRA eTIMS / TIMS\nReal-time Fiscal Signing"]
    end

    FrontOffice --> ORD
    ORD --> APP
    APP --> INV
    INV --> DISP
    DISP --> RIDER
    RIDER --> AR
    PROC --> VEND
    VEND --> INV
    AR --> GL
    BANK --> GL
    ORD --> KRA
```

The Six Operational Pillars

1. Omnichannel Demand Capture: Unifies online storefront transactions, walk-in counter sales, telephone orders, and negotiated B2B proforma quotes into a single consolidated order stream.
2. Immutable Inventory & Location Ledger: Tracks stock movements across central warehouses, regional retail stores, and mobile delivery vans in real time using true double-entry stock transactions with zero silent overrides.
3. End-to-End Route Logistics & Door Collections: Manages localized delivery zones, dynamic delivery fee calculation (by distance, weight, or flat fee), rider dispatch runs, and reconciled Cash-on-Delivery (COD) or M-Pesa collections at the doorstep.
4. Controlled Procurement & Accounts Payable: Streamlines requests for quotation (RFQs), purchase orders, Goods Received Notes (RGN), landed cost apportionments, and automated 3-way invoice matching.
5. Core Financial Ledger & Receivables: Posts financial journals, customer payments, credit balances, foreign currency revaluations, and trial balances without batch delays.
6. Regulatory Compliance by Design: Native integration with Kenya Revenue Authority (KRA) TIMS and eTIMS protocols, ensuring every invoice and credit note is cryptographically signed and QR-coded before dispatch.

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2. Complete Module Directory

Manual	Module Name	Primary Administrative Route	Core Business Function
Guide 01 (01-business-overview-and-dashboard.md	Business Overview & Dashboard	/admin	Executive KPI monitoring, real-time revenue, order velocities, branch rankings, and profit telemetry.
Guide 02 (02-company-setup-and-storefront.md	Company Setup & Brand Identity	/admin/company	Legal identity, KRA PIN, document layouts, brand themes, promotional banners, and customer guarantees.
Guide 03 (03-sales-quotes-and-customer-orders.md	Sales Quotes & Customer Orders	/admin/sales	Proforma quotation generation, order stage tracking, customer master cards, credit limits, and SLA monitoring.
Guide 04 (04-pricing-engine-promotions-and-margins.md	Pricing Engine & Promotions	/admin/pricing	Customer group pricing, tiered volume breaks, coupons, rebates, loyalty points, margin floors, and price simulators.
Guide 05 (05-products-catalog-and-inventory-control.md	Products Catalog & Inventory	/admin/catalog, /admin/inventory	SKU catalog, units of measure, kits/hampers, multi-warehouse stock, in-transit transfers, and live cycle counts.
Guide 06 (06-delivery-logistics-and-field-dispatch.md	Delivery Logistics & Field Dispatch	/admin/delivery	Polygon delivery zones, delivery times (slots) with capacity, dispatch boards, rider run sequencing and phone-number-and-PIN rider sign-in, COD reconciliation, rider pay and delivery costs, late-delivery alerts by screen and mail, connected couriers with courier-invoice matching, and delivery…
Guide 07 (07-users-roles-and-access-governance.md	Users, Roles & Access Governance	/admin/users	Staff directory, branch bindings, role templates, atomic permissions, and contextual ABAC security policies.
Guide 08 (08-projects-and-job-costing.md	Projects & Job Costing	/admin/projects	Job budgeting, material/labor tracking, milestone invoicing, customer sign-offs, and project profitability.

Guide 09 (09-vendor-management-procurement-and-payables.md	Vendor Management & Procurement	/admin/vendors	Supplier prequalification, RFQs, purchase orders, 3-way invoice matching, landed costs, and WHT certificates.
Guide 10 (10-finance-general-ledger-and-accounts-receivable.md	Finance, General Ledger & AR	/admin/finance	Chart of accounts, general journals, posting matrix, trial balance, customer receipts, store credit wallets, and FX
Guide 11 (11-banking-cash-and-payment-gateways.md	Banking, Cash & Payment Gateways	/admin/finance/bank-accounts, /admin/payments	Bank accounts, statement CSV reconciliations, cashier till counts, M-Pesa Bill Manager, and gateway audit logs.
Guide 12 (12-approvals-and-internal-controls.md	Approvals & Internal Controls	/admin/approvals, /admin/settings/approvals	Runtime workflow builder, threshold limits, substitute approvers, scenario switches, and stuck request reassignment.
Guide 13 (13-settings-branches-numbering-and-kra-etims.md	Settings, Branches & KRA eTIMS	/admin/settings	Operating branches, dimensions, gapless number series, stage SLA targets, and KRA eTIMS tax classification
Guide 14 (14-external-portals-customer-rider-vendor.md	External & Field Portals	/(store), /rider, /vendor	Branded web shop checkout, rider mobile run sheet & PIN verification, and supplier quotation/invoice self-service.

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3. End-to-End Implementation Project Lifecycle

When rolling out QSell for a new client organization, implementors must follow this strict 7-stage deployment sequence. Configuring modules out of order leads to blocked transactions and missing lookup dependencies.

```
sequenceDiagram
    autonumber
    actor Imp as Solutions Implementor
    participant Fnd as 1. Foundation & Legal Identity
    participant Str as 2. Locations, Tax & Numbers
    participant Mst as 3. Master Catalog & Pricing
    participant Log as 4. Delivery & Logistics
    participant Fin as 5. Finance & Postings
    participant Gov as 6. Users & Approvals
    participant Cut as 7. Cutover & Go-Live

    Imp->>Fnd: Configure Company Profile, Legal Name, KRA PIN, Bank Accounts & Brand Theme
    Imp->>Str: Set up Branches, Base Units of Measure, Tax Classifications & Number Series
    Imp->>Mst: Import Categories, Products, Kits, Customer Groups & Base Price Lists
    Imp->>Log: Define Delivery Zones, Rates, Dispatch Methods, Slots & Map Services
    Imp->>Fin: Set Chart of Accounts, Bank Accounts, Payment Providers & Posting Matrix
    Imp->>Gov: Define Staff Roles, Assign Branches, Configure Approval Chains & Thresholds
    Imp->>Cut: Perform Inventory Physical Count Cutover, Open Fiscal Period & Launch Pilot
```

Stage 1: Foundation & Legal Identity

- Actions: Complete Company Profile (Guide 02 (02-company-setup-and-storefront.md), upload high-resolution logo, set legal trading name, register corporate KRA PIN, address, telephone, email, and primary bank details.
- Verification: Preview invoice and receipt document layouts to ensure legal headers format correctly.

Stage 2: Locations, Tax & Numbering Series

- Actions: Define physical operating branches (Guide 13 (13-settings-branches-numbering-and-kra-etims.md). Configure Units of Measure (Guide 05 (05-products-catalog-and-inventory-control.md). Map KRA Tax Classifications (16% General VAT, 0% Zero-rated, 8% Fuel, Exempt). Establish gapless Number Series for Orders, Invoices, Receipts, and POs.
- Verification: Test number series generation to ensure sequence formatting matches company conventions (e.g., INV-2026-00001).

Stage 3: Master Catalog, Stock Locations & Pricing

- Actions: Create stock locations (warehouses, shops, dispatch vehicles). Establish category trees. Import SKUs with barcodes, weights, and dimensions. Build Kits & Hampers. Establish Customer Groups (Retail, Wholesale, Trade). Build Price Lists with quantity breaks and minimum price floors (Guide 04 (04-pricing-engine-promotions-and-margins.md).
- Verification: Run the Price Check Simulator (/admin/pricing/check) on sample products to confirm margin floors and quantity discounts calculate properly.

Stage 4: Delivery Zones & Logistics Setup

- Actions: Configure delivery zones using postal towns or map polygon boundaries (Guide 06 (06-delivery-logistics-and-field-dispatch.md). Configure rate cards (flat, weight-based, distance-based). Define delivery time slots and daily capacity caps. Connect courier APIs or create in-house rider profiles.
- Verification: Enter a test address in checkout or counter order to verify the exact zone, fee, and delivery time slot are assigned.

Stage 5: Finance, Banking & Gateways

- Actions: Set up Chart of Accounts (Guide 10 (10-finance-general-ledger-and-accounts-receivable.md). Configure General Ledger Posting Setup matrix. Set up commercial bank accounts and cash drawers (Guide 11 (11-banking-cash-and-payment-gateways.md). Integrate Safaricom M-Pesa Daraja APIs (Paybill/Till) and card payment gateways.
- Verification: Execute a test KES 10.00 M-Pesa STK Push payment and confirm automatic transaction ledger recording and settlement allocation.

Stage 6: Users, Security & Approval Controls

- Actions: Configure user accounts for staff members (Guide 07 (07-users-roles-and-access-governance.md). Assign specific branches and roles (Cashier, Storekeeper, Dispatcher, Sales Rep, Accountant, Branch Manager). Build Approval Workflows (Guide 12 (12-approvals-and-internal-controls.md) for high-value purchase orders, below-floor price overrides, and customer refunds.
- Verification: Submit a purchase order exceeding a test user's signing limit and verify that the document halts in "Pending Approval" state and routes to the designated manager's inbox.

Stage 7: Cutover, Ledger Initialization & Go-Live

- Actions: Lock legacy systems. Execute the Stock Ledger Cutover wizard (Guide 05 (05-products-catalog-and-inventory-control.md), importing physical on-hand inventory balances location by location. Post opening customer debit balances and vendor accounts payable balances. Open the active fiscal period.
- Verification: Print the Trial Balance and Stock Valuation reports and reconcile them to the client's certified opening balance sheet.

---

4. Sales Onboarding Curriculum (2-Week Mastery Track)

New commercial sales representatives and solutions consultants must complete this 10-day structured onboarding program to achieve fluency in pitching, demoing, and defending QSell against competing software.

Week 1: Core Commerce, Inventory & Logistics Operations

- Day 1: Platform Vision & The Modern African Business Reality
- Read: Guide 01 (Dashboard) (01-business-overview-and-dashboard.md and Guide 02 (Company Setup) (02-company-setup-and-storefront.md.
- Key Topic: Why traditional international ERPs fail in East Africa (lack of native M-Pesa, complex KRA eTIMS integrations, poor road network support, lack of door cash reconciliation).
- Practice: Configure a mock company profile, logo, and document layout in under 15 minutes.

- Day 2: The Omnichannel Sales Journey & Order Blueprint
- Read: Guide 03 (Sales & Orders) (03-sales-quotes-and-customer-orders.md.
- Key Topic: Walk prospects through the journey from Web Storefront / Counter Order $\rightarrow$ Order Confirmed $ \rightarrow$ Picking $\rightarrow$ Packing $\rightarrow$ Dispatched $\rightarrow$ Delivered.
- Practice: Issue a proforma quote, email it to a prospect, convert it into a confirmed order, and explain the timestamped stage audit trail.
- Day 3: Intelligent Pricing, Margin Protection & Promotions
- Read: Guide 04 (Pricing Engine) (04-pricing-engine-promotions-and-margins.md.
- Key Topic: How QSell eliminates rogue discounting and protects operating margins using price floors and role-based override limits.
- Practice: Use the Price Check Simulator to explain a complex pricing waterfall (Customer Group + Quantity Break + Active Promo Code - Margin Floor Lock).
- Day 4: Inventory Control, Multi-Location WMS & Stock Integrity
- Read: Guide 05 (Products & Inventory) (05-products-catalog-and-inventory-control.md.
- Key Topic: How QSell stops inventory theft and phantom stockouts across multiple branches and mobile vans.
- Practice: Execute a stock transfer from Central Warehouse to a Van, demonstrate in-transit tracking, and conduct a cycle count that continues while cashiers sell.
- Day 5: Delivery Logistics, Dispatch & Cash on Delivery (COD)
- Read: Guide 06 (Delivery) (06-delivery-logistics-and-field-dispatch.md and Guide 14 (Rider Portal) (14-external-portals-customer-rider-vendor.md.
- Key Topic: How QSell solves the last-mile delivery nightmare: pin-on-map zoning, rider run sheets, doorstep PIN verification, and door payment reconciliation.
- Practice: Plan a 3-stop delivery run on the Dispatch Board, open the Rider Mobile App, complete a dropoff, and reconcile cash collected at the counter.

Week 2: Procurement, Finance, Compliance & Executive Pitching

- Day 6: Vendor Management & 3-Way Match Procurement
- Read: Guide 09 (Vendor Management) (09-vendor-management-procurement-and-payables.md.
- Key Topic: Preventing procurement fraud through public RFQ bidding, Purchase Order approvals, Goods Received Notes (RGN), and automated 3-way matching.
- Practice: Issue an RFQ, submit a vendor bid through the Vendor Portal, generate a Purchase Order, receive partial goods, and book the supplier invoice.
- Day 7: Double-Entry Financial Accounting & Receivables
- Read: Guide 10 (Finance & AR) (10-finance-general-ledger-and-accounts-receivable.md and Guide 11 (Banking & Payments) (11-banking-cash-and-payment-gateways.md.
- Key Topic: Why CFOs love QSell: real-time general ledger postings, automated sales/stock sync, customer store credit wallets, and bank CSV reconciliation.
- Practice: Post a general journal, receive a customer payment, allocate it across three invoices, and inspect the real-time Trial Balance.
- Day 8: Regulatory Compliance: KRA eTIMS & Tax Architecture
- Read: Guide 13 (Settings & KRA eTIMS) (13-settings-branches-numbering-and-kra-etims.md.
- Key Topic: Turning KRA compliance into a sales superpower: zero external fiscal boxes required, automated digital invoice signing, and item synchronization.

- Practice: Explain the difference between TIMS (hardware device) and eTIMS (cloud digital signing) and demonstrate how an invoice receives its cryptographic signature.
- Day 9: Governance, Approvals & Enterprise Access Control
- Read: Guide 07 (Users & Roles) (07-users-roles-and-access-governance.md and Guide 12 (Approvals) (12-approvals-and-internal-controls.md.
- Key Topic: Corporate governance: separation of duties, tiered financial limits, substitute approvers during leave, and complete audit trail.
- Practice: Configure an approval rule for orders over KES 100,000, trigger an approval request, reject with a reason, and show the audit trail.
- Day 10: The Master Executive Pitch & Competitive Objections
- Read: Guide 08 (Projects) (08-projects-and-job-costing.md and review all Value Proposition sections.
- Key Topic: Delivering the 30-minute C-suite pitch to a CEO and CFO; handling objections regarding data migration, staff training, and software reliability.
- Final Exercise: Deliver a full end-to-end platform demo and answer 5 live prospect objections.

Interactive version with live navigation: /admin/docs/index

CHAPTER 01

Guide 01: Business Overview & Dashboard (`/admin`)

Category: Executive & Analytics | Reading Time: ~10 min | Words: 1,949

*The **Business Overview** dashboard is the operational and financial cockpit of QSell. It aggregates company-wide commercial performance in real time across all sales channels, retail branches, warehouse locations, and payment gateways.*

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1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- Blind Operations: Traditional businesses wait until the end of the day or month for accountants to balance physical receipt books and compile spreadsheets, leaving management blind to intraday inventory shortages or slumping sales.
- Disconnected Branch Visibility: Multi-location retail and wholesale operations struggle to compare branch performance fairly, resulting in slow-moving stock trapped in one branch while another suffers stockouts.
- Unreconciled Cash Leakage: Sales numbers often disguise collection failures. A company may show KES 1,000,000 in sales on paper, but only KES 400,000 in actual bank deposits, with the balance lost to uncollected credit or delayed mobile-money settlements.
- Invisible Margins: Heavy discounting by sales staff often inflates top-line revenue while wiping out bottom-line profit.

Target Client Profiles

- Multi-Branch Retail Chains: Supermarkets, electronics stores, and clothing boutiques needing centralized branch rankings and till balances.
- High-Velocity Wholesalers & FMCG Distributors: Businesses needing real-time dispatch velocities, receivables ageing, and payment method breakdowns.
- Executive Directors & C-Suite: CEOs and CFOs requiring instant access to true gross margin, tax collected, and critical items requiring attention.

Commercial Sales Pitch & Talk Track

> "With QSell, you never have to ask your store managers for an end-of-day sales report. The moment an order is punched into a till in Kisumu or checked out on your Nairobi website, your executive dashboard updates instantly. You see not just your total revenue, but the exact cash and M-Pesa collected, your true realized profit margin after product costs, and which specific branch or product is driving your growth today."

Measurable ROI & Safeguards

- 100% Intraday Telemetry: Zero reporting delay between order placement and executive awareness.
- Direct Collection Audit: Instantly compares Total Sales vs. Money Collected, exposing payment collection deficits before they turn into bad debts.
- Realized Margin Tracking: Monitors gross margin percentage in real time, alerting management when discounts erode product profitability.

---

2. Prerequisites & Configuration Sequence

Before the Business Overview dashboard can display complete figures, the following foundational setup must be completed:

```
graph TD
    B[1. Operating Branches Configured] --> P[2. Products & Landed Cost Defined]
    P --> G[3. Payment Providers Connected]
    G --> U[4. User Account Assigned to Branch/Scope]
    U --> DASH[Dashboard Populates Metrics]
```

1. Operating Branches: At least one branch must be active under Settings > Branches.
2. Product Cost Prices: Products must have their unit purchase cost entered under Catalog > Products for the Gross Profit KPI to calculate.
3. Payment Gateways: At least one payment provider (M-Pesa, Cash Till, Bank, or Card) must be configured under Settings > Payments.
4. Staff Scope & Permissions:

- The user must possess the dashboard:read permission.
- The user must be assigned to either a specific branch (to see localized data) or granted company-wide access (to see consolidated data and the branch filter).

---

3. Module Architecture & Telemetry Data Flow

The dashboard dynamically queries multiple operational domains in parallel and calculates real-time comparisons:

```
graph LR
    subgraph Transactions["Operational Ingestion"]
        ORD[Sales Orders Placed]
        PAY[Payment Gateways Confirmed]
        DEL[Delivery Statuses]
        STK[Inventory Thresholds]
    end

    subgraph Aggregator["Reporting Engine"]
        PERIOD[Date Period & Comparison Calculator]
        SCOPE[Branch Security & Currency Filter]
        FLOW[Order Flow Classifier]
    end

    subgraph DashboardView["Executive Presentation"]
        CARDS[KPI Cards & Sparklines]
        CHART[Trend Charts vs Previous Period]
        BRANCH[Branch Ranking Table]
        ATTN[Actionable Attention Queue]
    end
    Transactions --> Aggregator
    Aggregator --> DashboardView
```

- Scope Boundary: If a staff member is assigned to the "Mombasa Branch", the system automatically restricts all queries to orders, payments, and stock originating from that branch.
- Currency Isolation: All values are reported in the company's Base Currency (e.g., KES). Transactions placed in secondary foreign currencies (USD, EUR) are flagged in an informational footnote with order counts.

---

4. Interface-by-Interface Walkthrough

The Dashboard consists of five integrated functional sections:

Section A: Top Filter & Date Range Bar

- Preset Period Selector: Quick-jump dropdown for rapid temporal slicing.

- Comparison Mode Selector: Configures the comparative baseline against which percentage growth/decline is measured.
- Branch Scope Selector: Dropdown to switch between company-wide consolidation or a specific branch.
- Last Updated & Refresh Indicator: Displays the exact timestamp the data was pulled and the active timezone (e.g., Africa/Nairobi).

Section B: Executive Headline & Attention Banner

- Headline Summary: Natural-language sentences summarizing current performance (e.g., "Sales are up 14% compared to the previous 30 days, driven by Central Store.").
- Attention Queue (AttentionList): Visual callout cards for items requiring immediate human intervention:
- Unpaid Orders Pending Delivery
- Failed or Timed-Out Gateway Payments
- Low Stock / Depleted SKUs
- Late Delivery SLA Breaches

Section C: Sales Performance & Order Flow (Card: Sales)

- Key Metric Tiles:
- Sales: Total gross revenue from valid orders placed in the selected period.
- Orders: Total number of qualifying orders.
- Average Order Value (AOV): Gross Sales divided by Order Count.
- Tax Included in Sales: Total VAT and consumption taxes included in the reported sales total.
- Revenue Trend Chart: Daily or hourly sales graph plotting the current period's velocity against the comparative prior period.
- Order Flow Breakdown: Visual status bar and breakdown table showing where placed orders ended up:
- Counted in Sales: Completed, In-Transit, Awaiting Fulfillment, Credit Sales.
- Excluded from Sales: Abandoned Checkouts, Cancelled by Customer, Payment Failed, Refunded.
- Sales by Branch Table: Multi-column comparison listing Branch Name, Revenue, Revenue Share %, Total Orders, and Growth % vs. Prior Period.
- Best Sellers Ranking: Top products by revenue, showing units sold and gross income.

Section D: Money Collected & Liquidity (Card: Money Collected)

- Collected Revenue: Total cleared funds received through payment gateways and cash registers.
- Collection Rate %: (Money Collected ÷ Gross Sales) × 100.
- Note: Can exceed 100% when credit customers settle invoices from previous months.
- In-Flight & Failed Payment Audit: Flags customer transactions currently processing vs. abandoned.
- Payment Method Distribution: Metered progress bars showing the percentage and shilling value collected via M-Pesa Paybill, Cash Till, Bank Transfer, and Credit Cards.
- Non-Cash Tenders Notice: Footnote disclosing value paid via Loyalty Points redemptions and Gift Cards.

Section E: Gross Profit & Margin Control (Card: Gross Profit)

- Gross Profit (KES): Realized profit calculated as Selling Price - Unit Cost Price.
- Gross Margin %: (Gross Profit ÷ Gross Sales) × 100.
- Uncosted Items Notice: Warns if products without an entered cost were sold, preventing distorted margin figures.

---

5. Granular Field-by-Field Reference

Control / Metric	Type	Purpose	Validation & Rules
Period Preset	Dropdown	Sets the time window for all metrics.	Options: Today, Yesterday, Last 7 Days, Last 30 Days, This Month, Last Month, Custom.
Compare Mode	Dropdown	Determines the comparison baseline.	Options: Previous Period (same duration immediately preceding), Previous Year (same dates last year).
Branch Scope	Dropdown	Filters metrics to a specific operational unit.	Disabled if the logged-in user lacks multi-branch view rights. Shows "All Branches" or specific branch names.
From Date	Date Input	Custom period start date.	Format YYYY-MM-DD. Must be earlier than or equal to To Date.
To Date	Date Input	Custom period end date.	Format YYYY-MM-DD. Cannot be a future date beyond current server time.
Sales KPI	Currency	Gross monetary value of valid orders.	Formatted in base currency (KES) with thousands separators. Excludes cancelled and refunded orders.
Collection Rate	Percentage	Ratio of cash received to sales booked.	Displayed to 1 decimal place. Highlights liquidity health.
Gross Margin	Percentage	Operating profitability ratio.	Color-coded: Green for healthy margins, Amber for marginal, Red for negative gross profit.

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6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Analyzing Today's Intraday Performance

1. Navigate to /admin in the sidebar.
2. Locate the Period dropdown in the top filter bar.
3. Click the dropdown and select Today.
4. The screen refreshes without full page reload.
5. Review the Sales tile to see today's revenue generated since 00:00.
6. Check the Money Collected tile to verify that cash/M-Pesa deposits match order revenue.
7. Click any item in the Attention Queue (e.g., "3 orders pending dispatch") to jump directly to the pre-filtered order processing screen.

Action 2: Comparing Branch Sales for Monthly Review

1. In the top filter bar, set Period to This Month.
2. Set Compare to Previous Period (or Previous Year for seasonal analysis).
3. Scroll down to the Sales by branch table.
4. Review the Share % column to identify your highest-volume branch.
5. Review the vs Previous Period column:

- Positive numbers indicate revenue growth.
- Negative numbers indicate underperformance requiring operational review.

6. Click the Branch dropdown in the top filter bar and select the specific underperforming branch to isolate its top sellers and order bottlenecks.

Action 3: Reviewing Payment Method Health & Failed Transactions

1. Locate the Money collected card.
2. Inspect the In-Flight & Failed Payments status banner.
3. If failed payment attempts are flagged, note the count.
4. Review the breakdown list of payment methods:

- Identify whether M-Pesa Paybill or In-Store Cash represents the primary liquidity source.
- Note any funds flagged under "unconfigured methods" (payments received on legacy tills no longer assigned to branches).

---

7. Conditions, Business Rules & Edge Cases

Order Qualification Rules (What Counts as a Sale?)

- Counted in Sales:
- Orders marked as Completed, Delivered, Dispatched, Processing, or Confirmed.
- Credit sales with approved payment terms (invoiced to customer account).
- Excluded from Sales:
- Abandoned online shopping carts where the customer exited before checkout.
- Payment attempts that returned Failed, Cancelled, or Expired from Safaricom or card processors.
- Orders formally cancelled or refunded.

Multi-Branch Permission Boundary

- If a user has branches:read and dashboard:read across the company, they see consolidated numbers and can toggle between branches.
- If a user is restricted to a single branch (e.g. "Westlands Branch"), the branch dropdown is locked, and the byBranch comparison table is hidden to prevent competitive data leakage among branch staff.

Zero-Activity State Handling

- If a newly launched shop or branch has zero orders in the selected period, the dashboard displays:
- A clean empty state: "No orders were placed in this period."
- A dynamic jump button: "Show the 30 days up to [Last Order Date]" allowing managers to quickly review the last active sales period without guessing dates.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create a test branch and assign products to it.	Branch appears in the dashboard branch filter dropdown.	[ ]
2	Place a test order for KES 5,000 using M-Pesa.	Sales KPI increases by KES 5,000; Orders KPI increments by 1.	[ ]

3	Confirm M-Pesa payment reception.	Money Collected tile updates by KES 5,000; Collection Rate reflects 100%.	[ ]
4	Log in as a Branch-Restricted Cashier.	Dashboard only shows data for the cashier's assigned branch.	[ ]
5	Verify Gross Margin calculation.	Product cost is subtracted from KES 5,000; accurate gross margin % is displayed.	[ ]

Interactive version with live navigation: /admin/docs/01-business-overview-and-dashboard

CHAPTER 02

Guide 02: Company Setup & Storefront Identity (`/admin/company`)

Category: Identity & Branding | Reading Time: ~11 min | Words: 2,046

*The **Company Setup** module establishes the legal, fiscal, operational, and visual identity of the business. It governs how the company presents itself to customers on the web storefront and how legal identity, tax information, and banking*

The Company Setup module establishes the legal, fiscal, operational, and visual identity of the business. It governs how the company presents itself to customers on the web storefront and how legal identity, tax information, and banking instructions appear on formal commercial documents (quotes, invoices, and receipts).

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- Inconsistent Branding Across Channels: Businesses frequently struggle with fragmented brand images—a web shop that looks different from printed counter receipts, and quotes that lack professional banking instructions.
- KRA Tax Invoicing Compliance Risks: Failing to display an exact legal name, physical address, and KRA PIN on invoices invalidates customer input tax deductions and incurs penalties from the Kenya Revenue Authority.
- Costly Custom Development for Simple Storefront Changes: Minor changes like changing a brand accent color, updating a customer service WhatsApp number, or posting a temporary delivery banner usually require hiring external web developers.

Target Client Profiles

- Branded Consumer Retailers & E-Commerce Merchants: Businesses requiring an online storefront that mirrors their physical store branding and corporate identity.
- B2B Suppliers & Corporate Wholesalers: Organizations issuing proforma quotations and credit invoices requiring legal terms, VAT registration numbers, and wire transfer bank details.

Commercial Sales Pitch & Talk Track

> "With QSell, you can launch a fully branded, legally compliant online store and ERP in minutes. You don't need a web developer to change your brand colors, update your customer service hotline, or customize your invoice footer notes. Enter your details once in the Company Setup module, and every quote, invoice, receipt, and web page instantly updates with your brand, your KRA PIN, and your direct bank details."

Measurable ROI & Safeguards

- Zero Developer Fees for Brand & Storefront Management: 100% no-code configuration of colors, grid density, banners, FAQs, and policies.
- 100% Tax Document Legality: Guarantees that every printed and emailed invoice satisfies Section 23 of the Kenyan Tax Procedures Act.

---

2. Prerequisites & Configuration Sequence

Company Setup is the Stage 1 Foundation of the entire QSell platform. It should be configured before adding products, creating users, or transacting.

```
graph TD
    P[1. Company Profile: Legal Name & KRA PIN] --> D[2. Document Layouts: Headers & Footers]
    D --> A[3. Appearance: Brand Colors & Density]
    A --> S[4. Support & Contacts: WhatsApp & FAQs]
    S --> B[5. Banners & Policies: Terms & Notices]
    B --> M[6. Module Renaming: Custom Labels]
```

- Required Staff Permission: Users must possess the company:read permission to view and company:write (or administrative role) to save changes.

---

3. Module & Sub-Module Architecture

Company settings are organized into seven distinct sub-modules under /admin/company:

```
graph LR
    subgraph CompanySetup["Company Setup Sub-Modules"]
        PROF["Profile\n(Identity, KRA PIN, Bank)"]
        DOCS["Document Layouts\n(Quote, Invoice, Receipt)"]
        APP["Appearance\n(Colors & Grid Density)"]
        MODS["Modules\n(Custom Sidebar Labels)"]
        SUPP["Support\n(Channels, FAQs, Guarantees)"]
        BAN["Banners\n(Top Notice Campaigns)"]
        POL["Policies\n(Privacy, Returns, Terms)"]
    end

    subgraph AppliedSurfaces["Where Settings Take Effect"]
        STORE["Web Storefront & Customer Portal"]
        PRINT["Printable PDFs & Fiscal Invoices"]
        SIDEBAR["Admin Navigation Sidebar"]
    end

    PROF --> STORE
    PROF --> PRINT
    DOCS --> PRINT
    APP --> STORE
    MODS --> SIDEBAR
    SUPP --> STORE
    BAN --> STORE
    POL --> STORE
```

---

4. Interface-by-Interface Walkthrough

Sub-Module 1: Profile (/admin/company/profile)

Configures the core identity of the business. Divided into five sections:

1. Brand & Logo: Live visual brand card preview and high-resolution logo upload.
2. Identity & Tax: Legal entity name, public trading name, KRA PIN, VAT number, and business registration certificate number.
3. Address & Location: Physical street address, city, county/state, postal code, country, and landmark location label.
4. Contact: Primary corporate email, mobile phone (M-Pesa enabled), landline telephone, and official website URL.
5. Banking: Primary bank name, branch name, account name, and bank account number for wire transfers.

Sub-Module 2: Document Layouts (/admin/company/documents)

Controls the appearance and custom text printed on customer-facing paper and PDF documents:

- Quotes: Header text, footer notes (e.g. "Valid for 30 days. Prices inclusive of 16% VAT"), and alignment.
- Invoices & Proforma Invoices: Payment instructions, bank details placement, tax breakdown notes, and late interest warnings.
- Receipts: Counter receipt footers (e.g. "Goods once sold cannot be returned without receipt"), cashier signature line, and QR code placement.

Sub-Module 3: Appearance (/admin/company/appearance)

Governs the visual design of the customer storefront:

- Primary Color: Interactive color picker setting buttons, links, and checkout action buttons.
- Accent Color: Color picker setting invoice brand marks, highlights, and promotional tags.
- Product Listing Density: Configures desktop product catalog grid columns (3, 4, or 5 products per row) with live responsive previews.

Sub-Module 4: Modules (/admin/company/modules)

Allows businesses to rename standard admin navigation modules to match company terminology (e.g. renaming "Sales Quotes" to "Estimates", or "Vendor Management" to "Purchasing").

Sub-Module 5: Support (/admin/company/support)

Configures the storefront customer helpdesk:

- Contact Channels: Interactive list of support options (WhatsApp Chat, Telephone, Email, Physical Office) with operational hours.
- Customer Guarantees: Trust badges shown across the web shop (e.g. "Same-Day Nairobi Delivery", "30-Day Money Back Guarantee", "Official Authorized Distributor").
- Frequently Asked Questions (FAQs): Accordion FAQ builder with drag-and-drop reordering, generating structured FAQ rich data for Google search.

Sub-Module 6: Banners (/admin/company/banners)

Manages site-wide announcement banners displayed across the top of the storefront (e.g., "Holiday Sale: 15% off all solar panels until Friday!"). Supports start/end scheduling and call-to-action buttons.

Sub-Module 7: Policies (/admin/company/policies)

Dedicated rich-text editors for mandatory legal and operational pages:

- Privacy Policy: Data protection compliance.
- Delivery Policy: Delivery zones, turnaround times, and customer obligations.
- Return & Refund Policy: Return windows, condition requirements, and refund methods.

---

5. Granular Field-by-Field Reference

Company Profile Fields (/admin/company/profile)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Logo	Brand mark on shop & invoices.	Image Upload	Optional	Accepts PNG, JPEG, WebP, SVG up to 2MB. Square or landscape.
Storefront Tagline	Subtitle below brand name.	Text (Max 120 chars)	Optional	e.g., "Solar PV & Electrical Supplies".
Legal Name	Official registered company name.	Text (Max 120 chars)	Required	Must match Certificate of Incorporation.
Trading Name	Public brand name seen by customers.	Text (Max 120 chars)	Required	Displayed in website header and titles.

KRA PIN	Corporate tax identification number.	Text (Monospace, 11 chars)	Required	Must match pattern: Letter (A or P), 9 digits, ending in a letter (e.g. P051234567Z).
VAT Number	Value Added Tax registration number.	Text (Max 40 chars)	Optional	Displayed on tax invoices when populated.
Registration Number	Company registrar certificate number.	Text (Max 40 chars)	Optional	Displayed in legal footer notes.
Invoice Prefix	Default sequence prefix for invoices.	Text (Max 10 chars, uppercase)	Required	Default INV. e.g. INV- yields INV-0001.
Street Address	Physical office or warehouse location.	Text (Max 160 chars)	Required	e.g. "Enterprise Road, Industrial Area, Godown #4".
City	Operating city.	Text (Max 80 chars)	Required	e.g. "Nairobi".
County / State	County or administrative state.	Text (Max 80 chars)	Required	e.g. "Nairobi County".
Postal Code	Postal delivery code.	Text (Max 20 chars)	Required	e.g. "00100".
Country	Operating country.	Text (Max 80 chars)	Required	e.g. "Kenya".
Location Label	Landmark description for customers.	Text (Max 120 chars)	Optional	e.g. "Opposite City Cabanas".
Email	Official corporate email.	Email format (Max 160 chars)	Required	Valid email address.
Phone (Mobile)	Primary telephone for customer calls.	Telephone (Max 25 chars)	Required	Safaricom/Airtel format e.g. +254 712 345 678.
Tel (Landline)	Fixed line phone.	Telephone (Max 25 chars)	Optional	e.g. +254 20 123 4567.
Website	Web address.	URL format (Max 200 chars)	Optional	Must include https://.
Bank Name	Commercial bank holding primary account.	Text (Max 120 chars)	Optional	e.g. "KCB Bank Kenya".
Branch	Bank branch name.	Text (Max 120 chars)	Optional	e.g. "Industrial Area Branch".
Account Name	Exact name on bank account.	Text (Max 120 chars)	Optional	Must match wire transfer instructions.
Account Number	Bank account digits.	Text (Max 40 chars)	Optional	Account number printed on invoices.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Uploading and Saving the Company Logo & Profile

1. Navigate to /admin/company/profile.
2. In the Brand & logo section, click Upload logo (or drag and drop an image file into the drop zone).
3. The uploaded logo previews immediately inside the Brand preview card.
4. Fill in the Legal name, Trading name, and KRA PIN fields.
5. In the Address & location section, enter the physical street address, city, and county.
6. In the Banking section, enter the bank name and account number where customer wire transfers should be directed.
7. Click the Save profile button in the top right header.
8. The button displays "Saving…", then returns to "Save profile".
9. A green confirmation banner displays: "Company profile saved."

Action 2: Customizing Document Header and Footer Notes

1. Navigate to /admin/company/documents.
2. Scroll to the Invoices and proforma invoices card.
3. In the Header text box, enter any special notice (e.g. "Tax Invoice — Original").
4. In the Footer text box, enter direct payment instructions:

```
Please pay via M-Pesa Paybill: 123456, Account: [Invoice Number]
Or Wire to KCB Bank, Account: 1122334455, Industrial Area Branch.
Goods remain property of the company until paid in full.
```

5. Click Save document layouts in the top header.
6. A success message confirms: "Document layouts saved." Every future invoice generated will include this footer.

Action 3: Setting Up a Customer Service WhatsApp Channel

1. Navigate to /admin/company/support.
2. Locate the Contact channels section and click Add channel.
3. In the Channel kind dropdown, select WhatsApp.
4. In the Label field, enter "Customer Care Desk".
5. In the Value field, enter the international WhatsApp phone number (e.g., +254712345678).
6. In the Hours field, enter "Mon - Sat: 8:00 AM - 6:00 PM".
7. Click the Save support settings button.
8. The WhatsApp floating chat widget on the web shop immediately activates with this number.

---

7. Conditions, Business Rules & Edge Cases

KRA PIN Format Validation

- The system enforces strict regular expression validation on the kraPin field:
- Must begin with letter A (individual/sole proprietor) or P (company/partnership).
- Must contain exactly 9 numeric digits.
- Must end with a single alphabetic letter.
- If an invalid PIN is entered (e.g. 12345678 or P05123456), the save action is rejected with the message: "KRA PIN must be a letter (A or P), nine digits and a letter."

Document Sizing and Print Ratios

- Uploaded logos are automatically resized on invoices to maintain a maximum height of 60px and maximum width of 240px, preventing document pagination overflow.
- If no logo is uploaded, the document header gracefully falls back to displaying the Trading Name in bold 18pt typography.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Enter company profile and valid KRA PIN.	Green banner confirms profile saved without validation errors.	[ ]

2	Upload corporate logo (PNG format).	Logo renders cleanly in storefront header and document preview.	[ ]
3	Change Brand Primary Color to company brand color.	Storefront buttons and active admin highlights reflect new color.	[ ]
4	Print a sample Sales Quote PDF.	Header displays trading name, KRA PIN, address, and custom footer notes.	[ ]
5	Verify Storefront Footer Links.	Privacy, Delivery, and Return policies render cleanly when clicked.	[ ]

Interactive version with live navigation: /admin/docs/02-company-setup-and-storefront

CHAPTER 03

Guide 03: Sales Quotes & Customer Orders (`/admin/sales`)

Category: Commercial Operations | Reading Time: ~12 min | Words: 2,206

*The **Sales & Fulfillment** module orchestrates the complete Order-to-Cash (O2C) lifecycle. It manages proforma sales quotations, omnichannel customer orders (online storefront, physical point-of-sale, and back-office credit sales), custome*

The Sales & Fulfillment module orchestrates the complete Order-to-Cash (O2C) lifecycle. It manages proforma sales quotations, omnichannel customer orders (online storefront, physical point-of-sale, and back-office credit sales), customer master cards, credit limit enforcement, newsletter subscribers, and product reviews.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The Delivery & Fulfillment Blame Game: When an order is delivered late or items go missing, sales reps blame the warehouse, warehouse staff blame the dispatcher, and dispatchers blame the courier. QSell eliminates this through an immutable, server-timestamped order timeline that tracks the exact second an order entered and exited every stage (Received $\rightarrow$ Confirmed $\rightarrow$ Reserved $\rightarrow$ Picked $\rightarrow$ Packed $\rightarrow$ Dispatched $\rightarrow$ Delivered).
- Uncontrolled Credit Overextension: Sales reps often accept orders from customers who have exceeded their credit limits or have 90-day overdue invoices. QSell enforces hard credit limit checks at order placement, automatically blocking unapproved credit orders.
- Friction in B2B Quotation Negotiations: B2B deals require multi-line formal proforma quotes with negotiated discounts and validity dates. In manual systems, quotes are made on spreadsheets and must be re-typed into the accounting system when won. QSell converts quotes to confirmed sales orders in a single click with zero data re-entry.

Target Client Profiles

- Wholesalers & B2B Trade Distributors: Managing formal quotations, corporate customer accounts, credit terms (Net 30, Net 60), and multi-branch delivery schedules.
- Omnichannel Retailers: Processing walk-in counter sales alongside high-volume e-commerce orders.
- Field Sales Teams: Generating field quotes on tablets that link directly to warehouse inventory.

Commercial Sales Pitch & Talk Track

> "With QSell, your sales team moves deals from quotation to cash with zero friction and 100% accountability. Every quote you generate can be converted into a live sales order with a single click. Every single order is tracked with an unalterable server timestamp at every milestone—so you know exactly how long an order took to be picked in the warehouse, when it was handed to the rider, and the exact second the customer signed for it at the door."

Measurable ROI & Safeguards

- Zero Double-Entry Overhead: Quotes instantly convert to Orders, Orders to Packing Slips, and Orders to KRA-signed Invoices.
- Elimination of Bad Debts: Automatic credit-ceiling enforcement blocks orders when customers exceed agreed limits.
- Order SLA Enforcement: Real-time alerts when orders languish in picking or packing beyond target thresholds.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    PROD[1. Products & Selling Prices Active] --> CUST[2. Customer Master Cards & Groups]
    CUST --> NUM[3. Number Series Bound: Quotes & Orders]
    NUM --> SLA[4. Order Stage Target Timers Set]
    SLA --> SALES[Sales Quotes & Orders Active]
```

1. Active Sellable Products: Products must be assigned a business classification marked as "Sold to Customers" with active selling prices.
2. Customer Master Cards: Cash or credit customer accounts must exist under /admin/customers.
3. Number Series: Document number series for Sales Quotes (QT-) and Sales Orders (ORD-) must be active under Settings > Number Series.
4. Order SLA Targets: Stage timeout targets should be defined under Settings > Order Targets to trigger late-stage notifications.
5. Staff Permissions:

- quotes:read, quotes:manage: Access and manage quotations.
- orders:read, orders:manage: Process orders and print packing slips.
- customers:read, customers:manage: Create and edit customer cards.

---

3. Module & Sub-Module Architecture

The Sales module coordinates five operational areas:

```
graph LR
    subgraph SalesSubModules["Sales Sub-Modules"]
        Q["Sales Quotes\n(/admin/quotes)"]
        O["Customer Orders\n(/admin/orders)"]
        C["Customers\n(/admin/customers)"]
        N["Newsletter\n(/admin/newsletter)"]
        R["Product Reviews\n(/admin/reviews)"]
    end

    subgraph FulfillmentFlow["Order-to-Cash Workflow"]
        CONF["Order Confirmed"]
        STK["Stock Reserved"]
        PICK["Pick & Pack"]
        SHIP["Dispatch & Deliver"]
        INV["KRA Tax Invoice"]
    end

    Q -- "Convert to Order" --> O
    C -- "Credit Check & Terms" --> O
    O --> CONF
    CONF --> STK
---    STK --> PICK
    PICK --> SHIP
    SHIP --> INV
```

4. Interface-by-Interface Walkthrough

Sub-Module 1: Sales Quotes (/admin/quotes)

- Quotes List View (QuotesTable): Paged table showing Quote Number, Customer Name, Issue Date, Expiration Date, Total Amount, and Status (Draft, Sent, Accepted, Converted, Expired, Rejected).
- Quote Creation & Editor (QuoteEditor):
- Customer selection with instant customer credit standing preview.

- Line-item editor: SKU selection, unit of measure, quantity, base price, discount %, and tax rate.
- Real-time calculation toolbar: Subtotal, total discount, VAT, and grand total.
- Validity expiration date picker and customer memo notes.
- Quote Detail Modal (QuoteDetails): Full proforma document view with actions to Print PDF, Send via Email, Accept, or Convert to Order.

Sub-Module 2: Customer Orders (/admin/orders)

- Orders List View: Filterable by Channel (Web Shop, Counter POS, Sales Rep, B2B Account), Status (Pending, Confirmed, Processing, Dispatched, Delivered, Cancelled), Payment Status (Unpaid, Paid, Partial), and Branch.
- Order Details Modal (OrderDetailsModal): Comprehensive multi-tab operational hub:

1. Order Summary Tab: Order lines, quantities, unit prices, applied coupons, delivery fees, and grand total.
2. Stage Timeline Panel (OrderTimelinePanel): Visual vertical timeline showing every stage event with exact server timestamps, actors, and duration spent in each stage.
3. Deliveries Panel (OrderDeliveriesPanel): Shows partial and complete dispatches, courier tracking numbers, and delivery run links.
4. Stock Allocation Panel (OrderStockPanel): Shows warehouse reservations, batches allocated, and goods issue status.
5. Collection Panel (OrderCollectionPanel): Payment method, gateway reference, M-Pesa receipt code, and door cash collection status.
6. KRA Invoice Preview (OrderInvoicePreview): Electronic tax invoice details, KRA digital signature code, and QR verification code.

Sub-Module 3: Customers (/admin/customers)

- Customer Directory: Tabular list showing Account Code, Customer Name, Account Type (Cash or Credit), Phone, Email, City, Balance Owed, and Status (Active, Suspended).
- Customer Card Details:
- Identity & Legal: Name, Trading Name, KRA PIN, Identity Document (National ID / Passport / Incorporation Cert).
- Credit Terms & Risk: Credit limit (e.g. KES 500,000), payment grace period (e.g. 30 days), and price group assignment (e.g. Wholesale Tier 1).
- Billing & Shipping Addresses: Multiple physical delivery locations with geo-coordinates.
- Dimension Bindings (DimensionFields): Cost center, sales territory, or industry vertical tags for financial reporting.

Sub-Module 4: Newsletter (/admin/newsletter)

- List of customer email subscribers captured via storefront footers and popups.
- Export to CSV action for email marketing campaigns (Mailchimp, Sendgrid).

Sub-Module 5: Product Reviews (/admin/reviews)

- Customer review moderation queue showing Product Name, Customer Name, Star Rating (1-5), Review Headline, Review Body, and Submission Date.
- Action buttons: Approve Review (publishes to storefront) or Reject / Archive (removes from public display).

---

5. Granular Field-by-Field Reference

Sales Quote Editor Fields (/admin/quotes/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Customer	Account receiving the quote.	Lookup Autocomplete	Required	Type 2+ characters to search customer cards.
Quote Date	Issue date.	Date (YYYY-MM-DD)	Required	Defaults to today's date.
Valid Until	Expiration date of quoted prices.	Date (YYYY-MM-DD)	Required	Must be $\ge$ Quote Date. Quotes cannot be converted after this date.
Branch	Fulfilling branch location.	Dropdown	Required	Determines inventory availability checks.
Product Line: SKU	Product item.	Lookup Autocomplete	Required	Must be an active, sellable product.
Product Line: Qty	Number of units quoted.	Decimal / Integer	Required	Must be > 0. Validated against unit of measure decimals.
Product Line: Price	Unit selling price.	Currency (KES)	Required	Defaults to price list. Cannot violate price floor.
Product Line: Discount	Line-item discount percentage.	Percentage (0-100%)	Optional	Capped by user's role override limit.
Notes / Terms	Commercial stipulations on quote.	Text (Max 500 chars)	Optional	Printed on proforma footer (e.g., "Lead time 7 days").

Customer Card Fields (/admin/customers/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Account Type	Determines payment requirements.	Radio Selector	Required	Cash (must pay before dispatch) or Credit (invoiced).
Customer Name	Full legal or business name.	Text (Max 120 chars)	Required	Displayed on tax invoices.
KRA PIN	Customer's tax registration.	Text (11 chars)	Optional for Retail, Required for B2B	Must be a valid KRA PIN format for B2B eTIMS transmission.
Phone Number	Primary mobile contact.	Phone (Max 25 chars)	Required	Used for delivery SMS notifications and M-Pesa STK push.
Email Address	Invoicing email.	Email (Max 160 chars)	Optional	PDF invoices and proformas are dispatched here.
Credit Limit	Maximum outstanding balance allowed.	Currency (KES)	Required if Credit	e.g. 250000.00. Orders blocked if balance + order $>$ limit.
Payment Terms	Grace period in days.	Integer (Days)	Required if Credit	e.g. 30 for Net 30 days.
Price Group	Custom pricing tier.	Dropdown	Optional	e.g. Retail, Wholesale, Contractor.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Creating and Converting a Proforma Sales Quote to an Order

1. Navigate to /admin/quotes and click New Quote in the top right.
2. In the Customer lookup field, search for and select the client (e.g. "Acme Construction Ltd").
3. Set the Valid Until date (e.g., 14 days from today).
4. Click Add Line Item.

5. In the product autocomplete, select the product (e.g., "500W Solar Panel").
6. Enter the Quantity (e.g. 20). The system automatically pulls the current customer group price.
7. Enter a negotiated discount (e.g. 5%). The system recalculates the subtotal and VAT in real time.
8. Click Save Quote. The quote status becomes Sent.
9. When the client approves the proposal, open the quote details and click the Convert to Order button.
10. An InlineConfirm prompt appears: "Convert Quote #QT-00142 into a confirmed Sales Order?"
11. Click Confirm Conversion.
12. The quote status updates to Converted, and the system redirects to the newly created Sales Order with all lines, prices, and customer details intact.

Action 2: Inspecting an Order's Timestamped Stage Timeline

1. Navigate to /admin/orders.
2. Locate the order using the search bar (by Order Number, Customer Name, or Phone).
3. Click the order row to open the Order Details Modal.
4. Click the Timeline tab.
5. The timeline renders chronologically:

- 09:14:02 — Order Placed (Channel: Web Shop, Actor: Customer).
- 09:14:05 — Payment Confirmed (M-Pesa STK Push: KES 14,500, Ref: QA712XYZ).
- 09:15:20 — Stock Reserved (Warehouse: Central Depot, 2 items locked).
- 09:40:12 — Pick Started (Actor: John Kamau - Storekeeper).
- 09:52:00 — Packed & Ready for Dispatch (Packing Slip #PS-1082 printed).
- 10:15:30 — Dispatched (Rider: Peter Ochieng, Run #RUN-04).
- 10:48:15 — Delivered (Customer Signature & Verification PIN Confirmed).

6. Hover over any stage to view duration spent (e.g., "Time in Packing: 11 minutes").

Action 3: Reviewing and Moderating Customer Product Reviews

1. Navigate to /admin/reviews.
2. The table displays pending reviews submitted by shoppers on the storefront.
3. Review the star rating, title, and comments.
4. To approve: Click Approve. The review immediately becomes visible on the product page and updates the product's aggregate star rating.
5. To reject: Click Reject. An inline text prompt asks for an internal reason (e.g., "Offensive language" or "Spam"). Click Confirm Rejection.

---

7. Conditions, Business Rules & Edge Cases

The Credit Limit Breach Rule

- When an order is placed on credit (either via counter order or quote conversion), the system checks:

$$\text{Current Unpaid Balance} + \text{New Order Total} \le \text{Customer Credit Limit}$$

- If the calculation exceeds the limit:

- If an Approval Workflow for credit override is configured, the order status halts in Pending Credit Approval and alerts the Credit Manager.
- If no approval workflow exists, order creation is blocked with the error: "Customer credit limit exceeded by KES [Amount]. Payment required before placement."

Stage Immutability & Reversals

- Once an order stage timestamp is recorded in the append-only event log, it can never be edited or deleted.
- If an order must be cancelled after dispatch:

1. The order status moves to Delivery Failed or Cancelled.
2. A return-to-stock movement is issued, recording the physical return of the goods to inventory.
3. A credit note is generated referencing the original invoice.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create a test Cash Customer and Credit Customer.	Both customers saved; credit limit reflected on Credit Customer.	[ ]
2	Issue a Sales Quote with a 10% line discount.	Proforma calculates discount and VAT accurately; status shows Draft/Sent.	[ ]
3	Convert Quote to Order.	Order created with identical lines; Quote status transitions to Converted.	[ ]
4	Attempt to place an order exceeding Credit Customer's limit.	System blocks or routes to Credit Approval queue as configured.	[ ]
5	Verify Order Timeline Events.	Placement, payment, and dispatch timestamps render accurately in order modal.	[ ]

Interactive version with live navigation: /admin/docs/03-sales-quotes-and-customer-orders

CHAPTER 04

Guide 04: Pricing Engine, Promotions & Margins (`/admin/pricing`)

Category: Commercial Operations | Reading Time: ~11 min | Words: 2,198

*The **Pricing Engine** is the commercial brain of QSell. It governs how base prices are determined across different customer tiers, retail channels, and branches, while enforcing strict margin protection rules, promotional coupons, retrospe*

The Pricing Engine is the commercial brain of QSell. It governs how base prices are determined across different customer tiers, retail channels, and branches, while enforcing strict margin protection rules, promotional coupons, retrospective rebates, loyalty points, and role-based discount caps.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- Erosion of Profitability Through Uncontrolled Discounting: In competitive sales environments, sales reps frequently offer deep discounts to close deals, inadvertently selling products below cost or wiping out gross margin.
- Accidental Discount Stacking: Businesses often run promotional coupons (e.g. 10% off) that unintentionally stack on top of existing wholesale volume discounts and loyalty redemptions, leading to crippling margin loss.
- Complexity in Multi-Tier B2B Pricing: Managing different price sheets for retail customers, wholesale stockists, contractors, and export clients on paper or spreadsheets causes billing errors and client disputes.
- No Explanation of How a Price Was Reached: When a customer asks "Why was I charged KES 1,850 instead of KES 1,700?", legacy ERPs cannot explain the calculation. QSell generates a deterministic Price Trace explaining every step in the waterfall.

Target Client Profiles

- Wholesalers & FMCG Distributors: Utilizing customer tiers (Wholesale A/B/C), minimum order quantities (MOQs), and tiered volume breaks (e.g. 1-10 units @ KES 1,000; 11-50 units @ KES 920; 51+ units @ KES 850).
- Multi-Branch Retailers: Charging localized prices based on branch geography or transport costs (e.g. Nairobi vs. Lodwar).
- High-Margin Hardware & Contractor Suppliers: Requiring cost-plus minimum price floors to ensure every sale yields at least a 15% gross margin.

Commercial Sales Pitch & Talk Track

> "With QSell, you will never accidentally sell a product at a loss again. Our pricing engine lets you configure sophisticated wholesale price lists, volume breaks, and seasonal promotional codes, while locking your bottom line with automated Price Floors. If a salesperson attempts to discount an item below your required margin floor, the system blocks the transaction or automatically routes it to the Managing Director for approval. Best of all, our Price Check simulator allows you to test any pricing rule before it goes live."

Measurable ROI & Safeguards

- 100% Margin Floor Lock: Hard stop prevents any quote or order from being sold below cost + minimum markup.
- Deterministic Price Trace: Complete auditability on every transaction line item.
- Controlled Discretionary Limits: Sales reps, supervisors, and managers each have defined discount authority ceilings.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    PROD[1. Products & Base Landed Costs Configured] --> GRP[2. Customer Groups Defined: Wholesale, Retail]
    GRP --> LIST[3. Price Lists Created with Scope & Validity]
    LIST --> FLOOR[4. Price Floors Set: Cost + Minimum %]
    FLOOR --> LIMIT[5. Staff Override Limits Assigned]
    LIMIT --> SIM[6. Price Check Simulator Verification]
```

1. Product Master Data: SKUs must exist with valid Base Selling Prices and Unit Cost Prices under Catalog > Products.
2. Customer Groups: Defined under /admin/pricing/groups (e.g. Retail Cash, Trade Tier 1, Government Tender).
3. Price Lists: Created with currency (KES), validity dates, and branch/customer scope.
4. Price Floors: Configured under /admin/pricing/floors to set non-negotiable minimum profit barriers.
5. Override Limits: Configured under /admin/pricing/limits to set role-based discount authority.
6. Required Permissions: pricing:read to view and pricing:manage to create or alter price rules.

---

3. Module & Sub-Module Architecture

The Pricing Engine operates as a deterministic 2-phase calculation waterfall:

```
graph TD
    subgraph Phase1["Phase 1: Base Price Resolution (Choose Winner)"]
        IN[Inputs: Customer, Group, Branch, SKU, Qty, Date]
        IN --> MATCH[Match Active Price Sources]
        MATCH --> RANK[Rank by Specificity: Customer > Group > Branch > Standard]
        RANK --> PRIO[Break Ties: Priority > Newest Effective Date]
        PRIO --> BASE[Base Price Chosen]
    end

    subgraph Phase2["Phase 2: Adjustments & Safeguards (Stack & Check)"]
        BASE --> DISC[Apply Eligible Discounts & Coupons]
        DISC --> REB[Accrue Retrospective Rebates]
        REB --> PROMO[Evaluate Loyalty Points & Gift Cards]
        PROMO --> FLOOR{Check Price Floor: Price >= Cost + Min Margin?}
        FLOOR -- "Compliant" --> WIN[Effective Final Price]
        FLOOR -- "Breached" --> APP{Within Staff Override Limit?}
        APP -- "Yes" --> WARN[Flag Margin Warning & Allow]
---     APP -- "No" --> BLOCK[Route to Approval / Block Sale]
    end
```

4. Interface-by-Interface Walkthrough

Sub-Module 1: Customer Groups (/admin/pricing/groups)

- Defines customer segmentation tiers (e.g., Walk-in Retail, Wholesale Bronze, Wholesale Platinum, Staff).
- Allows assigning customers in bulk or binding automatic qualifying criteria.

Sub-Module 2: Price Lists (/admin/pricing)

- Table of active and scheduled price lists showing Code, Name, Currency, Priority, Effective Dates, and Scope.
- Line Editor: Set prices per SKU, Unit of Measure, and Quantity Breaks (e.g. Buy 10+, Buy 50+, Buy 100+).

Sub-Module 3: Discounts & Coupons (/admin/pricing/discounts)

- Configures promotional campaigns:
- Type: Percentage discount (e.g. 10% off), Fixed amount (e.g. KES 500 off), or Free Item.
- Trigger: Coupon code entry, minimum cart value threshold (e.g. Spend KES 10,000+), or specific product bundles.

- Stacking Rules: Whether this discount can combine with other active promotions.

Sub-Module 4: Rebates (/admin/pricing/rebates)

- Long-term customer loyalty agreements:
- Tracks cumulative purchase volume over a calendar quarter or year.
- Accrues rebate percentages (e.g. "Buy KES 2,000,000 of cement in Q1, receive 3% credit rebate").
- Automatically generates Accounts Receivable Credit Memos upon milestone attainment.

Sub-Module 5: Loyalty Points (/admin/pricing/loyalty)

- Point accrual engine: Set earn rate (e.g. 1 Point per KES 100 spent) and redemption rate (e.g. 1 Point = KES 1.00 store credit).
- Expiration rules and minimum points required to redeem.

Sub-Module 6: Gift Cards (/admin/pricing/gift-cards)

- Prepaid voucher management: Generate digital gift card codes, assign balances, track partial redemptions, and monitor expiration dates.

Sub-Module 7: Price Book (/admin/pricing/book)

- Printable and exportable matrix showing what every product sells for in any combination of Customer Group, Branch, and Quantity tier. Ideal for distributing physical price sheets to field sales agents.

Sub-Module 8: Pricing Insights (/admin/pricing/insights)

- Executive analytics dashboard displaying:
- Average realized gross margin across all orders.
- Total revenue surrendered to promotional discounts and coupon codes.
- Frequency of manual price overrides by sales representatives.

Sub-Module 9: Price Floors (/admin/pricing/floors)

- The ultimate margin defense system:
- Configures minimum selling prices based on:
- Cost + Percentage Markup (e.g. Cost + 15%).
- Cost + Fixed Amount (e.g. Cost + KES 200).
- Hard Absolute Price Floor (e.g. Never sell below KES 1,500).
- Scoped by specific Product, Product Group, or Category.

Sub-Module 10: Override Limits (/admin/pricing/limits)

- Governs how far a staff member may manually lower a price or apply a discretionary discount:
- Junior Sales Rep: Max 3% discount, cannot breach price floor.
- Senior Sales Account Manager: Max 10% discount.
- Branch Manager: Max 18% discount.
- Commercial Director: Unlimited override authority.

Sub-Module 11: Price Check / Simulator (/admin/pricing/check)

- The interactive sandbox where implementors and sales reps can test pricing scenarios:

- Input: Select a Customer, Branch, Product SKU, Quantity, and Date.
- Output: Instant live calculation with full waterfall trace displaying every considered price list, why one won, and which discounts applied.

Sub-Module 12: Price Rules (/admin/pricing/rules)

- Master resolution settings configuring the relative weights and specificity rankings of competing price sources.

---

5. Granular Field-by-Field Reference

Price List Line & Quantity Break Fields (/admin/pricing/[id])

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Product / SKU	Target item being priced.	Lookup Autocomplete	Required	Must be an active sellable product.
Unit of Measure	Selling package.	Dropdown	Required	e.g. Piece, Box of 24, Kilogram.
Minimum Quantity	Tier threshold (From Qty).	Integer / Decimal	Required	Tier 1 begins at 1. Subsequent tiers must be higher.
Tier Price	Unit price for this quantity tier.	Currency (KES)	Required	Must be > 0. Format: 2 decimal places.
Pricing Method	How tier applies.	Radio Selector	Required	All Units (retroactive to unit 1) or Incremental (bracketed).
Start Date	Effective start date.	Date (YYYY-MM-DD)	Optional	If blank, active immediately upon publishing.
End Date	Expiration date.	Date (YYYY-MM-DD)	Optional	Must be $\ge$ Start Date. Automatically retires at 23:59:59.

Price Floor Fields (/admin/pricing/floors/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Scope Type	Level of floor application.	Dropdown	Required	Company-Wide, Category, Product Group, Specific SKU.
Target Entity	Selected category or product.	Lookup Autocomplete	Required if not Company-Wide	Selects the entity to protect.
Floor Method	How the minimum price is calculated.	Dropdown	Required	Cost + Markup %, Cost + Fixed Amount, Fixed Minimum Price.
Floor Value	The rate or amount.	Decimal / Currency	Required	e.g. 15.0 for 15% markup, or 250.00 for KES 250 margin.
Enforcement	Behavior upon violation.	Radio Selector	Required	Hard Stop (Block Sale) or Require Manager Approval.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Setting Up a Tiered Wholesale Price List

1. Navigate to /admin/pricing and click New Price List.

2. Enter List Name: "Wholesale Platinum Tier 1".
3. Set Currency: KES.
4. Set Scope: In the Customer Group dropdown, select "Wholesale Platinum".
5. Set Priority: 10 (higher priority wins over generic lists).
6. Click Save & Add Products.
7. In the product lines table, click Add Product.
8. Select the SKU (e.g., "Cement 50kg Bag").
9. In Tier 1: Set From Qty: 1, Price: 750.00.
10. Click Add Tier: Set From Qty: 50, Price: 710.00.
11. Click Add Tier: Set From Qty: 200, Price: 680.00.
12. Click Save Price List.
13. Status updates to Active. Any Wholesale Platinum customer ordering 50+ bags instantly receives KES 710.00.

Action 2: Protecting Margins by Configuring a Price Floor

1. Navigate to /admin/pricing/floors and click New Price Floor.
2. Set Scope Type to Category.
3. Select Category: "Power Tools & Machinery".
4. Set Floor Method: Cost + Markup %.
5. Enter Floor Value: 20.0 (meaning Cost + 20%).
6. Set Enforcement: Hard Stop (Block Sale).
7. Click Activate Price Floor.
8. A green confirmation banner displays: "Price floor activated. 48 products protected." Any attempt to sell a power tool below cost + 20% is now impossible across all channels.

Action 3: Simulating a Transaction in the Price Check Sandbox

1. Navigate to /admin/pricing/check.
2. In the Customer lookup, select "QuickFix Contractors".
3. In the Product lookup, select "Inverter 5kVA".
4. In the Quantity box, type 5.
5. In the Branch dropdown, select "Nairobi Westlands".
6. Click the Simulate Price button.
7. The Waterfall Trace displays:

- Standard Retail Price: KES 120,000.00
- Customer Group Match (Trade): KES 108,000.00 (Wins by Specificity)
- Quantity Break (5+ units): KES 102,000.00
- Active Coupon SAVE5: - KES 5,100.00 (Applied)
- Price Floor Check: Landed Cost KES 85,000.00 + 15% Floor (KES 97,750.00).
- Floor Result: PASS (Net KES 96,900 is below floor? Triggers warning/approval).

8. The tool displays the exact formula and reason string that will be written to the order.

---

7. Conditions, Business Rules & Edge Cases

Specificity Hierarchy (Who Wins When Prices Compete?)

When multiple active price sources match an order line, the winner is determined strictly by the Specificity Hierarchy:

1. Contract Locked Price (Explicit legal agreement with client).
2. Customer-Specific Price (Negotiated price sheet for this single account).
3. Customer Group Price (Tiered wholesale or trade group).
4. Route / Van Price (Mobile distribution route).
5. Branch / Location Price (Regional store price).
6. Standard Product Price (Base catalog price).

Ties are broken by Price List Priority integer, then Newest Effective Date.

Preventing Coupon Cannibalization

- If a discount has the flag allow_stacking = false, applying it disables all other promotional discounts on that order line.
- The system automatically compares the single unstacked discount against the sum of competing stacked discounts and awards the customer the better of the two deals without double-dipping.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create Customer Group "Wholesale" and assign a test customer.	Customer profile reflects the new group.	[ ]
2	Create a Price List for Wholesale with quantity break at 10 units.	Buying 9 units quotes Base Price; buying 10 units quotes Tier Price.	[ ]
3	Configure a 20% Cost-Plus Price Floor on a test product.	Selling below Cost + 20% is blocked with explicit floor warning.	[ ]
4	Test Staff Override Limit for a Junior Sales Rep.	Entering a manual discount above 5% is blocked.	[ ]
5	Run Price Check Simulator on 3 complex scenarios.	Trace displays correct winning source, applied discounts, and margin check.	[ ]

Interactive version with live navigation: /admin/docs/04-pricing-engine-promotions-and-margins

CHAPTER 05

Guide 05: Products Catalog & Inventory Control (`/admin/catalog`, `/admin/inventory`)

Category: Supply Chain & Inventory | Reading Time: ~12 min | Words: 2,242

*The **Products & Inventory Control** module governs the entire physical life of products—from SKU master definition and multi-level category classification to multi-location warehouse management, units of measure conversions, kits and hampe*

The Products & Inventory Control module governs the entire physical life of products—from SKU master definition and multi-level category classification to multi-location warehouse management, units of measure conversions, kits and hampers, two-step stock transfers, and continuous physical cycle counts.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The "Single Stock Number" Blind Spot: In basic systems, a product has one global stock number (e.g. "50 units on hand"). Management has no idea if those 50 units are in the Nairobi central depot, the Mombasa branch, or on a rider's delivery van, leading to false promises to customers.
- Lost Goods in Transit: Moving stock between warehouses or out on delivery vans is where theft and shrinkage occur. Traditional spreadsheets write off stock immediately upon dispatch, losing all accountability while goods are on the road. QSell introduces an In-Transit Ledger where goods remain on the company's books until formally acknowledged and received at the destination.
- Stock Counts That Force Business Closures: Businesses often shut down operations for an entire weekend to conduct physical inventory counts. In QSell, Continuous Cycle Counting allows warehouse staff to count shelves while retail cashiers and the web shop continue selling without discrepancy errors.
- Unit of Measure Confusion: Selling a product in pieces, 12-packs, and 24-box cartons typically causes inventory chaos. QSell enforces strict base-unit decimal conversions (e.g., kilograms, litres, pieces).

Target Client Profiles

- Multi-Branch Retailers & Wholesalers: Operating central distribution hubs, regional branch stores, and mobile van-sales fleets.
- Hardware, Agro-Dealers & Building Suppliers: Selling goods by weight, length, volume, or pack sizes (cement, paint, fertilizer, cabling).
- Gift, Hamper & Food Distributors: Assembling multi-item gift hampers and combo kits priced as a single unit but fulfilled and taxed as individual components.

Commercial Sales Pitch & Talk Track

> "With QSell, you always know exactly what you have, exactly where it sits, and what it is worth. Your inventory isn't just a number on a spreadsheet—it is an immutable ledger. You can see real-time stock across every warehouse, branch shop, and delivery van. When you move stock between locations, it is tracked in transit. And best of all, you can perform physical stock audits on your shelves during business hours while your cashiers keep ringing up sales."

Measurable ROI & Safeguards

- Zero Shrinkage in Transit: Two-step transfers require sender dispatch and receiver sign-off with variance reporting.

- Zero Interrupted Trading: Continuous cycle counts eliminate the need for costly weekend store closures.
- Kit Component Explosion: Bundles automatically deduct stock and calculate KRA tax for individual underlying components.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    UOM[1. Units of Measure: Pcs, Kg, Box Defined] --> CLAS[2. Business Classifications: Stocked, Service]
    CLAS --> LOC[3. Stock Locations Created: Warehouses, Stores, Vans]
    LOC --> CAT[4. Categories & Tax Classes Mapped]
    CAT --> PROD[5. SKUs & Kits Configured]
    PROD --> CUT[6. Stock Ledger Opening Balance Cutover]
```

1. Units of Measure: Defined under /admin/settings/units (Base units and conversion multipliers).
2. Business Classifications: Defined under /admin/item-types (Determining whether an item is physical stock, service, or consumable).
3. Stock Locations: Defined under /admin/inventory/locations (At least one default warehouse or shop).
4. Categories: Created under /admin/categories with mapped KRA tax classifications.
5. Staff Permissions:

- products:read, products:manage: Create and manage catalog SKUs.
- stock:read, stock:manage: View stock levels, issue transfers, and conduct counts.
- stock:cutover: Execute initial opening balance migration.

---

3. Module & Sub-Module Architecture

The Products & Inventory module is divided into two operational domains:

```
graph TD
    subgraph CatalogDomain["Catalog Definition (/admin/catalog)"]
        PROD["Products Master & SKUs"]
        CAT["Category Hierarchy"]
        CLAS["Business Classifications"]
        UOM["Units of Measure & Decimals"]
        KITS["Kits & Hampers (Bundles)"]
    end

    subgraph InventoryDomain["Warehouse Management (/admin/inventory)"]
        LOC["Stock Locations (Depot, Shop, Van)"]
        BAL["Stock Balances: On-Hand vs Reserved"]
        TRANS["Stock Transfers & In-Transit"]
        COUNTS["Physical Cycle Counts & Audits"]
        LEDGER["Immutable Stock Ledger"]
    end

    CatalogDomain --> InventoryDomain
```

---

4. Interface-by-Interface Walkthrough

Sub-Module 1: Products Master (/admin/products)

- Product Directory: Server-paged table with instant filtering by Category, Stock Status (In Stock, Low Stock, Out of Stock), Classification, and Channel visibility.
- Product Editor (/admin/products/[id]):
- Basic Info: Title, SKU, Barcode (EAN-13/UPC), Brand, Category, Short Description, and HTML Rich-Text Body.

- Pricing & Tax: Selling Price, Cost Price (for gross margin calculation), KRA Tax Classification (16% VAT, Zero-Rated, Exempt).
- Stock & Units: Unit of Measure picker, Minimum Reorder Threshold, Safety Stock, and Track Inventory toggle.
- Web Shop Flags: Show on Web Shop toggle, SEO title/description, and high-resolution gallery image uploader.

Sub-Module 2: Trash (/admin/trash/products)

- Soft-delete recovery screen. Products removed from the catalog can be Restored as Drafts or permanently purged (only if no historic transaction or stock ledger records exist).

Sub-Module 3: Business Classifications (/admin/item-types)

- Establishes product accounting behavior:
- Stocked Goods: Physical inventory tracked in warehouses.
- Non-Stocked Services: Labor, installation, delivery fees, consulting.
- Consumables / Packaging: Boxes, tape, branded bags used at checkout.
- Controls whether the item is "Sold to Customers" across quotes, orders, and storefront.

Sub-Module 4: Units of Measure (/admin/settings/units)

- Global table of measurement units (e.g., PCS, KG, LTR, BOX, MTR).
- Enforces decimal precision (e.g. Kilograms allow 3 decimals 1.425 kg, while Pieces allow 0 decimals 1 pc).
- Defines conversion ratios (e.g., 1 Box = 24 Pieces).

Sub-Module 5: Categories (/admin/categories)

- Multi-level tree view for organizing products into departments, categories, and subcategories. Drives storefront mega-menus and automated KRA tax inheritance.

Sub-Module 6: Kits and Hampers (/admin/pricing/kits)

- Bundle creator: Defines combo packages (e.g., "Solar Home Kit 1kVA" or "Executive Christmas Hamper").
- Line component mapping: Links 4 individual panels, 1 inverter, 2 batteries, and 1 mounting kit into a single sellable SKU.
- Component Tax Handling: Automatically apportions VAT across components with differing tax rates.

Sub-Module 7: Stock Locations (/admin/inventory/locations)

- Defines physical locations:
- Warehouses: Central distribution godowns.
- Retail Stores: Front-office shops where point-of-sale orders draw inventory.
- Delivery Vans: Mobile stock locations assigned to riders for run dispatch.
- Quarantine / Damaged: Holding locations for defective customer returns.

Sub-Module 8: Stock by Location (/admin/inventory/by-location)

- Real-time stock matrix displaying product rows across location columns, showing:
- On-Hand: Total physical units on shelves.
- Reserved: Units committed to active customer orders awaiting dispatch.
- Available: Uncommitted stock available to promise (On-Hand - Reserved).

Sub-Module 9: Stock Transfers (/admin/inventory/transfers)

- Two-step inter-location workflow:

1. Dispatch Transfer: Source location picks and dispatches goods. Stock moves from source On-Hand to In-Transit.
2. Receive Transfer: Destination location inspects physical delivery and accepts goods. Stock moves from In-Transit to destination On-Hand.
3. Discrepancy Reporting: Records damaged or missing items during transit.

Sub-Module 10: Stock Counts (/admin/inventory/counts)

- Cycle counting engine:
- Create a count session for a specific warehouse, aisle, or category.
- Staff scan barcodes and record physical counts on mobile or desktop.
- System freezes an inventory snapshot at count initiation: Cashiers keep selling!
- Reconciliation view highlights positive/negative variances with KES financial impact.
- Manager approves count, posting adjustment entries to the stock ledger.

Sub-Module 11: Stock Ledger & Cutover (/admin/inventory/ledger)

- The single source of truth: An immutable, append-only log of every physical movement (receipts, sales dispatches, transfers, adjustments, returns, write-offs).
- Includes the Cutover Wizard used during client onboarding to import verified opening inventory balances.

---

5. Granular Field-by-Field Reference

Product SKU Master Fields (/admin/products/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Product Title	Name displayed on shop & quotes.	Text (Max 160 chars)	Required	Must be unique within company.
SKU Code	Unique internal stock identifier.	Text (Alphanumeric, Max 40)	Required	Uppercase recommended (e.g. SLR-PAN-500W).
Barcode	EAN/UPC barcode for POS scanners.	Text (Max 48 chars)	Optional	Must be unique across all active products.
Category	Organizational department.	Dropdown	Required	Determines tax rate and storefront navigation.
Unit of Measure	Selling unit.	Dropdown	Required	Select from active UOMs (e.g. Piece, Box).
Selling Price	Base retail price.	Currency (KES)	Required	Inclusive or exclusive of VAT per company setting.
Cost Price	Landed purchase cost.	Currency (KES)	Required	Used for margin floors and gross profit KPI.
Tax Classification	KRA eTIMS tax rate.	Dropdown	Required	General Rate (16%), Zero-Rated (0%), Exempt.
Track Inventory	Enables warehouse stock tracking.	Toggle (On/Off)	Required	Set to Off for services and digital goods.
Low Stock Threshold	Alert trigger limit.	Integer / Decimal	Optional	Generates dashboard attention warning when stock drops below.
Show on Web Shop	Customer storefront visibility.	Toggle (On/Off)	Required	If Off, product only sellable via POS or B2B Quotes.

Stock Transfer Fields (/admin/inventory/transfers/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Transfer Number	Document sequence ID.	Text (Auto-generated)	System	e.g. TRF-2026-0042.
Origin Location	Source warehouse or shop.	Dropdown	Required	Must have sufficient available stock.
Destination Location	Receiving warehouse or van.	Dropdown	Required	Cannot equal Origin Location.
Product Line: SKU	Item being transferred.	Lookup Autocomplete	Required	Must exist at Origin Location.
Product Line: Qty	Number of units to move.	Decimal / Integer	Required	Cannot exceed Origin's available stock.
Carrier / Driver	Person or courier moving goods.	Text (Max 120 chars)	Optional	Staff driver or courier tracking code.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Executing an Inter-Branch Stock Transfer

1. Navigate to /admin/inventory/transfers and click New Transfer.
2. Select Origin Location: "Central Depot - Industrial Area".
3. Select Destination Location: "Westlands Retail Store".
4. In the lines section, search for the product (e.g., "100Ah Deep Cycle Battery").
5. Enter Quantity: 10. The system verifies that 10 units are available at Central Depot.
6. Click Dispatch Transfer.
7. An InlineConfirm prompt appears: "Dispatch 10 units into In-Transit status?" Click Confirm Dispatch.
8. Print the Transfer Delivery Note for the driver.
9. When the truck arrives at Westlands, the storekeeper opens /admin/inventory/transfers, selects the transfer, and clicks Receive Goods.
10. If all 10 arrived safely, click Accept All into Stock.
11. If 1 battery was cracked during transport, enter Received: 9, Damaged: 1.
12. Click Confirm Receipt. The system places 9 batteries into Westlands stock and automatically routes the 1 damaged battery into the Quarantine location.

Action 2: Conducting a Cycle Count While Trading Continues

1. Navigate to /admin/inventory/counts and click Start New Count.
2. Select Location: "Mombasa Road Showroom".
3. Select Scope: "Lighting & Fixtures" category.
4. Click Initialize Count Session.
5. The system captures an immutable snapshot of current book stock. Staff take barcode scanners to the shelves and scan physical items.
6. Even if a customer buys a light bulb at the counter while the count is underway, the system's timestamped calculation accounts for the transaction without throwing off the audit.
7. Click Review Variances.
8. The screen displays discrepancies (e.g. Book Stock: 42, Physical Scanned: 40, Variance: -2 units, Financial Loss: KES 1,600).

9. Enter an explanation reason: "Damaged showroom display units written off".
10. Click Approve & Post Adjustments.
11. Stock balances are instantly aligned with physical reality, and an inventory write-off entry is posted to the General Ledger.

---

7. Conditions, Business Rules & Edge Cases

The Three Stock Numbers (On-Hand, Reserved, Available)

$$\text{Available Stock} = \text{On-Hand Stock} - \text{Active Unfulfilled Reservations}$$

- On-Hand: Physical boxes on the shelf.
- Reserved: Goods promised to orders currently being packed or awaiting dispatch.
- Available: What the storefront and sales reps can sell. An order for 10 units will be refused if only 8 units are Available, even if 15 units are physically On-Hand.

Deactivating a Stock Location

- A location cannot be deleted or deactivated if:

1. It holds physical On-Hand stock > 0.
2. It has open in-flight stock transfers.
3. It has active orders with unfulfilled reservations.

- The system blocks deactivation with the message: "Cannot deactivate location: 14 units remain on hand. Transfer or write off stock first."

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create Base Units of Measure (PCS with 0 decimals, KG with 3 decimals).	Decimal constraints enforced accurately during entry.	[ ]
2	Create Central Warehouse, Retail Shop, and Van locations.	Locations appear in Stock by Location matrix.	[ ]
3	Execute Opening Balance Cutover via Stock Ledger Wizard.	On-hand quantities reflect physical counts accurately.	[ ]
4	Perform a 2-step stock transfer from Warehouse to Retail Shop.	Stock moves to In-Transit upon dispatch, and to destination upon receipt.	[ ]
5	Build a Kit Product and place a test order.	Master kit is sold; individual component stock quantities are deducted.	[ ]

Interactive version with live navigation: /admin/docs/05-products-catalog-and-inventory-control

CHAPTER 06

Guide 06: Delivery Logistics & Field Dispatch (`/admin/delivery`)

Category: Supply Chain & Field | Reading Time: ~43 min | Words: 8,503

*The **Delivery & Field Logistics** module orchestrates the physical journey of goods from the dispatch warehouse to the customer's doorstep. It governs polygon delivery zoning, dynamic rate calculation (by zone, weight, or distance), visual*

The Delivery & Field Logistics module orchestrates the physical journey of goods from the dispatch warehouse to the customer's doorstep. It governs polygon delivery zoning, dynamic rate calculation (by zone, weight, or distance), visual dispatch boards, rider run sequencing, Cash-on-Delivery (COD) custody reconciliation, third-party courier APIs, and scheduled on-site installation visits.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The "No Postcode" Reality in Africa: Standard international e-commerce software assumes structured postal codes (ZIP codes). In Kenya, addresses are descriptive (e.g., "Westlands, Muthithi Road, opposite Unga House, 3rd Floor"). QSell allows customers to either select localized zones (County $\rightarrow$ Town $\rightarrow$ Estate) or drop a GPS pin on the map, finding the exact delivery zone and fee automatically.
- Cash on Delivery (COD) Reconciliation Black Holes: Riders collect cash and direct M-Pesa payments at customer doorsteps. In manual systems, days pass before riders hand in cash, leading to lost money, disputed balances, and theft. QSell treats Rider Cash Custody as a living ledger: every shilling collected is recorded against the rider's profile, and the cashier must formally accept the cash handover to clear the rider.
- Unverified Third-Party Courier Billing: Couriers invoice merchants at the end of the month, and most businesses pay these invoices blindly. In QSell, what each courier charged for each parcel is recorded against that parcel as it happens. When the courier's invoice arrives, the office types it in and matches it to those recorded charges; the screen says plainly whether the two agree, or the invoice is higher or lower, and by how much. Charges the courier has made that no invoice has yet covered are listed so nothing is forgotten, and a charge can sit on one invoice only, so the same parcel cannot be paid twice.
- Overpromising and Missed Delivery Windows: Dispatchers accept more orders than their vehicle fleet can carry. QSell enforces Delivery Slots with Capacity Limits (and a weight limit if wanted), closing a slot once it is full, and refusing times that are too soon for the goods to be ready. The shopper chooses a day and time at checkout; the promise is shown on the dispatch board and on the customer's tracking page.
- Finding out a delivery is late only when the customer complains: Every delivery promised a time window is watched. It is flagged At risk before the window ends and Late after it, and a delivery that stays late goes up to the supervisor. The dispatchers and supervisors can be mailed automatically, and every flagged delivery is acknowledged with a note saying what is being done about it.
- Riders who share one login, or whose phones are not secure: A rider with an account of their own signs in with their phone number and a 6-digit PIN, chosen by the rider and known only to them. Wrong guesses shut the rider out for a while, so a lost or borrowed phone is not a way in. A rider can see their own earnings and what they have been paid, which ends the daily argument about pay.
- Not knowing whether delivery makes money: The delivery report and the performance screen show what customers paid for delivery against what it cost (rider pay, courier charges, fuel, vehicle costs), and how often deliveries are on time, succeed at the first try, and how full each delivery time runs.

Target Client Profiles

- Omnichannel Retailers & E-Commerce Brands: Operating in-house motorcycle fleets or hybrid fleets (own riders for local delivery, couriers for upcountry).
- FMCG & Beverage Distributors: Managing multi-drop truck and van delivery runs with cash settlement upon delivery.
- Solar PV, Electronics & Equipment Suppliers: Requiring technical crews to visit customer premises, complete installation checklists, and capture digital customer sign-offs.

Commercial Sales Pitch & Talk Track

> "With QSell, delivery is no longer an operational headache that eats away your profit. Whether you use your own motorcycle riders, third-party couriers, or have customers collect at your counter, QSell handles it seamlessly. Your customers can drop a pin on a map or pick their estate; your dispatch board clusters orders into optimized delivery runs; riders confirm deliveries using the customer's delivery code; and any cash collected at the door is locked in a digital custody ledger until handed to your cashier."

Measurable ROI & Safeguards

- 100% Cash-on-Delivery Accountability: Zero unrecorded door payments; riders cannot start new runs until previous cash floats are reconciled.
- 15–20% Reduction in Last-Mile Delivery Spend: Automated distance/weight rate cards prevent undercharging customers for shipping.
- Protection Against Courier Overbilling: Each courier invoice is checked against the charges recorded for that courier's parcels, and a charge cannot be counted on two invoices.
- Fewer Late Deliveries Going Unnoticed: Every delivery promised a window is raised at risk, late and still-late, once each, with a record of who acknowledged it and what they said.
- Fewer Rider Pay Disputes: Pay is recorded as each delivery is made, taken back only with a written reason, and paid out in one payout that is kept for good.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    LOC[1. Stock Locations & Dispath Origins Set] --> ZONE[2. Delivery Zones & Polygons Defined]
    ZONE --> METH[3. Delivery Methods Enabled per Branch]
    METH --> CHRG[4. Delivery Rate Cards & Policies Configured]
    CHRG --> RID[5. Riders & Couriers Registered]
    RID --> MAP[6. Map Service Selected: OSM vs Google]
```

1. Dispatch Origins: Stock locations must have physical street addresses and GPS coordinates under Inventory > Stock Locations.
2. Delivery Zones: Defined under /admin/delivery/zones with mapped towns and estates.
3. Delivery Methods: Enabled under /admin/delivery/methods per branch (Own Fleet, Courier, Click & Collect / Pickup, Scheduled Visit).
4. Delivery Charges: Rate cards configured under /admin/delivery/charges.
5. Riders & Couriers: Choose which kinds of rider the company uses under /admin/delivery/riders (own staff, riders with an account of their own, or both); add riders there. A rider who will sign in with a phone number and PIN needs an account of their own and a Kenyan mobile number (for example 0712 345 678). Couriers are connected under /admin/delivery/couriers.
6. Delivery Times (optional): Slots, days with no slots and lead times are set under /admin/delivery/slots; a branch needs at least one slot for shoppers to be offered a delivery time.
7. Rider Pay (optional): Set what a rider earns for each delivery and for each day on a run under /admin/delivery/costs. To have rider pay appear in the accounts, choose the two accounts under In the books: the account rider pay is booked to, and the account that holds what is owed to riders (the books must be live; otherwise pay is still recorded and paid out, but nothing is posted).

8. Mail for Lateness (optional): Type the mail addresses of the dispatchers and supervisors under /admin/delivery/slots (When a delivery counts as late).
9. Required Permissions: deliveries:read, deliveries:dispatch (dispatch board, late deliveries, delivery performance), riders:manage (riders and their PINs), delivery_slots:manage (slots, closed days, lead times and the lateness settings), couriers:manage (connecting couriers), delivery_costs:manage (costs, rider pay, courier invoices, the delivery report), door_payments:manage.

---

3. Module & Sub-Module Architecture

The Delivery module coordinates the entire fulfillment chain across 19 sub-modules:

```
graph TD
    subgraph Planning["1. Planning & Setup"]
        ZONE["Delivery Zones & Polygons"]
        METH["Methods & Invoicing Mode"]
        CHRG["Rate Cards & Surcharges"]
        SLOT["Delivery Time Slots & Caps"]
        MAP["Map Services (OSM / Google)"]
    end

    subgraph Operations["2. Dispatch & Execution"]
        DEL["Deliveries (Partial & Backorders)"]
        BOARD["Visual Dispatch Board"]
        RUN["Delivery Runs & Route Order"]
        APP["Rider Mobile App (/rider)"]
        VISIT["Field Installation Visits"]
    end

    subgraph Control["3. Financial Reconciliation & SLA"]
        DOOR["Door Payments & Cash Custody"]
        COST["Rider Pay & Delivery Margin"]
        LATE["Late Delivery SLA Alerts"]
        CINV["Courier Invoice Reconciliation"]
        REP["Delivery Performance Reports"]
    end

    Planning --> Operations
    Operations --> Control
```

---

4. Interface-by-Interface Walkthrough

1. Deliveries (/admin/delivery/deliveries)

- Master directory of fulfillment parcels. Filter by Status (Draft, Released, Picking, Packed, Ready, On Run, Delivered, Failed), Method, and Branch. Supports Partial Dispatches (sending in-stock items while back-ordering the rest).

2. Delivery Zones (/admin/delivery/zones)

- Geographic zone builder: Group counties, towns, and estates (e.g. "Nairobi Central", "Westlands & Parklands", "Thika & Environs"). Supports polygon geofencing on an interactive map.

3. Dispatch Board (/admin/delivery/dispatch)

- Visual operations kanban:
- Ready for Dispatch: Orders packed and waiting for transport.
- Assigned to Run: Grouped by rider or vehicle.
- Out for Delivery: Live monitoring of parcels currently on the road.
- With couriers: Parcels handed to a courier, with its tracking number and last news.
- A delivery promised a time shows a chip such as Promised Fri 6 Nov, 09:00 to 12:00, turning At risk and then Late using the company's own warning time (see Late Deliveries).

- A parcel whose booking with the courier's system failed (or whose cancellation failed) shows a Courier not booked chip.

4. Delivery Runs (/admin/delivery/runs)

- Multi-drop trip management: Sequenced list of stops for a rider. Calculates total run weight, departure time, estimated arrival at each stop, and starting cash float.

5. Riders (/admin/delivery/riders)

- Who rides: choose whether the company uses its own staff as riders, riders with an account of their own, or both. A kind cannot be switched off while people of that kind are still working.
- Proof at the door: what a rider must show to mark a delivery delivered (nothing, the customer's code, a photo, or either).
- Add a member of staff as a rider (the staff member keeps signing in with their own account) or Make an account for a rider (an account that opens the rider app and nothing else).
- Each rider's card: kind, email, phone, vehicle, branch, runs still to do, and status (Riding, Paused, Left); buttons to pause, resume, mark as left, and (for riders with their own account) give a new first password.
- Sign in with a phone number and PIN (riders with their own account only): the rider's PIN panel shows No PIN, Waiting for the rider to choose their own, PIN set or Shut out for ..., the number they sign in with, when they last signed in, and any wrong tries in a row. Buttons: Give a PIN, Give a new PIN, Let them try again, Turn off the PIN. See Action 4.

6. Door Payments (/admin/delivery/door-payments)

- Cash custody ledger: Shows cash and M-Pesa collected by each rider at customer doors.
- Reconciles rider cash against the counter till during evening debrief.

7. Delivery Costs (/admin/delivery/costs)

- What riders are paid: what a rider earns for each delivery made and for each day on a run on which something was delivered (once a day, however many runs). Either can be nothing. Pay already earned never changes when the rules change.
- Fuel and vehicle costs: type in fuel, a vehicle cost (repairs, service, parking, tolls) or another cost, with what it was for, and optionally the run it belongs to. See Action 8.
- Costs: every cost, newest first, filtered by kind and by days. Rider pay is recorded as deliveries are made; a courier's charge is typed on the delivery it belongs to. A mistake is never edited: it is taken back with a written reason, and the original stays on record.
- In the books: the two accounts rider pay is posted to. Pay is saved first and posted afterwards, so a mistake in the accounts never stops a rider's delivery; what is waiting is counted and can be posted at once.
- Owed to riders / Paid to riders: what each rider is owed (pay earned less anything taken back, not yet paid) and the payouts made, each with its number.

8. Late Deliveries (/admin/delivery/late)

- Every delivery promised a window that is At risk or Late now, the most overdue first, showing how late, the highest level reached (At risk: the dispatcher is told, Late: the dispatcher is told, Still late: the supervisor is told), when it was raised, and how many mails were sent about it (sent, waiting, could not be sent).
- Each has an Acknowledge box (What is being done about it) that records who answered, when and what they said. One acknowledgement covers every level raised for that window.

9. Delivery Slots (/admin/delivery/slots)

- Slots: the delivery times a branch offers (for example Morning 09:00 to 12:00), the days of the week they run, how many deliveries a day each can take, how many minutes before the start an order must be placed, and optionally the most weight it can carry in a day. A slot can be stopped (deliveries already promised it keep it).
- Days with no slots: close a day for all branches or one (public holidays, stock-taking), with the reason.

- How long the goods need (lead times): the least time from order to ready, for a delivery zone or for an item (for example fresh or made-to-order goods). A delivery cannot be promised a slot that starts before the longest lead time that applies.
- When a delivery counts as late: how many minutes before the end of the window a delivery is At risk, after how many minutes late it goes to the supervisor, and who is mailed at each level.
- At checkout the shopper sees the days and times that are open; one that fills before the order is made is never a reason to refuse the sale. See Conditions and Rules.

10. Couriers (/admin/delivery/couriers)

- The courier companies the business sends parcels with. A courier is either by hand (the office types its tracking number and news) or connected to its own system over the internet: the system then books each parcel, follows it, and cancels it with the courier by itself. A courier's own system is connected through settings (its web address, the header that carries your key, the paths used to book, ask about and cancel a parcel, and where its answers keep the tracking number and status); no programming is needed for couriers whose system speaks the common web format, and a specific courier's own connection can be added later.
- The key the courier gave you is kept sealed and never shown again. The secret that signs the courier's news (so that only the real courier can report to you) is shown once, together with the address the courier is to send its news to.
- A courier can be stopped and started, edited (two people editing at once are warned), and given a new secret.

11. Visits (/admin/delivery/visits)

- Technical installation crew scheduling: Assigns technicians to site visits (e.g. solar PV installation, air-conditioner servicing). Provides digital checklists and captures customer signatures on site.

12. Delivery Report (/admin/delivery/reports)

- Executive analytics: Delivery success rates, failure reason distributions (e.g. customer unavailable, wrong address), and zone profitability.

13. Courier Invoices (/admin/delivery/courier-invoices)

- Charged to us, on no invoice yet: by courier, how many charges and how much have not been put on any invoice.
- Type in an invoice: the courier, the invoice number, the day on the invoice, the invoice total and an optional note. The invoice is typed in; it is not read from an uploaded file.
- Each invoice shows its total, the charges matched to it and the difference, with a status: Nothing matched yet, Agrees with our charges, We have fewer charges than the invoice, or We have more charges than the invoice. Match charges lists that courier's charges that are on no invoice, to tick; matched charges can be taken off. A wrong invoice is voided with a reason.
- Matching changes no charge and posts nothing; paying the invoice is done in the usual way.

14. Delivery Performance (/admin/delivery/performance)

- For the days chosen (the last 30 by default, at most a year): Overall (deliveries made, % on time, % at the first try, average time from going out to delivered, how late the late ones were, release-to-delivered time), By rider, By courier (parcels sent and what became of them, including parcels that could not be booked), and Delivery times (places offered, booked, days full and share used for each slot; how many times shoppers chose at checkout and how many had gone by delivery).
- On time means delivered by the end of the window the customer was promised; deliveries promised no window are left out of that rate. First try means no failed attempt before. Rates are marked good (95% and over), to watch (80% and over) or poor.

15. Delivery Charges (/admin/delivery/charges)

- Rate card matrix: Configure pricing rules:
- Flat Rate per Zone (e.g. Nairobi CBD: KES 250).
- Weight-Tier Matrix (e.g. 0-5 kg: KES 300, KES 50 per additional kg).
- Distance Formula (e.g. Base KES 200 + KES 35/km from branch pin).

- Free Shipping Thresholds (e.g. Free delivery for orders over KES 10,000).

16. Delivery Methods (/admin/delivery/methods)

- Per-branch toggle enabling or disabling specific fulfillment modes (Own Fleet, Courier, In-Store Collection, Installation Visit).

17. Delivery Invoicing (/admin/delivery/invoicing)

- KRA Tax compliance setting: Choose whether delivery charges appear on the Same Tax Invoice as products, or on a Separate Delivery Invoice signed independently for KRA eTIMS.

18. Stock Timing (/admin/delivery/stock)

- Configures the exact moment stock is deducted from the physical ledger:
- Option A: When order is paid.
- Option B: When packed into dispatch vehicle.
- Option C: When customer confirms receipt at the door.

19. Map Service (/admin/delivery/maps)

- Provider configuration: Toggle between free OpenStreetMap (Nominatim) or Google Maps Platform for address autocomplete and distance calculations, comparing cost vs. accuracy tradeoffs.

20. The Rider App (/rider)

- Sign in (/rider/sign-in): a rider with an account of their own enters their Phone number and PIN. A link underneath offers Sign in with email and password instead (used by staff who ride, and still available to any rider).
- Choose your own PIN (/rider/pin): shown the first time after the office gives a PIN, and later from Change my PIN. Until the rider has chosen their own PIN, every other page of the app brings them back here.
- My runs (/rider): runs still to do and the last few finished; a notice of the cash the rider holds from doors; a link to My pay; the runs kept on the phone open without signal.
- A run and its stops: each stop with the customer's name, address, phone, items and links to call, chat and navigate; buttons to mark delivered or failed; cash and M-Pesa collected at the door; finishing the run.
- My pay (/rider/pay): Owed to me now (earned and not yet paid), Earned this month, What I earned (the latest 30 entries, each marked Paid or Not paid yet; pay the office took back shows as Taken back by the office, without the office's reason) and What was paid to me (the latest 10 payouts with number, day, how it was paid and reference). A rider sees only their own.
- Sign out at the top, back to the rider sign-in.

21. What the Customer Sees

- At checkout: for a delivery, the days and times that are open (with places left), and the earliest time; the shopper's choice is kept with the order. If the chosen time has filled by the time the delivery is made, the sale and the delivery go ahead and the customer is sent a message saying the time could not be kept; the office sees what the customer chose and what became of it on the order.
- On the tracking page: the promised window until it is delivered, and, for a parcel with a courier that gave a page, a link Follow it on their page.

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5. Granular Field-by-Field Reference

Delivery Zone Fields (/admin/delivery/zones/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints

Zone Code	Internal zone identifier.	Text (Max 20 chars, uppercase)	Required	Unique code e.g. ZONE-NBI-WEST.
Zone Name	Name shown to dispatchers.	Text (Max 80 chars)	Required	e.g. "Westlands & Parklands".
County	Primary administrative county.	Dropdown	Required	e.g. Nairobi, Mombasa, Kiambu.
Included Towns / Estates	List of covered estates.	Multi-Select Tag Input	Required	e.g. Westlands, Parklands, Highridge, Spring Valley.
Map Polygon Boundary	GPS geofence coordinates.	Map Drawing Tool	Optional	Polygon coordinates defining exact boundary.
Active Status	Controls zone availability.	Toggle (On/Off)	Required	Inactive zones cannot be selected at checkout.

Delivery Rate Policy Fields (/admin/delivery/charges/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Policy Name	Reference name.	Text (Max 120 chars)	Required	e.g. "Standard Nairobi Zone Rates 2026".
Calculation Method	How fees are calculated.	Dropdown	Required	Zone Rate Card, Distance from Branch, Flat Weight.
Base Fee	Minimum starting charge.	Currency (KES)	Required	e.g. 250.00.
Per-Kilometer Fee	Variable distance charge.	Currency (KES)	Required if Distance	e.g. 35.00 per km beyond base radius.
Per-Kilogram Extra	Heavy parcel surcharge.	Currency (KES)	Optional	Charge per kg exceeding base weight limit.
Free Delivery Threshold	Order value qualifying for free shipping.	Currency (KES)	Optional	Orders $\ge$ this value receive 100% shipping discount.

Rider Account Fields (/admin/delivery/riders, Make an account for a rider)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Name	The rider's name.	Text (Max 120)	Required	Shown on runs and pay.
Email to sign in with	The rider's account address.	Email	Required	Must not already belong to another account.
Phone	The number the rider signs in with when using a PIN, and the number customers can call.	Phone	Required	For a PIN: a Kenyan mobile number (07.. or 01.., with or without +254).
First password	The password the rider changes at their first email sign-in.	Text	Required	Must meet the password rules.
Vehicle (optional)	Motorcycle, van or plate number.	Text (Max 80)	Optional	Shown on the rider's card.
Branch	The branch the rider belongs to.	Lookup	Optional	Only active branches.

Rider PIN Fields

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Phone number (rider sign-in)	Identifies the rider.	Kenyan mobile number	Required	Written any way (0712 345 678, +254712345678, 712345678). A number belongs to one rider at…
PIN (rider sign-in)	Proves it is the rider.	6 digits	Required	Digits only.

The PIN the office gave you / The PIN you have now	The PIN the rider is replacing.	6 digits	Required	A wrong one counts as a wrong try.
Your new PIN	The rider's own PIN.	6 digits	Required	Not one digit repeated, not digits in a row (123456, 987654), not a common choice, not the end of the rider's own phone number, and not…
Your new PIN again	Confirms the new PIN.	6 digits	Required	Must match.

Fuel and Vehicle Cost Fields (/admin/delivery/costs)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
What kind of cost	Groups the cost in the report.	Choice: Fuel, Vehicle cost (repairs, service, parking, tolls), Other cost	Required	Rider pay and courier charges are recorded by the system, not typed here.
How much (KSh)	The amount.	Currency	Required	More than nothing, to the cent, at most 10 million.
Run it was for (optional)	Ties the cost to a run (and so to its rider).	Lookup of the company's runs	Optional	
What it was for	The words kept with the cost.	Text (3 to 300 characters)	Required	For example the vehicle and the trip.

Rider Pay Fields (/admin/delivery/costs, What riders are paid)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
For each delivery (KSh)	Earned by the rider for each delivery made.	Currency	Required	Nothing (0) is fine. A delivery that could not be made earns nothing.
For each day on a run (KSh)	Earned once a day for a day on a run on which something was delivered.	Currency	Required	Nothing (0) is fine.

Cost Take-Back and Payout Fields

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Why it is being taken back	The reason kept with the reversing entry.	Text	Required	At least a few words; a cost can be taken back once.
Account rider pay is booked to	The cost account rider pay is posted to.	Lookup of accounts	Required to post	Must differ from the other account.
Account that holds what is owed to riders	The payable account.	Lookup of accounts	Required to post	
Paid by	How the rider is paid.	Payment method (cash, M-Pesa or bank)	Required	A method that does not pay in cash, M-Pesa or through the bank is refused.
Reference (M-Pesa code, cheque or slip number)	The proof of payment.	Text	Optional	Kept with the payout.

Slot, Closed Day and Lead Time Fields (/admin/delivery/slots)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Branch	Where the deliveries leave from.	Lookup	Required	
Name	The slot's name shown to shoppers.	Text	Required	Unique for the branch.

Days	The days of the week the slot runs.	Day buttons (Monday to Sunday)	Required	At least one.
Starts (Nairobi) / Ends (Nairobi)	The window promised to the customer.	Time (HH:MM)	Required	The end must be after the start.
Deliveries a day	How many deliveries the slot can take each day.	Whole number	Required	1 to 10,000.
Closes (minutes before)	How long before the start orders stop being taken for it.	Whole number	Required	0 to 10,080.
Most it can carry in a day, in kg (empty for no limit)	Weight limit for the day.	Number	Optional	More than nothing.
Day / Branch (all if empty) / Why (Close a day)	Closes a day to all branches or one.	Date, lookup, text	Required (day and why)	A day already closed is refused.
Minutes the zone needs / Minutes the item needs (Lead times)	The least time before goods are ready.	Whole number	Required	A zone or an item is chosen first.

Lateness Settings Fields (/admin/delivery/slots, When a delivery counts as late)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
At risk this many minutes before the window ends	When a promised delivery is flagged At risk (also once the window has begun and it is not	Whole number	Required	0 to 1,440; default 60.
Tell the supervisor after this many minutes late	When a late delivery goes up to the supervisor.	Whole number	Required	1 to 1,440; default 60.
Mail the dispatchers when a delivery is at risk or late	Who is mailed for the first two levels.	Mail addresses separated by commas	Optional	At most ten; each must look like an address; shown back in lower case, once each.
Mail the supervisors when it has been late too long	Who is mailed for the third level.	Mail addresses separated by commas	Optional	As above. Empty means no mail; the Late deliveries screen shows everything either way.
What is being done about it (acknowledging)	The answer kept with the delivery.	Text (5 to 300 characters)	Required	

Courier Fields (/admin/delivery/couriers, Add a courier)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Name	The courier's name; also what is shown on deliveries and invoices.	Text	Required	Unique among the company's couriers.
How it is connected	By hand, or connected to the courier's own system.	Choice	Required	A courier by hand needs nothing more.
The courier's web address	Where the company's system talks to the courier's.	Web address	Required if connected	Must start with https and be a public address on the internet.
Key the courier gave you	Proves the company's requests to the courier.	Secret text	Required if connected	Kept sealed; never shown again. Left empty when editing, the old key is kept.
Header that carries the key	The name of the header the courier expects the key in.	Text	Optional	The courier's documents say which.
Path to book a parcel / Path to ask about one / Path to cancel one	The parts of the courier's address used for each job.	Text	Optional	Asking and cancelling need the courier's tracking number in the path.

Where to find things in the courier's answers (JSON, optional)	Where the courier's answers keep the tracking number, status, who signed and proof.	Text	Optional	Left empty, the common defaults are used.

Courier Invoice Fields (/admin/delivery/courier-invoices)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Courier	Whose invoice it is.	Choice of the couriers that have charged or are connected	Required	Only that courier's charges can be matched to it.
Invoice number	The number printed on the courier's invoice.	Text (Max 60)	Required	Unique for the courier (any case); a voided invoice frees its number.
Day on the invoice	The invoice date.	Date	Required	Not in the future.
Invoice total (KSh)	What the invoice adds up to.	Currency	Required	More than nothing.
Note (optional)	Anything worth keeping.	Text (Max 300)	Optional	
Why it is being voided	The reason kept with a voided invoice.	Text	Required to void	At least a few words.

Delivery Performance Fields (/admin/delivery/performance)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
From day / To day	The days to look at.	Date	Required	The first cannot be after the last; at most a year. Default: the last 30 days.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Planning and Dispatching a Multi-Drop Delivery Run

1. Navigate to /admin/delivery/dispatch.
2. The board lists orders in Ready for Dispatch status.
3. Click Create Delivery Run.
4. Select Fulfilling Branch: "Nairobi Industrial Area".
5. Select Assigned Rider: "David Mutua (Motorcycle #KMDF 123A)".
6. In the orders list, select the 4 customer orders destined for Westlands.
7. Click Add to Run.
8. In the Stop Order list, drag and drop the stops into optimal street delivery sequence.
9. Click Finalize & Dispatch Run.
10. The system prompts: "Confirm dispatch of 4 parcels? Stock will move to David Mutua's mobile custody."
11. Click Confirm Dispatch.
12. The run status updates to Out for Delivery, and the parcels appear immediately on David Mutua's mobile phone at /rider.

Action 2: Customer Handover & Verification PIN Confirmation

1. The rider arrives at the customer's address and opens the /rider mobile web app.
2. Selects the active stop: "Order #ORD-10492 - Jane Wanjiku".
3. The rider asks for the 6-digit delivery code the customer was sent in the "on its way" message (by email) when the delivery left. The customer gives it only when the parcel arrives. (If the company's rule at the door is a photo, or nothing, the app

asks for that instead; see Proof at the door on the Riders screen.)

4. The rider types the code (for example 048213) into the app.
5. If the order is Cash on Delivery, the rider selects Payment Collected:

- Types cash amount received: KES 3,500.00.

6. Click Confirm Delivery.
7. The system validates the code instantly, marks the delivery as Delivered, updates the customer order status, and adds KES 3,500.00 to David Mutua's cash custody ledger.

Action 3: Reconciling Evening Cash Handover from a Rider

1. At the end of the shift, the rider returns to the cashier desk.
2. The cashier navigates to /admin/delivery/door-payments.
3. In the rider list, select David Mutua.
4. The screen displays: "Total Cash Custody: KES 14,200.00 (from 4 deliveries today)".
5. The rider hands the physical cash to the cashier.
6. The cashier counts KES 14,200.00, selects the destination till ("Main Cash Till #1"), and clicks Accept Cash Handover.
7. An InlineConfirm prompt appears: "Accept KES 14,200.00 from David Mutua into Main Cash Till #1?"
8. Click Confirm Handover.
9. The rider's custody balance drops to zero, and a cash receipt journal entry is posted to the till ledger.

Action 4: Giving a Rider a PIN, and the Rider's First Sign-In

1. Open /admin/delivery/riders. The rider must have an account of their own, a Kenyan mobile number and be riding or paused.
2. On the rider's card, in Sign in with a phone number and PIN, click Give a PIN.
3. A yellow box shows six digits, for example 482915, with the number the rider signs in with. This is the only time the PIN is shown; the system does not keep it. Give it to the rider in person or by a call. Click I have given it to them.
4. The rider opens /rider; being signed out, they are taken to the sign-in. They enter their phone number and the PIN and click Sign in.
5. They are taken to Choose your own PIN. They enter the PIN the office gave, then a new PIN twice. Weak PINs are refused with the reason shown under the box. Click Save my PIN.
6. They arrive at My runs. The rider's card now shows PIN set and when they last signed in.
7. To give a new PIN (a forgotten PIN), click Give a new PIN and confirm. The old PIN stops working and any phone signed in with it is signed out within a minute; the rider repeats steps 4 to 6.
8. To stop the PIN, click Turn off the PIN and confirm. The rider can then sign in only with email and password.

Action 5: When a Rider Is Shut Out

1. After five wrong tries in a row the rider sees Too many wrong tries. Try again in 15 minutes. Further wrong tries lengthen the wait (30 minutes, 1 hour, up to 4 hours). Even the right PIN does not work until the wait is over.
2. The rider's card shows Shut out for .... Click Let them try again to clear it at once, or leave it to run out.
3. A wrong number and a wrong PIN get the same answer, so someone guessing cannot learn which numbers belong to riders.

Action 6: Handling a Late Delivery

1. Open /admin/delivery/late (or watch the mail the dispatchers receive, if mail addresses are set).
2. Find the delivery: its card shows how late it is and the highest level reached.

3. Act on it (call the rider, tell the customer, move the delivery to another run).
4. In What is being done about it type what you did (a few words) and click Acknowledge. The answer, your name and the time are kept; the card then shows who acknowledged it.
5. The delivery leaves the screen once it is delivered, cancelled or returned.

Action 7: Connecting a Courier and Sending a Parcel

1. Open /admin/delivery/couriers and fill Add a courier: the Name, How it is connected, and, for a connected courier, the web address, the key and the paths from the courier's documents. Click Add the courier.
2. For a connected courier, a box shows the address the courier sends its news to and the secret that signs its news. Copy both now and give them to the courier; the secret is not shown again. Click I have copied it.
3. On a ready delivery that goes by courier, choose the courier under Courier from your list (or type a name for a courier by hand and a tracking number) and dispatch it. Dispatch happens at once.
4. The system books the parcel with the courier within a minute. If the courier cannot be reached it tries again after 2, 4, 8, 16 and 32 minutes, then shows the failure on the delivery and the board; Try the courier again tries at once. A courier being down never holds up a dispatch.
5. The system asks the courier about the parcel while it is on the road (every 15 minutes at first, then less often); Ask the courier now does it at once. News the courier sends to the address is accepted only if correctly signed.
6. When the courier reports delivered, the delivery is completed with who signed and a link to the proof. A problem reported by the courier becomes a failed attempt. A cancelled delivery is cancelled with the courier by the system.
7. To stop using a courier, click Stop using and confirm; Make a new secret replaces the signing secret (the old one stops working).

Action 8: Recording Fuel and Other Running Costs

1. Open /admin/delivery/costs. In Fuel and vehicle costs choose What kind of cost, type How much (KSh) and What it was for, and optionally pick the Run it was for.
2. Click Save the cost. A confirmation appears and the form empties; clicking twice saves it once.
3. The cost appears in the list and counts in the delivery report. It is not posted to the accounts here: record the receipt as an expense or a supplier invoice in the usual way, so it is not counted twice.
4. If it was typed wrongly, type Why it is being taken back on its row and confirm Take it back.

Action 9: Typing In and Matching a Courier Invoice

1. Open /admin/delivery/courier-invoices. Under Type in an invoice choose the courier, type the number, the day and the total, and click Record the invoice.
2. The invoice appears as Nothing matched yet. Click Match charges to ....
3. Tick the courier's charges that appear on the invoice and click the Match button (it shows how many and how much).
4. The invoice shows what was matched and the difference: Agrees with our charges, or fewer or more charges than the invoice. Decide what to do about a difference (ask the courier, or find the missing charge).
5. To correct a mistake, click Take off beside a charge and confirm, or void the whole invoice by typing Why it is being voided and confirming Void it.

Action 10: Paying a Rider

1. Open /admin/delivery/costs, Owed to riders.
2. Find the rider and click Pay. Choose Paid by and, if you have one, type the Reference, then click the pay button (it shows the amount).
3. The payout is numbered, the rider's balance clears, and (when the books are live and the accounts are chosen) the accounts are posted. The rider sees it under What was paid to me.

---

7. Conditions, Business Rules & Edge Cases

Handling a Failed Delivery Attempt

- If a rider arrives and the customer is unreachable:

1. The rider clicks Report Failed Attempt in the mobile app.
2. The system forces selection of a mandatory Failure Reason Code:

- Customer Phone Unreachable
- Customer Not at Premises
- Customer Refused / Cash Shortage
- Inaccessible Road / Weather Block

3. The rider must enter a short memo note.
4. The delivery transitions to Attempt Failed. Stock remains in the rider's custody until the run is closed, where it must be physically returned to the warehouse storekeeper.

Invoicing Modes for Delivery Charges (KRA Compliance)

Under /admin/delivery/invoicing, the business configures one of two fiscal modes:

- Combined Invoicing: Delivery is printed as a service line item on the primary product tax invoice, taxed at the standard 16% VAT rate.
- Separate Delivery Invoicing: Products are billed on Invoice A, while the delivery fee is billed on a separate Delivery Invoice B. This is required when delivery is handled by an affiliated logistics entity with a separate KRA PIN.

Rider PIN Sign-In: Rules

- Only riders with an account of their own sign in with a PIN. A member of staff who rides always signs in with their own account: a PIN must never open a back-office account.
- A PIN sign-in opens the rider app only, whatever the account holds.
- The system never keeps a PIN, only a scrambled form made with a secret of the server's own. The office sees a PIN once, when it is given.
- A rider who is paused, has left, whose account is switched off, or whose kind of rider has been switched off cannot sign in, even with the right PIN.
- Changing a rider's phone number turns their PIN off (a PIN belongs to the number it was given for); a rider who leaves loses it, freeing the number for someone else. A phone number belongs to one rider at a time.
- Sign-in attempts are also limited by device address and by phone number, so many guesses from many places run out as quickly as many from one. Someone repeatedly trying a rider's number can shut that rider out for a while; the office's Let them try again is the remedy.
- A phone signed in with an old PIN is signed out when the PIN is changed, replaced or turned off (within about a minute).

Delivery Slots, Cut-Offs and Customer Choice

- A shopper is offered a day and time only if the slot runs that day, the day is not closed, places are left, the weight limit is not passed, the cut-off has not passed and the goods can be ready in time (the longest lead time that applies).
- The shopper's cut-off is judged at the moment they chose; a day that has already passed is refused.
- The delivery is booked the slot when the delivery is made. If the slot has filled since, neither the sale nor the delivery is held up: the customer's request is marked as not kept, and the customer is sent a message in the same step.
- A booked window is kept as promised, even if the slot is later changed or stopped.

Lateness

- A delivery promised a window is On track, At risk (within the warning time of the end of the window, or the window has begun and it is not on its way), Late (past the end), On time or Delivered late. Cancelled and returned deliveries are not watched.
- At risk, Late and Still late: the supervisor is told are each raised once for a delivery and a window; a delivery given a new window is watched afresh. A background task checks every minute.
- Mail is queued in the same step that raises a level, one mail for each address; if the mail server is down the mail is tried again after 5, 10, 20 and 40 minutes and then left as could not be sent. A mail never carries an address, phone number or payment detail. Changing the list later does not change who was already told.

Couriers

- Calls to a courier go only to secure (https) public addresses; the courier's own error text is never shown to staff, only a plain reason.
- A booking that fails because the courier refused the request fails at once; one that fails because the courier could not be reached is retried. The same piece of news from a courier (pushed or asked) counts once.
- Cancelling a delivery cancels its parcel with the courier; a parcel never booked is simply closed.

Costs, Pay and Courier Invoices

- Costs are never edited or deleted; a mistake is taken back by an opposite entry with a reason, once. A cost taken back cannot be matched to a courier invoice.
- Rider pay for a delivery is counted once however many times the delivery is retried. Pay for a day is counted once for a rider and a day.
- What is owed to a rider is the pay earned less anything taken back, not yet paid out. If more was taken back than earned, nothing can be paid and the balance is set against their next pay.
- Fuel and vehicle costs reduce the delivery margin in the report but are not posted to the accounts.
- A courier charge is on at most one invoice; matching a set of charges is all or nothing; a voided invoice frees its charges and its number; an invoice's words cannot be changed once typed.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Configure 3 Delivery Zones with towns and estates.	Zones appear in storefront checkout dropdown.	[ ]
2	Configure a Distance-based Rate Card (KES 200 base + KES 30/km).	Checkout pin-drop calculates exact distance and fee accurately.	[ ]
3	Make an account for a rider with a Kenyan mobile number, then Give a PIN.	The PIN is shown once; the rider signs in at /rider with the number and PIN and is made to choose their own PIN before anything	[ ]
4	Plan a Delivery Run and dispatch 2 test orders.	Parcels appear on rider run sheet; stock moves to mobile custody.	[ ]
5	Execute Door Payment collection and Cashier Handover.	Rider custody ledger updates and clears to zero upon cashier sign-off.	[ ]
6	Try five wrong PINs for the test rider.	The rider is shut out (even for the right PIN) and the card shows Shut out for ...; Let them try again restores sign-in.	[ ]

7	Give the test rider a new PIN while a phone is signed in with the old one.	The old PIN fails and the old phone is signed out within a minute.	[ ]
8	Set rider pay (for example KES 50 for each delivery) and complete a test delivery.	The rider's My pay shows the entry as Not paid yet; after Pay, it shows Paid and the payout.	[ ]
9	Record a fuel cost against a run.	It appears in the cost list and in the delivery report's cost line, and is taken back with a reason on request.	[ ]
10	Add a slot with room for one delivery a day and book two orders.	The second shopper is not offered the slot; a shopper whose chosen slot filled still completes the order and is sent	[ ]
11	Give a delivery a window that has already ended, with mail addresses set.	Within a minute the delivery appears on Late deliveries, the dispatchers are mailed, and acknowledging it records your	[ ]
12	Add a courier by hand and a connected courier (test key).	The connected courier's news secret is shown once; the board shows Courier not booked if the test address cannot be reached,	[ ]
13	Type a courier invoice and match two charges.	The invoice shows Agrees with our charges when the total equals the charges, or the difference when it does not; a	[ ]
14	Open Delivery performance after a few test deliveries.	On-time, first-try, rider, courier and delivery-time figures appear, and show nothing (not zero percent) where no delivery was	[ ]

Common Client Setup Pitfalls

- A rider has a landline or foreign number: a PIN cannot be given. Correct the phone number on the rider first; changing it turns an existing PIN off.
- Two riders share one phone number: the second cannot be given a PIN. Each rider needs their own number.
- Slots exist but shoppers are never offered a time: the branch has no slot on that day of the week, the day is closed, the cut-off has passed, or a lead time makes the goods too late.
- Lateness mail never arrives: no addresses were typed in the lateness settings, or the company's mail sending is not configured; the Late deliveries screen still shows everything.
- Rider pay is recorded but not in the accounts: the books are not live, or the two accounts under In the books were not chosen. What is waiting is counted on the screen.
- A courier invoice does not agree: a charge was never typed on the delivery, was taken back, or is already on another invoice. The difference shows how much is missing or extra.

Interactive version with live navigation: /admin/docs/06-delivery-logistics-and-field-dispatch

CHAPTER 07

Guide 07: Users, Roles & Access Governance (`/admin/users`)

Category: Governance & Security | Reading Time: ~8 min | Words: 1,580

*The **Users & Access Governance** module provides enterprise-grade identity, authentication, Role-Based Access Control (RBAC), and Attribute-Based Access Control (ABAC). It ensures that every staff member has precisely the permissions requi*

The Users & Access Governance module provides enterprise-grade identity, authentication, Role-Based Access Control (RBAC), and Attribute-Based Access Control (ABAC). It ensures that every staff member has precisely the permissions required for their operational responsibilities while enforcing strict separation of duties across cash handling, inventory management, procurement, and financial reporting.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The "Admin Password Shared by Everyone" Disaster: In many businesses, staff share a single master administrator login. When money goes missing, orders are deleted, or prices are discounted, management cannot hold any individual accountable. QSell mandates individual staff accounts with immutable audit logging for every action.
- Internal Fraud & Collusion: If the person receiving goods in the warehouse can also create purchase orders and approve supplier invoices, fake deliveries and embezzlement inevitably follow. QSell enforces strict Separation of Duties (SoD) through granular atomic permissions.
- Data Leakage Between Competing Branches: In multi-branch retail, store managers often compete for performance bonuses. Giving all staff global access causes friction and leakage of customer contact lists. QSell's Branch-Scoped Access restricts staff visibility strictly to their assigned operating location.
- After-Hours System Tampering: Dishonest staff may log into the system late at night or from home to manipulate records. QSell's Access Policies (ABAC) allow management to enforce contextual constraints (e.g. "Logins only permitted Monday-Saturday between 7:00 AM and 7:00 PM from authorized company network IPs").

Target Client Profiles

- Medium to Large Enterprises & Retail Chains: Employing 10 to 500+ staff across sales, logistics, warehousing, accounting, and executive leadership.
- Franchise & Multi-Branch Operators: Requiring independent branch staff permissions with centralized corporate oversight.
- Audit-Sensitive Corporations: Requiring proof of strict internal controls for external financial auditors and board compliance.

Commercial Sales Pitch & Talk Track

> "With QSell, you have total control over who sees what and who can do what in your business. You can give a counter cashier the exact ability to ring up sales and print receipts, without letting them see your product cost prices or company profit. Your storekeepers can manage stock movements without touching invoices. And you can ensure that cashiers can only log into their till during business hours while physically at the store."

Measurable ROI & Safeguards

- 100% Personal Accountability: Every single transaction, stage movement, and document edit is stamped with the authenticated user's ID.

- Zero Margin Leakage to Frontline Staff: report_profit:read permission separates top-line sales figures from sensitive product cost prices.
- Contextual Perimeter Defense: Time-of-day and IP-based access policies block unauthorized remote logins.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    BRAN[1. Operating Branches Configured] --> ROLES[2. Standard Roles Configured & Permissions Bound]
    ROLES --> POL[3. Optional ABAC Contextual Policies Created]
    POL --> USERS[4. User Accounts Created & Bound to Roles/Branches]
```

1. Operating Branches: Defined under Settings > Branches so users can be assigned to physical operating units.
2. Roles Defined: Built under /admin/users/roles with explicit permission bindings.
3. Staff Account Creation: Users invited or created under /admin/users.
4. Required Permissions: Only staff with users:read and users:manage can access this module.

---

3. Module & Sub-Module Architecture

Access control is built upon a 3-layer security model:

```
graph TD
    USER["User Account (Email, Password, Assigned Branch)"]
    ROLE["Roles (e.g. Cashier, Storekeeper, Accountant)"]
    PERM["Atomic Permissions (e.g. orders:read, stock:manage)"]
    ABAC["Access Policies (Context: Time, IP, Device, Amount)"]

    USER --> ROLE
    ROLE --> PERM
    USER --> ABAC
    PERM --> ACCESS{"Grant or Deny Action?"}
    ABAC --> ACCESS
```

1. Layer 1: Identity & Branch Scope: The authenticated user identity bound to a primary operating branch (or "All Branches" for corporate directors).
2. Layer 2: Role-Based Permissions (RBAC): Role templates grouping atomic permissions across 10 system categories.
3. Layer 3: Contextual Policies (ABAC): Dynamic rules that evaluate environmental attributes (time of day, network IP, document amount) before permitting an action.

---

4. Interface-by-Interface Walkthrough

Sub-Module 1: Users Directory (/admin/users)

- Master list of staff accounts showing Name, Email, Assigned Branch, Role(s), MFA Status, and Account State (Active, Suspended).
- Actions: Invite User, Edit Account, Reset Password, Impersonate (Super Admin only), and Deactivate User.

Sub-Module 2: Roles & Permissions (/admin/users/roles)

- Role catalog view showing pre-packaged and custom roles (e.g. Super Administrator, Branch Manager, Senior Accountant, Sales Representative, Storekeeper, Counter Cashier, Dispatcher).
- Role Permission Matrix: Interactive accordion grouped into the 10 core permission categories:

1. General (Dashboard, Analytics export).

2. Orders & Sales (Read, create, edit, cancel orders, quotes, customers).
3. Projects (Job costing, milestone approval).
4. Vendor Management (RFQs, POs, RGN receipts, supplier invoices).
5. Customers (Credit limit overrides, pricing group assignments).
6. Products & Inventory (Catalog edits, stock transfers, cycle counts, ledger cutover).
7. Finance & Payments (General journals, bank accounts, receipts, fiscal period closing).
8. Reports & Insights (Sales, collections, gross profit, receivables ageing).
9. Company & Settings (Brand profile, document layouts, payment gateways, KRA eTIMS).
10. Users & Access (User creation, role configuration, access policies).

Sub-Module 3: Access Policies (/admin/users/policies)

- Contextual policy designer:
- Create rules combining attributes: If User Role = Cashier AND Current Time NOT BETWEEN 08:00 AND 18:00 THEN DENY LOGIN.
- IP Whitelisting: If Action = Post General Journal AND User IP NOT IN [Head_Office_IPs] THEN REQUIRE MFA.

---

5. Standard Role Templates (Implementor Reference)

Role Template	Primary Target Persona	Key Granted Permissions	Key Restricted / Denied Permissions
Counter Cashier	Front-office retail staff.	orders:create, orders:read, receipts:create, pos:cash_sale	Denied: report_profit:read, gl_journals:post, stock:manage, pricing:manage
Warehouse Storekeeper	Godown and inventory staff.	stock:read, stock:manage, inventory:counts, inventory:transfers, rgn:read, rgn:receive	Denied: orders:delete, gl_journals:post, company:write, supplier_invoices:read
Sales Representative	Field and counter sales reps.	quotes:manage, orders:create, customers:read, pricing:read	Denied: pricing:override_unlimited, gl_journals:post, report_profit:read
Logistics Dispatcher	Fleet coordinator.	deliveries:read, deliveries:dispatch, riders:manage, delivery_runs:manage	Denied: receipts:delete, finance:manage, gl_accounts:write
Senior Accountant	Back-office finance team.	finance:manage, gl_journals:post, bank_accounts:read, receipts:read,	Denied: users:manage, system:configure, stock:physical_count
Branch Manager	Operational head of a store.	Scoped to assigned branch: all sales, stock, delivery, and branch report permissions.	Denied: reports:all_branches, company:profile_write, tax_compliance:manage
Super Administrator	Company directors and IT head.	Full global access across all 10 categories, all branches, and all settings.	None.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Onboarding a New Cashier Account

1. Navigate to /admin/users and click New User.
2. Enter Full Name: "Faith Mwangi".
3. Enter Corporate Email: "faith.mwangi@company.co.ke".
4. In the Primary Branch dropdown, select "Nairobi Westlands Store".
5. In the Role selection list, check the box for Counter Cashier.
6. Check the box Require Password Reset on First Login.
7. Click Create User.
8. The system dispatches an activation email to Faith Mwangi containing a secure, single-use login link.

Action 2: Creating a Custom Restricted Role for Sales Interns

1. Navigate to /admin/users/roles and click New Role.
2. Enter Role Name: "Sales Intern".
3. Enter Description: "Can create draft quotes and view customers; cannot view costs or confirm orders."
4. In the permissions list:

- Expand Orders & Sales: Check quotes:read, quotes:create, customers:read.
- Leave orders:confirm and orders:cancel unchecked.
- Expand Reports & Insights: Leave report_profit:read unchecked.

5. Click Save Role.
6. A success message confirms: "Role 'Sales Intern' created with 3 active permissions."

Action 3: Enforcing Operating Hours via an Access Policy

1. Navigate to /admin/users/policies and click New Access Policy.
2. Enter Policy Name: "Cashier Operating Window".
3. In the Target Roles picker, select Counter Cashier.
4. In the Conditions builder:

- Select Attribute: Current Time (Local).
- Select Operator: Outside Hours.
- Enter Time Range: 07:30 to 19:30.

5. In the Action selector, choose Deny Access with Notice.
6. Enter Customer Message: "System access is restricted outside operational retail hours. Please contact your Branch Manager."
7. Click Activate Policy.
8. Any cashier attempting to log in at 10:00 PM is immediately blocked with the configured message.

---

7. Conditions, Business Rules & Edge Cases

Preventing Accidental Lockout (Super Admin Safeguard)

- The system prevents deactivating or deleting the last active user holding the Super Administrator role.
- If an administrator attempts to remove the users:manage permission from their own account, the system rejects the operation: "Cannot remove administrative access from your own active session."

Multi-Branch Visibility Isolation

- Users with the reports:all_branches permission can view company-wide consolidated analytics.
- Users lacking this permission have their queries automatically filtered at the database level by branch_id = user.assigned_branch_id.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create user accounts for each staff member with correct branch bindings.	Users successfully receive activation emails.	[ ]
2	Assign "Counter Cashier" role to a test user.	User can create orders but cannot access General Ledger or Gross Profit.	[ ]
3	Log in as Branch Manager assigned to Branch B.	Dashboard and order lists strictly show Branch B data; Branch A is invisible.	[ ]
4	Test Time-of-Day Access Policy.	Login outside permitted hours is blocked with clean error banner.	[ ]
5	Verify Super Admin protection rule.	System refuses to delete or deactivate the primary Super Admin account.	[ ]

Interactive version with live navigation: /admin/docs/07-users-roles-and-access-governance

CHAPTER 08

Guide 08: Projects & Job Costing (`/admin/projects`)

Category: Commercial Operations | Reading Time: ~8 min | Words: 1,512

*The **Projects & Job Costing** module manages complex, milestone-based customer deliverables, technical installations, and internal business initiatives. It tracks tasks, billable materials, direct labor expenses, customer milestone billing*

The Projects & Job Costing module manages complex, milestone-based customer deliverables, technical installations, and internal business initiatives. It tracks tasks, billable materials, direct labor expenses, customer milestone billings, and actual gross profitability per job.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The "Profitable on Paper, Loss in Reality" Trap: Service contractors and equipment installers (e.g. Solar PV installers, HVAC technicians, machinery distributors) often win projects with a high contract value, only to discover at completion that unexpected labor hours, extra cabling, and sub-contractor fees wiped out all profit. QSell tracks Budget vs. Actual Costs in real time as materials are dispatched and hours are logged.
- Delayed Milestone Billing: In multi-stage projects (e.g. 30% Advance Deposit, 50% Delivery of Equipment, 20% Final Commissioning), businesses frequently forget to issue milestone invoices promptly, choking cash flow. QSell binds project tasks directly to Accounts Receivable billing schedules.
- Premature Project Closure: Jobs are often marked "finished" while open vendor invoices or unreturned equipment remain unaccounted for. QSell enforces a strict state machine (draft $\rightarrow$ open $\rightarrow$ on_hold $\rightarrow$ completed $\rightarrow$ closed), preventing a project from being formally closed until all costs and billings reconcile.

Target Client Profiles

- Solar PV & Renewable Energy Contractors: Managing residential and commercial solar installations requiring site assessments, panel mounts, inverters, wiring, and commissioning.
- Commercial Equipment & Fit-Out Contractors: Installing industrial kitchen equipment, backup diesel generators, security CCTV systems, and retail shop fittings.
- Internal Corporate Project Teams: Managing internal capital expenditure projects (e.g. opening a new retail branch or refurbishing a warehouse).

Commercial Sales Pitch & Talk Track

> "With QSell, you can quote, execute, and bill complex installation jobs with total financial clarity. You can build a project budget with materials from your warehouse and estimated labor costs, track every milestone task, issue customer milestone invoices, and watch your true gross margin update with every screw and hour spent. You will never finish a project wondering where your profit went."

Measurable ROI & Safeguards

- Real-Time Job Profitability: Realized gross margin monitored continuously against initial quotation budgets.
- Strict State-Machine Integrity: Projects cannot be prematurely closed with unbilled milestones or open tasks.
- Zero Double-Keying: Direct quote-to-project-to-invoice workflows eliminate re-typing.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    CUST[1. Customer Account Created] --> CURR[2. Currency & Price Lists Defined]
    CURR --> PROD[3. Inventory SKUs & Labor Services Cataloged]
    PROD --> USR[4. Project Manager Assigned Permission: projects:read/manage]
    USR --> PROJ[5. Project Created & Opened]
```

1. Customer Master Card: A customer must be selected under /admin/customers (unless the project is an internal company project).
2. Catalog SKUs: Physical items (e.g. solar panels, inverters) and labor services (e.g. technical installation per day) must exist in the catalog.
3. Staff Permissions: projects:read to view and projects:manage to create, edit, change status, and add tasks.

---

3. Module & Sub-Module Architecture

The Project Lifecycle is governed by a deterministic, non-reversible state machine:

```
stateDiagram-v2
    [*] --> draft: Create Project
    draft --> open: Open Project (Approve Budget)
    draft --> cancelled: Cancel
    open --> on_hold: Place On Hold (Client Delay)
    on_hold --> open: Resume Work
    open --> completed: Complete (All Tasks Done)
    on_hold --> completed: Complete
    completed --> closed: Close (Financials Reconciled)
    completed --> open: Reopen (Defect Rectification)
    open --> cancelled: Cancel
    on_hold --> cancelled: Cancel
    closed --> [*]
---    cancelled --> [*]
```

4. Interface-by-Interface Walkthrough

1. Projects Directory (/admin/projects)

- Filterable list displaying Project Code, Title, Customer Name, Status (Draft, Open, On Hold, Completed, Closed, Cancelled), Budget Amount, Total Billed, and Target Completion Date.

2. Project Creation Form (/admin/projects/new)

- Header: Project Title, Project Code (auto-generated e.g. PRJ-2026-0012), and Description.
- Customer Assignment: Bind to a commercial client or mark as an Internal Company Project.
- Dates & Currency: Estimated Start Date, Target Completion Date, and Contract Currency (KES, USD).
- Budget Formulation: Target contract billing amount and estimated expense ceiling.

3. Project Detail Workspace (/admin/projects/[id])

- Executive KPI Tiles:
- Contract Budget: Total agreed revenue from customer.
- Invoiced to Date: Amount billed via milestone invoices.
- Materials Cost: Value of inventory items issued to the project.
- Realized Gross Margin: Profit percentage earned on the project.

- Action Toolbar: Contextual state buttons based on current status (Open Project, Put On Hold, Resume, Mark Completed, Close Project).
- Task & Milestone Manager:
- Interactive task checklist (e.g. Site Assessment, Mounting Rails Installation, Inverter Wiring, Testing & Commissioning).
- Task state badges: Open, Completed, Cancelled.
- Linked Documents Panel: Direct links to generated Sales Quotes, Sales Orders, Material Requisitions, and Customer Invoices.

---

5. Granular Field-by-Field Reference

Project Workspace Fields (/admin/projects/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Project Title	Descriptive job name.	Text (Max 160 chars)	Required	e.g. "10kW Hybrid Solar Installation - Karen Residence".
Project Code	Unique identifier.	Text (Auto-generated)	System	e.g. PRJ-2026-0004.
Customer	Billable client.	Lookup Autocomplete	Optional	If blank, treated as an internal CapEx project.
Currency	Invoicing currency.	Dropdown	Required	Default KES. Locked once transactions exist.
Contract Budget	Total agreed deal value.	Currency (KES)	Required	Must be > 0 for customer projects. Format: 2 decimals.
Start Date	Project kick-off date.	Date (YYYY-MM-DD)	Required	Cannot be later than Target Completion Date.
Target Completion	Promised hand-over date.	Date (YYYY-MM-DD)	Required	Must be $\ge$ Start Date. Used for late milestone alerts.
Description / Scope	Detailed scope of work.	Rich Text (Max 2,000 chars)	Optional	Outlines deliverables and milestone criteria.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Creating and Opening a Turnkey Project

1. Navigate to /admin/projects and click New Project.
2. Enter Project Title: "Solar PV Installation - 5kVA System".
3. In the Customer lookup, select "Dr. Evans Mutiso".
4. Enter Contract Budget: KES 650,000.00.
5. Set Start Date: Tomorrow's date; Target Completion: 10 days later.
6. Click Save Draft. Status becomes Draft.
7. In the Tasks section, click Add Task:

- Task 1: "Site Survey & Structural Roof Check".
- Task 2: "Solar Panels Mounting & Conduit Piping".
- Task 3: "Inverter & Lithium Battery Wiring".

- Task 4: "KPLC Net-Metering Commissioning & Client Sign-Off".

8. In the top action bar, click Open Project.
9. The status transitions to Open. Team members can now issue stock and record time against the project.

Action 2: Completing Tasks and Milestone Invoicing

1. When technicians complete the roof mounting, open /admin/projects/[id].
2. Locate Task 2: "Solar Panels Mounting".
3. Click the checkmark icon to Mark Completed.
4. The task status updates to Completed.
5. Click Generate Milestone Invoice in the action bar.
6. The system opens the invoice builder pre-populated with the client's details, referencing Project #PRJ-2026-0012.
7. Enter Milestone 2 amount: KES 325,000.00 (50% payment).
8. Post the invoice. The project workspace immediately reflects: "Invoiced to Date: KES 325,000.00 (50%)".

Action 3: Closing a Fully Reconciled Project

1. When all tasks are finished and the client signs the completion certificate:
2. In the top action bar, click Mark Completed.
3. Status moves to Completed.
4. Review the final financial summary card:

- Total Invoiced: KES 650,000.00
- Actual Material Costs: KES 412,000.00
- Direct Labor Expenses: KES 45,000.00
- Net Gross Profit: KES 193,000.00 (29.7% Margin).

5. Click the Close Project button.
6. An InlineConfirm prompt appears: "Formally close project PRJ-2026-0012? No further expenses or lines may be posted."
7. Click Confirm Closure. The status transitions to Closed.

---

7. Conditions, Business Rules & Edge Cases

State Machine Restrictions

- A project can only be Closed if its current status is Completed. An Open or On Hold project cannot jump directly to Closed.
- If defects or post-installation warranty claims arise after completion, an authorized manager may click Reopen Project, moving the status from Completed back to Open.
- A project in Closed status is immutable: no new tasks, material requisitions, or invoices can be attached.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create a project in Draft status.	Saved successfully; editable capabilities enabled.	[ ]

2	Add 3 milestone tasks.	Tasks render in sequence with status Open.	[ ]
3	Transition project from Draft to Open.	Status badge updates to green Open; tasks become actionable.	[ ]
4	Attempt to close an Open project directly.	Action blocked with message: "Complete the project before closing it."	[ ]
5	Complete all tasks and close the project.	Status transitions to Closed; edit controls lock.	[ ]

Interactive version with live navigation: /admin/docs/08-projects-and-job-costing

CHAPTER 09

Guide 09: Vendor Management, Procurement & Payables (`/admin/vendors`)

Category: Supply Chain & Procurement | Reading Time: ~11 min | Words: 2,151

*The **Vendor Management & Procurement** module orchestrates the complete Procure-to-Pay (P2P) lifecycle. It encompasses vendor prequalification, sealed competitive quotation rounds (RFQs), purchase orders, Goods Received Notes (RGN), automa*

The Vendor Management & Procurement module orchestrates the complete Procure-to-Pay (P2P) lifecycle. It encompasses vendor prequalification, sealed competitive quotation rounds (RFQs), purchase orders, Goods Received Notes (RGN), automated 3-way invoice matching, landed cost capitalization, vendor payment vouchers, Accounts Payable ageing, anti-fraud bank account controls, and KRA Withholding Tax certificates.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- Procurement Fraud & Inflated Supplier Invoices: In traditional businesses, suppliers submit invoices that charge higher prices than originally quoted, or bill for 100 units when only 80 arrived. Accounts departments often pay these blindly. QSell enforces strict Automated 3-Way Matching: a supplier invoice cannot be posted unless its line quantities and prices match the original Purchase Order (PO) and the physical Goods Received Note (RGN) within configured tolerance limits.
- Vendor Bank Account Impersonation Fraud: The single most common corporate cybercrime occurs when a fraudster hacks a supplier's email and requests that pending invoice payments be sent to a "new bank account". QSell eliminates this through Mandatory Bank Change Quarantine: changing a vendor's bank details triggers a phone call-back verification protocol, secondary managerial approval, and a 72-hour payment cool-off period.
- Hidden Import & Freight Costs: Importing goods involves sea freight, port clearance, customs import duty, and local transport. If these expenses are treated as general overhead rather than capitalized into the product's unit cost, the business underprices its goods and loses money. QSell's Landed Cost Apportionment wizard automatically divides freight and duty across product unit costs based on weight or value.
- Statutory Withholding Tax (WHT) Penalties: Kenyan tax law requires companies to withhold 2% VAT or 3%/5% Income Tax from vendor payments and remit it to KRA. QSell calculates withholding tax automatically on every payment voucher and generates monthly KRA Withholding Certificates in one click.

Target Client Profiles

- Importers & Large Distributors: Managing container shipments, customs clearing agents, and multi-supplier product lines.
- Manufacturing & Assembly Plants: Procuring raw materials, packaging, and sub-components via competitive RFQs and blanket contracts.
- Corporate Enterprises & Institutions: Requiring formal procurement compliance (requisitions, multi-quote comparisons, separation of duties).

Commercial Sales Pitch & Talk Track

> "With QSell, you take total control of your company's spending. You eliminate procurement fraud before it happens. Your buyers can run sealed competitive quote rounds through our Vendor Self-Service Portal; your warehouse receives goods with digital Goods Received Notes; and your finance team never pays a single shilling until our system proves that the invoice matches the purchase order and the warehouse delivery note. Furthermore, our anti-fraud bank change security protects you against CEO and supplier wire fraud."

Measurable ROI & Safeguards

- 100% 3-Way Match Assurance: Eliminates over-billing and paying for missing goods.
- Automated Landed Costing: True inventory valuation ensures selling prices maintain required profit margins.
- Zero KRA Withholding Tax Fines: Automated calculation and certificate generation ensure full compliance with the Tax Procedures Act.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    CAT[1. Procurement Categories & Document Types Configured] --> VEND[2. Vendors Registered & Prequalified]
    VEND --> RULE[3. Ordering & Payment Rules Established]
    RULE --> POST[4. Vendor Posting Groups & AP Accounts Mapped]
    POST --> SEC[5. Anti-Fraud Bank Change Call-Back Policy Activated]
```

1. Procurement Categories: Defined under /admin/vendors/categories (e.g. Raw Materials, Office Supplies, Logistics).
2. Document Types: Compliance mandates set under /admin/vendors/document-types (e.g. KRA Tax Compliance Certificate, Business Permit).
3. Vendor Accounts: Created or invited under /admin/vendors.
4. Payment Rules & Tax Codes: Withholding VAT and Income Tax percentages set under /admin/vendors/payment-rules.
5. Vendor Posting Groups: Linked to General Ledger Accounts Payable under Finance > Posting Setup.
6. Required Permissions: vendors:read, purchase_orders:read/write, rgn:read/write, supplier_invoices:read/write, vendor_payments:read/write.

---

3. Module & Sub-Module Architecture

The Procure-to-Pay (P2P) pipeline spans 25 operational sub-modules:

```
graph TD
    subgraph Sourcing["1. Sourcing & Qualification"]
        PRE["Vendor Prequalification & Review Queue"]
        RFQ["Requests for Quotation (Sealed Rounds)"]
        CONT["Blanket Contracts & Price Locks"]
        DOC["Expiring Compliance Documents"]
    end

    subgraph Purchasing["2. Ordering & Goods Receipt"]
        REQ["Purchase Requisitions"]
        PO["Purchase Orders (Draft -> Approved -> Sent)"]
        RGN["Received Goods Notes (RGN / GRNI Accrual)"]
        RTV["Goods Sent Back (Return to Vendor)"]
    end

    subgraph Payables["3. Invoicing & Settlement"]
        INV["Supplier Invoices (3-Way Matching)"]
        MEMO["Credit Memos & Vendor Refunds"]
        VOUCH["Payment Vouchers (Dual Approval)"]
        WHT["Withholding Tax Certificates"]
        BANK["Anti-Fraud Bank Account Controls"]
    end

    Sourcing --> Purchasing
    Purchasing --> Payables
```

---

4. Interface-by-Interface Walkthrough

1. Vendors Directory (/admin/vendors)

- Master supplier cards: Company Name, KRA PIN, Primary Contact, Category, Prequalification Tier, Credit Terms (e.g. Net 30), and Balance Owed.

2. Requests for Quotation / RFQs (/admin/vendors/rfqs)

- Competitive bidding engine:
- Create an RFQ for specific items or services.
- Invite prequalified vendors to submit sealed bids through the Vendor Portal.
- Comparison Sheet: Side-by-side bid matrix comparing unit prices, delivery lead times, and payment terms.
- Award the winning vendor and auto-generate the Purchase Order in one click.

3. Purchase Orders (/admin/vendors/orders)

- Legally binding purchase commitments:
- Header: Vendor, Delivery Warehouse, Target Delivery Date, Currency.
- Lines: SKU, Quantity, Agreed Price, Tax Class.
- Lifecycle: Draft $\rightarrow$ Pending Approval $\rightarrow$ Released $\rightarrow$ Sent to Vendor $\rightarrow$ Confirmed $\rightarrow$ Partially Received $\rightarrow$ Fully Received $\rightarrow$ Closed.

4. Received Goods / RGN (/admin/vendors/receipts)

- Goods Receipt Note (RGN):
- When the delivery truck arrives, warehouse staff inspect physical boxes.
- Record line-by-line acceptance (e.g. Ordered: 100, Received Good: 95, Damaged: 5).
- Automatically posts the accounting accrual: Dr Inventory / Cr Goods Received Not Invoiced (GRNI).

5. Supplier Invoices (/admin/vendors/invoices)

- The 3-Way Matcher:
- Vendor submits their invoice via the Vendor Portal (or staff type it in).
- System automatically links the invoice lines to the corresponding RGN and PO lines.
- Flags price variances, quantity discrepancies, or duplicate invoice numbers.

6. Landed Costs (/admin/vendors/landed-costs)

- Import cost capitalization: Apportions shipping freight, marine insurance, and customs duty across received product lines by weight, volume, or value.

7. Payments to Vendors (/admin/vendors/payments)

- Payment voucher management:
- Select approved due invoices.
- Automatically deducts Withholding Tax (WHT).
- Routes voucher for Dual Managerial Approval before initiating bank EFT transfer or M-Pesa B2B payout.

8. What We Owe / Payables Ageing (/admin/vendors/payables)

- Accounts Payable ageing report: Current, 1-30 days, 31-60 days, 61-90 days, 90+ days overdue, with printable vendor statements.

9. Vendor Scorecards (/admin/vendors/scorecards)

- Automated supplier rating based on: Delivery on-time %, quality acceptance rate (% goods rejected at RGN), and price stability.

10. Anti-Fraud Bank Changes (/admin/vendors/bank-changes)

- Controlled quarantine for vendor bank account edits:
- Flags any request to update payment details.
- Requires telephone call-back verification to the vendor's primary registered landline.
- Requires 2nd senior manager sign-off.
- Enforces a mandatory 72-hour payment cool-off quarantine before funds can be disbursed to the new account.

11. Petty Cash Floats (/admin/vendors/petty-cash)

- Office expense float management: Imprest float balance, petty cash expense vouchers with scanned receipts, float replenishment workflows, and GL account allocation.

---

5. Granular Field-by-Field Reference

Purchase Order Form Fields (/admin/vendors/orders/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Vendor	Selected supplier.	Lookup Autocomplete	Required	Must be an active, approved vendor.
PO Number	Sequence code.	Text (Auto-generated)	System	e.g. PO-2026-0038.
Delivery Location	Destination warehouse.	Dropdown	Required	Warehouse receiving the physical shipment.
Promised Date	Expected arrival date.	Date (YYYY-MM-DD)	Required	Used for vendor scorecard on-time tracking.
Payment Terms	Credit agreement.	Dropdown	Required	e.g. Advance Payment, Cash on Delivery, Net 30.
Product Line: SKU	Item to purchase.	Lookup Autocomplete	Required	Must be an active product or raw material.
Product Line: Qty	Ordered quantity.	Decimal / Integer	Required	Must be > 0.
Product Line: Unit Cost	Purchase price per unit.	Currency (KES)	Required	Pre-populated from vendor contract or last price.
Product Line: Tax	VAT rate on purchase.	Dropdown	Required	16% Input VAT, Zero-Rated, or Exempt.

Supplier Invoice Matching Fields (/admin/vendors/invoices/[id])

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Vendor Invoice No	Number on vendor's bill.	Text (Max 60 chars)	Required	Unique per vendor. Duplicates blocked at entry.
Invoice Date	Date on vendor's paper.	Date (YYYY-MM-DD)	Required	Cannot be later than current server date.
Linked RGN	Matching Goods Receipt Note.	Lookup Autocomplete	Required	Pulls received quantities into match grid.

Price Tolerance	Acceptable variance %.	Decimal	System	Defined in settings (default 0.5%). Breaches require manager approval.
Withholding Tax Code	Applicable tax deduction.	Dropdown	Optional	e.g. WHT 2% (VAT) or WHT 5% (Management/ Services).

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Receiving Goods and Generating a Goods Received Note (RGN)

1. Navigate to /admin/vendors/receipts and click New Goods Receipt.
2. Select the Purchase Order (e.g. PO-2026-0038 from "SolarTech Kenya Ltd").
3. The lines auto-populate with the ordered items: "100 units of 500W Panels".
4. Enter the vendor's physical Delivery Note Number (e.g. DN-88192).
5. In the Received Qty column, type 100.
6. If 2 panels are shattered upon inspection, enter Accepted: 98, Rejected: 2, and select Rejection Reason: "Physical Transit Damage".
7. Click Post Goods Receipt (RGN).
8. The system updates warehouse stock immediately by 98 units, posts the GRNI accrual in the general ledger, and generates formal PDF RGN documentation for the delivery driver.

Action 2: 3-Way Matching and Booking a Supplier Invoice

1. Navigate to /admin/vendors/invoices and click New Supplier Invoice.
2. Select Vendor: "SolarTech Kenya Ltd".
3. Type the Vendor's Invoice Number: INV-99410.
4. Link the invoice to RGN #RGN-2026-0024.
5. The system performs the 3-Way Match:

- PO Price: KES 12,000.00 per panel.
- Invoice Price: KES 12,000.00 per panel. $\rightarrow$ MATCH
- RGN Accepted Qty: 98 units.
- Invoice Billed Qty: 98 units. $\rightarrow$ MATCH
- Tax Rate: 16% VAT. $\rightarrow$ MATCH

6. Green badges confirm: "3-Way Match Passed (0% variance)".
7. Click Post Supplier Invoice.
8. The document transitions to Posted, moves into Accounts Payable, clears the GRNI accrual, and books the liability to Vendor Payables.

Action 3: Processing an Anti-Fraud Vendor Bank Account Change

1. Navigate to /admin/vendors/bank-changes.
2. A pending request appears: "SolarTech Kenya Ltd submitted new bank details: Equity Bank, Acc #011223344".
3. The system locks all pending payment vouchers for SolarTech Kenya Ltd.
4. The Procurement Officer dials the vendor's registered landline telephone (taken from the original prequalification certificate, not from the email request).

5. After confirming the legitimacy of the change with the vendor's Managing Director, the officer enters the Call-Back Verification Code and notes.
6. The Finance Director opens the request and clicks Approve Bank Change.
7. The system initiates the 72-Hour Cool-Off Period. A countdown timer displays on screen: "Payments to this account unlock in 71h 59m", preventing immediate wire transfers.

---

7. Conditions, Business Rules & Edge Cases

The Automated 3-Way Matching Tolerance Rule

- If the price billed on a supplier invoice exceeds the purchase order price:
- If variance $\le \text{Configured Tolerance}$ (e.g. 0.5% or KES 50), the system permits booking with a price adjustment log.
- If variance $> \text{Configured Tolerance}$, booking is blocked until the procurement manager issues an approved Purchase Order Amendment or the vendor submits a corrected invoice.

Duplicate Invoice Protection

- The system enforces a strict composite database unique constraint: (vendor_id, LOWER(TRIM(vendor_invoice_number))).
- If an accountant attempts to enter an invoice number that already exists for that vendor (even with extra spaces or different letter casing), the system blocks the entry with the message: "Invoice [Number] has already been booked for this vendor on [Date]."

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create a Vendor Master Card with KRA PIN and Credit Terms.	Vendor saved and visible in purchasing lookup lists.	[ ]
2	Issue a Purchase Order and receive goods via RGN.	Stock on hand increases; GRNI ledger accrual posted.	[ ]
3	Enter a Supplier Invoice with mismatched quantities.	System flags quantity discrepancy and blocks posting until reconciled.	[ ]
4	Test Anti-Fraud Bank Change Workflow.	Change triggers telephone call-back requirement and 72-hour cool-off.	[ ]
5	Post a Payment Voucher with Withholding Tax (WHT).	2% VAT or 5% WHT deducted; KRA certificate generated.	[ ]

Interactive version with live navigation: /admin/docs/09-vendor-management-procurement-and-payables

CHAPTER 10

Guide 10: Finance, General Ledger & Accounts Receivable (`/admin/finance`)

Category: Finance & Accounting | Reading Time: ~12 min | Words: 2,240

*The **Core Finance & Accounts Receivable** module provides a complete, double-entry financial accounting system with the discipline of Microsoft Dynamics 365 Business Central. It encompasses the Chart of Accounts, General Journals, immutabl*

The Core Finance & Accounts Receivable module provides a complete, double-entry financial accounting system with the discipline of Microsoft Dynamics 365 Business Central. It encompasses the Chart of Accounts, General Journals, immutable G/L Entries and Registers, Real-Time Trial Balances, Fiscal Periods, the Posting Setup Matrix, Customer Receipts Allocation, Customer Wallets (Store Credit), Safeguarded Refunds, Foreign Exchange Revaluation, and Debtor Ageing.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The Month-End "Disappearing Debtors" Hunt: In traditional systems, sales invoices, cashier receipts, and warehouse dispatches are recorded in separate tables that frequently disagree with the general ledger. Accountants spend weeks chasing discrepancies between the Accounts Receivable sub-ledger and the G/L balance. QSell guarantees Sub-Ledger and G/L Invariant Harmony: every invoice, receipt, return, and wallet credit is posted together with its balancing G/L double entries in the exact same atomic transaction.
- Accidental Back-Dated Posting Chaos: A rogue or untrained user edits an invoice from three months ago, throwing off previously certified and tax-audited financial statements. In QSell, Posted Means Immutable. Past journal lines can never be edited or deleted. Errors must be corrected by formal reversals or credit memos with mandatory reason logging. Furthermore, closed fiscal periods are cryptographically locked.
- Overpayment & Customer Credit Chaos: When a customer pays KES 10,000 for an KES 8,000 order, most systems treat the extra KES 2,000 as a negative debt or loose money. QSell features a native Customer Wallet Sub-Ledger: unallocated overpayments or return credits are securely parked in the customer's digital store wallet as a clean liability, ready to automatically offset future purchases.
- Refund Embezzlement & Fraud: Internal staff frequently issue unauthorized cash refunds to personal phone numbers. QSell enforces a Separation of Duties Refund Framework: refunds require dual managerial approval, mandatory upload of supporting evidence, phone-number velocity limits, and daily payout caps.

Target Client Profiles

- Corporate CFOs, Financial Controllers & Chief Accountants: Demanding IFRS-compliant double-entry accounting, audit-ready journal registers, and instant drill-down from financial statements to source transactions.
- High-Volume Retail & Wholesale Distributors: Managing thousands of customer invoices, partial payments, credit notes, and multi-currency transactions.
- B2B Trade Merchants: Managing credit customers with strict 30/60/90-day payment terms and debt collection schedules.

Commercial Sales Pitch & Talk Track

> "With QSell, you get the financial rigor and posting discipline of tier-one enterprise ERPs like Microsoft Dynamics Business Central, built specifically for modern African businesses. There are no batch delays and no month-end reconciliation surprises. The second a cashier rings up a sale or an M-Pesa payment confirms, your General Ledger, Accounts Receivable sub-ledger, and Trial Balance update simultaneously. Your books are permanently audit-ready, your customer store wallets are automated, and your refund payouts are locked with anti-fraud limits."

Measurable ROI & Safeguards

- Zero Reconciliation Hunting: General Ledger control accounts and Accounts Receivable balances match at all times.
- 100% Audit Integrity: Append-only ledger registers record Who, What, When, Why, and From Where for every shilling.
- Guaranteed Fraud Protection: Segregation of duties prevents the person creating a credit memo or refund from approving or disbursing the payout.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    COA[1. Chart of Accounts Defined] --> POST[2. General Posting Setup Matrix Configured]
    POST --> PER[3. Fiscal Calendar Periods Opened]
    PER --> WAL[4. Customer Credit & Wallet Policy Set]
    WAL --> LIMIT[5. Refund Anti-Fraud Limits Assigned]
    LIMIT --> TRANS[6. Financial Invoicing & Postings Active]
```

1. Chart of Accounts: Defined under /admin/finance/accounts (Assets, Liabilities, Equity, Revenue, COGS, Expenses).
2. Posting Setup Matrix: Configured under /admin/finance/posting-setup (Mapping which G/L accounts receive sales revenue, inventory, tax, and receivables postings).
3. Fiscal Calendar: Periods opened under /admin/finance/periods (At least one active open accounting period).
4. Customer Credit & Wallet Policy: Defined under /admin/finance/receivables/wallet-policy.
5. Refund Limits: Defined under /admin/finance/refunds/limits.
6. Required Permissions: gl_accounts:read, journals:read/post, ledger_registers:read, trial_balance:read, receipts:read/create, refunds:read/act.

---

3. Module & Sub-Module Architecture

The Finance architecture coordinates the General Ledger and Accounts Receivable sub-ledgers:

```
graph TD
    subgraph OperationalSources["Operational Triggers"]
        SALE[Sales Invoices & POS]
        PMT[Customer Payments & M-Pesa]
        RET[Goods Returns & Credit Notes]
        STK[Warehouse Cost Issues]
    end

    subgraph PostingEngine["Core Financial Engine (src/server/domains/finance)"]
        MATRIX[Posting Setup Matrix]
        VAL[Fiscal Period & Balance Validation]
        POST[Atomic Multi-Ledger Commit]
    end

    subgraph PermanentBooks["Immutable Financial Ledgers"]
        GL[General Ledger Entries: gl_entries]
        REG[Posting Registers: ledger_registers]
        AR[Customer Sub-Ledger: customer_detailed_entries]
        WAL[Customer Wallet Liability: wallet_entries]
        TB[Real-Time Trial Balance]
    end

    OperationalSources --> PostingEngine
    PostingEngine --> MATRIX
    MATRIX --> VAL
    VAL --> POST
    POST --> GL
    POST --> REG
    POST --> AR
    POST --> WAL
```

```
    GL --> TB
```

---

4. Interface-by-Interface Walkthrough

Part A: General Ledger (/admin/finance)

1. Chart of Accounts (/admin/finance/accounts)

- Master list of accounts organized by category: 1000 Assets, 2000 Liabilities, 3000 Equity, 4000 Revenue, 5000 Cost of Goods Sold, 6000 Operating Expenses.
- Account configuration: Account Code, Account Name, Account Type (Posting, Heading, Total), Direct Posting Toggle, Blocked Status, and Balance Sheet / Income Statement mapping.

2. General Journals (/admin/finance/journals)

- Manual journal preparation workspace:
- Create balanced debit and credit entries with Document Number, Posting Date, G/L Account, Description, Amount, and Dimension Sets (Branch, Department).
- Posting Preview: Runs the complete posting simulation in memory and renders every entry that would be created before writing to the database.

3. G/L Entries (/admin/finance/entries)

- The immutable general ledger: Search and filter millions of ledger lines by Account, Date, Document No, or Register ID. Shows Debit, Credit, and Source Document link.

4. Postings / Registers (/admin/finance/registers)

- The master audit log: Each row represents one atomic posting event, recording Register No, Starting/Ending Entry ID, Posting Date, User ID, Source Document, and Reason Code.
- Reverse Transaction Action: Reverses an entire register with an equal-and-opposite posting in one click with mandatory reason capture.

5. Trial Balance (/admin/finance/trial-balance)

- Real-time trial balance report: Filter by Date Range, Dimension (Branch/Cost Center), or Account Group. Shows Opening Balance, Period Net Movement (Debits/Credits), and Closing Balance. Total Debits strictly equal Total Credits.

6. Fiscal Periods (/admin/finance/periods)

- Fiscal calendar management: Configure 12 monthly periods per fiscal year. Set period status (Open, Soft-Closed, Closed, Locked). Prevents back-dated postings into audited periods.

7. Posting Setup (/admin/finance/posting-setup)

- The business-to-accounting translation matrix: Maps product business classifications, customer groups, and payment methods to specific G/L accounts (e.g. Retail Hardware Sales $\rightarrow$ Account 4100; Wholesale Cement Sales $ \rightarrow$ Account 4200).

Part B: Accounts Receivable (/admin/finance/receipts, /admin/finance/receivables)

8. Receipts (/admin/finance/receipts, /new)

- Customer payment recording:
- Select Customer, Payment Method (Cash, Bank Transfer, Cheque, M-Pesa), Amount Paid, and Payment Reference.
- Invoice Allocation Grid: Displays customer's open unpaid invoices. Allows auto-allocating payment to oldest invoices first, or manually selecting specific invoices to settle.

9. Customer Wallets (/admin/finance/receivables/wallets)

- Customer store credit liability sub-ledger:
- Displays customer wallet balances.
- History of wallet funding (overpayments, returned goods, cash advances) and redemptions against orders.

10. Credit Adjustments (/admin/finance/receivables/credit-adjustments)

- Manual adjustments to customer balances (bad debt write-offs, courtesy credits):
- Requires mandatory 2nd-person managerial approval before affecting the general ledger.

11. Customer Refunds & Limits (/admin/finance/refunds, /limits)

- Payout governance:
- Multi-tier refund request workflow (Request $\rightarrow$ Verification & Evidence $\rightarrow$ Approval $\rightarrow$ Disbursement).
- Anti-Fraud Limits: Sets maximum daily refund amount per customer phone number, per cashier, and per branch.

12. Receivables Ageing (/admin/finance/receivables/ageing)

- Debtor analysis dashboard: Buckets outstanding customer debt into Current, 1-30 Days, 31-60 Days, 61-90 Days, and 90+ Days. Drill down to individual overdue invoices to trigger dunning notices or pause credit accounts.

13. Foreign Exchange & Revaluation (/admin/finance/exchange-rates)

- Multi-currency rate table and month-end FX revaluation wizard: Automatically calculates and posts unrealized foreign exchange gains or losses on foreign currency receivables (USD, EUR, GBP) to the G/L.

---

5. Granular Field-by-Field Reference

Chart of Accounts Fields (/admin/finance/accounts/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Account Code	Unique accounting number.	Text (Alphanumeric, Max 20)	Required	Must be unique (e.g. 1100, 4100).
Account Name	Description of the account.	Text (Max 120 chars)	Required	e.g. "Accounts Receivable - Trade".
Account Type	Role in the financial chart.	Dropdown	Required	Posting (holds entries), Heading, Total (summary).
Category	Primary financial statement classification.	Dropdown	Required	Asset, Liability, Equity, Revenue, COGS, Expense.
Direct Posting	Allows manual journal entries.	Toggle (On/Off)	Required	Set to Off for control accounts (AR, AP, Stock) to prevent manual tampering.
Blocked Status	Prevents future postings.	Toggle (On/Off)	Required	Blocked accounts reject all new journal postings.

Customer Receipt Fields (/admin/finance/receipts/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Customer	Account making the payment.	Lookup Autocomplete	Required	Loads customer's open invoices and current wallet balance.

Receipt Date	Date funds cleared.	Date (YYYY-MM-DD)	Required	Must fall within an open fiscal period.
Payment Method	Ingress payment channel.	Dropdown	Required	e.g. M-Pesa Paybill, KCB Bank Transfer, Cash Till.
Amount Received	Monetary sum paid.	Currency (KES)	Required	Must be > 0. Format: 2 decimal places.
Reference / Trx ID	Bank or M-Pesa transaction code.	Text (Max 60 chars)	Required	e.g. QA812739182 or Cheque number.
Overpayment Handling	What to do with excess cash.	Dropdown	Required	Credit to Customer Wallet, Leave as Unapplied Receipt.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Recording and Allocating a Customer Payment

1. Navigate to /admin/finance/receipts and click New Receipt.
2. In the Customer lookup, select "Nairobi Hardware Wholesalers".
3. Enter Amount Received: KES 150,000.00.
4. In the Payment Method dropdown, select "Bank Transfer - KCB".
5. In the Reference box, enter the bank deposit slip number: DEP-99412.
6. The screen displays the customer's unpaid invoices:

- Invoice #INV-1002: Balance KES 80,000.00.
- Invoice #INV-1008: Balance KES 50,000.00.
- Invoice #INV-1014: Balance KES 40,000.00.

7. Click Auto-Allocate (Oldest First).
8. The system allocates KES 80,000 to INV-1002 (paid in full), KES 50,000 to INV-1008 (paid in full), and the remaining KES 20,000 to INV-1014 (partial payment, KES 20,000 remaining).
9. Click Post Receipt.
10. The receipt is posted. General ledger entries are generated instantly (Dr KCB Bank / Cr Accounts Receivable), and official payment receipts are generated.

Action 2: Posting a General Journal with Posting Preview

1. Navigate to /admin/finance/journals and open the "General Operations" batch.
2. Click Add Journal Line:

- Account: 6100 - Office Electricity & Water.
- Side: Debit, Amount: KES 18,500.00.
- Description: "March Power Bill - Head Office".

3. Click Add Balancing Line:

- Account: 1120 - Petty Cash Float.
- Side: Credit, Amount: KES 18,500.00.

4. Click the Posting Preview button in the header.
5. The preview modal renders showing the exact G/L entries that will be written, proving debits equal credits.

6. Click Confirm & Post Journal.
7. The register number is generated (REG-2026-0182), entries are written to gl_entries, and the batch clears.

Action 3: Reversing an Erroneous Financial Posting

1. Navigate to /admin/finance/registers.
2. Locate the register created in error (e.g. REG-2026-0182).
3. Click View Register Details.
4. In the top action bar, click Reverse Register.
5. The system displays a confirmation modal requiring a Mandatory Reversal Reason:

- Type reason: "Incorrect electricity bill amount entered; will re-enter with official KPLC receipt."

6. Click Confirm Reversal.
7. The system creates a new balancing register (REG-2026-0183) containing exact equal-and-opposite debits and credits, permanently linking it to the original register in the audit trail.

---

7. Conditions, Business Rules & Edge Cases

Direct Posting Lock on Control Accounts

- Control accounts (such as Accounts Receivable 1200, Accounts Payable 2100, and Inventory at Cost 1300) have direct_posting = false.
- If a user attempts to create a manual general journal line posting directly to Account 1200, the system blocks the action with the error: "Direct posting is disabled for Account 1200 (Accounts Receivable). Post customer transactions through customer invoices, receipts, or credit memos to preserve sub-ledger integrity."

Fiscal Period Locking Rules

- If an accounting period is set to Closed or Locked (e.g. December 2025):
- Any transaction dated within that period is strictly rejected.
- Transactions must have their posting date moved into the earliest open fiscal period.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Import or configure standard Chart of Accounts.	Accounts appear in numerical order with categories assigned.	[ ]
2	Configure Posting Setup matrix for Sales, Inventory, and Tax.	Test invoice posts accurately without missing account errors.	[ ]
3	Execute Posting Preview on a manual General Journal.	Preview modal displays balanced debits and credits.	[ ]
4	Post a customer receipt and verify Trial Balance.	Accounts Receivable decreases; Bank Account balance increases; TB balances.	[ ]
5	Attempt to post a journal into a Closed Fiscal Period.	System rejects entry with explicit period closed warning.	[ ]

Interactive version with live navigation: /admin/docs/10-finance-general-ledger-and-accounts-receivable

CHAPTER 11

Guide 11: Banking, Cash & Payment Gateways (`/admin/finance/bank-accounts`, `/admin/payments`, `/admin/currencies`)

Category: Finance & Accounting | Reading Time: ~11 min | Words: 2,048

*The **Banking, Cash & Payment Gateways** module governs the physical and digital cash flow of the business. It encompasses commercial bank accounts, physical cashier cash drawers, Safaricom M-Pesa merchant integrations (STK Push, Paybill, T*

The Banking, Cash & Payment Gateways module governs the physical and digital cash flow of the business. It encompasses commercial bank accounts, physical cashier cash drawers, Safaricom M-Pesa merchant integrations (STK Push, Paybill, Till, and Bill Manager), automated bank statement CSV reconciliations, multi-currency exchange rates, and raw integration audit logs.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The "Missing M-Pesa Payment" Nightmare: In typical businesses, customers send money to a Paybill number, but network drops or customer typos mean the order remains "Unpaid" on the screen while funds are deducted from the customer's phone. Staff argue with customers and manually check Safaricom statement portals. QSell provides Real-Time Payment Activity Telemetry: every inbound webhook, STK prompt, and transaction payload is captured live. If an orphan payment arrives, the Reconciliation Workspace matches it to the pending order with a single click.
- Cash Till Shortages & Till Theft: In busy retail stores, cashiers collect physical cash notes all day. Without strict controls, till shortages are discovered days later when it is impossible to identify which cashier was responsible. QSell mandates End-of-Day Cash Drawer Counts: cashiers count their physical notes (KES 1,000s, 500s, 200s, 100s, 50s, coins) against the system's expected book balance, and variances are automatically flagged and posted.
- Manual Bank Statement Reconciliation: Financial controllers spend days each month printing bank statements and ticking off line items against customer deposits. QSell features an Automated Bank Statement CSV Reconciler: upload a standard bank statement, and the system automatically matches deposits to open invoices and customer ledger entries based on reference numbers, dates, and amounts.
- Exposing API Keys in Plaintext: Many platforms store payment provider API keys and consumer secrets in unencrypted configuration files. QSell encrypts all API credentials at rest in the database with hardware-grade cryptographic sealing; keys are never exposed on screen or printed in server logs.

Target Client Profiles

- High-Velocity Omnichannel Retailers: Processing hundreds of M-Pesa and card transactions hourly across in-store POS and web shop.
- Multi-Branch Store Networks: Operating independent bank accounts and cash drawers per branch location.
- Cross-Border Regional Traders: Invoicing and collecting in multiple currencies (KES, USD, EUR, UGX, TZS) with daily exchange rate revaluations.

Commercial Sales Pitch & Talk Track

> "With QSell, you will never lose track of a single payment. We integrate natively with Safaricom M-Pesa (both STK Push and Bill Manager), major Kenyan banks, and card processors. When a customer pays, your books update immediately. If a network blip occurs, our payment activity dashboard shows you the raw gateway response in real time. And at the end of every day, your cashiers conduct digital till counts that verify physical cash against book sales down to the last five-shilling coin."

Measurable ROI & Safeguards

- 100% Payment Ingress Visibility: Zero lost mobile-money transactions; orphan payments matched in seconds.
- Instant Till Shortage Detection: Physical cash variances highlighted before cashiers leave the premises.
- Zero API Credential Exposure: Sealed cryptographic storage ensures bank and M-Pesa secrets cannot be stolen.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    CURR[1. Currencies Configured: KES Base, USD, EUR] --> BANK[2. Bank Accounts & Cash Drawers Created]
    BANK --> PROV[3. Payment Providers Configured: M-Pesa, Banks, Cards]
    PROV --> METH[4. Payment Methods Bound to Operating Branches]
    METH --> RECON[5. Daily Bank Statements & Till Counts Active]
```

1. Currencies Active: Base currency (KES) and foreign currencies established under /admin/currencies.
2. Bank Accounts Created: Defined under /admin/finance/bank-accounts (Commercial banks and physical tills).
3. Payment Providers Configured: Credentials entered under Settings > Payments.
4. Public Base URL Set: Company or branch must have a valid public HTTPS URL configured so gateway webhooks can reach the server.
5. Staff Permissions: bank_accounts:read/write, cash_counts:read/create, payment_activity:read, transactions:reconcile.

---

3. Module & Sub-Module Architecture

The Banking and Payment ecosystem is structured into three operational layers:

```
graph TD
    subgraph Ingress["1. Ingress & Settlement Channels"]
        STK["M-Pesa STK Push (Online / POS)"]
        C2B["Paybill / Till C2B Inbound"]
        BILL["M-Pesa Bill Manager (E-Invoices)"]
        CARD["Credit / Debit Cards (Pesapal)"]
        CASH["Cashier Physical Till Drawers"]
    end

    subgraph Telemetry["2. Activity & Audit Telemetry"]
        ACT["Payment Activity Webhook Stream"]
        AUDIT["Integration Audit Log (Payload Inspector)"]
        TRANS["Raw Transaction Ledger"]
    end

    subgraph Reconciliation["3. Financial Reconciliation"]
        RECON["Payment Reconciliation Matcher"]
        STMT["Bank Statement CSV Import"]
        TILL["End-of-Day Cash Till Counts"]
        GL["General Ledger Bank Posting"]
    end
```

```
    Ingress --> Telemetry
    Telemetry --> Reconciliation
    Reconciliation --> GL
```

4. Interface-by-Interface Walkthrough

1. Bank Accounts (/admin/finance/bank-accounts)

- Master directory of corporate liquid funds:
- Commercial Bank Accounts (e.g. KCB, Equity Bank, Standard Chartered).

- Mobile Money Settlement Accounts (Safaricom M-Pesa Paybill / Buy Goods Till).
- Physical Cash Drawers (e.g. Till #1 - Westlands, Till #2 - Mombasa).
- Displays real-time Book Balance, Last Reconciled Date, Currency, and Status.

2. Bank Statements & Reconciliation (/admin/finance/bank-statements)

- Reconciles bank reality with general ledger books:
- Upload official bank statement CSVs.
- Interactive split screen: Statement Lines on the left; System Bank Ledger Entries on the right.
- Auto-Match Button: Matches transactions by Reference ID, Date, and Shilling Amount.
- Unmatched transactions can be posted directly as bank charges, interest income, or direct customer deposits.

3. Cash Counts (/admin/finance/cash-counts)

- End-of-shift retail till verification:
- Select Branch and Cash Drawer.
- Enter physical denomination counts (Quantity of KES 1,000, 500, 200, 100, 50, 40, 20, 10, 5, 1).
- Displays Expected Balance vs. Counted Balance, calculating exact Shortage or Overage.
- Posts variance adjustment entries with mandatory supervisor approval.

4. Currencies & FX (/admin/currencies)

- Multi-currency management: Base currency (KES) and foreign currencies (USD, EUR, GBP, UGX, TZS).
- Maintain daily spot exchange rates and view monthly FX revaluation logs.

5. Transactions Ledger (/admin/transactions/ledger)

- Low-level payment record: Shows every payment attempt, gateway transaction ID, channel, customer phone number, monetary value, and settlement status (Pending, Paid, Failed, Reconciled).

6. Payment Activity (/admin/payments/activity)

- Real-time developer & operations log: Displays live stream of incoming gateway webhook notifications from Safaricom Daraja, banks, and card processors.
- Essential for customer care desks investigating delayed or disputed payments.

7. M-Pesa Bill Manager Invoices (/admin/payments/bills)

- Push electronic bills directly to customer Safaricom M-Pesa apps:
- Select customer invoice.
- Send M-Pesa Bill Manager notification.
- The customer receives an official Safaricom notification on their phone displaying invoice details and a single "Pay Now" prompt.
- Receipting is 100% automated upon PIN entry.

8. Reconciliation (/admin/reconciliation)

- The orphan payment matching engine:
- Lists payments received by M-Pesa or bank transfer that could not be automatically matched to an order (e.g. customer entered wrong account number).
- Search pending customer orders and click Manually Match & Settle to instantly confirm the order.

9. Integration Audit Log (/admin/integration-audit)

- Deep forensic payload viewer: Inspect the complete HTTP headers, request payloads, and response bodies for every outbound API call and inbound webhook.

---

5. Granular Field-by-Field Reference

Bank Account Master Fields (/admin/finance/bank-accounts/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Account Code	Internal accounting ID.	Text (Max 20 chars, uppercase)	Required	Unique e.g. BANK-KCB-01.
Account Name	Descriptive label.	Text (Max 120 chars)	Required	e.g. "KCB Operating Account - Industrial Area".
Account Type	Nature of the account.	Dropdown	Required	Commercial Bank, Cash Drawer (Till), Mobile Money Settlement.
Bank Name	Institution name.	Text (Max 120 chars)	Required for Bank	e.g. "KCB Bank Kenya".
Account Number	Official account number.	Text (Max 40 chars)	Required for Bank	Account number at the bank.
Currency	Monetary denomination.	Dropdown	Required	e.g. KES, USD. Locked after transactions occur.
Linked G/L Account	General Ledger posting account.	Lookup Autocomplete	Required	Maps to Asset account in Chart of Accounts (e.g. 1110 Bank).

Cash Drawer Count Fields (/admin/finance/cash-counts/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Cash Drawer	Target till being audited.	Dropdown	Required	Select from active tills at user's branch.
Count Date & Time	Timestamp of audit.	DateTime	System	Recorded at server time of count submission.
Denominations	Physical note and coin counts.	Integer per denomination	Required	Enter quantity of notes: KES 1000, 500, 200, 100, 50, coins.
Supervisor Witness	Witnessing staff member.	Lookup Autocomplete	Required	Staff member co-signing the physical count.
Variance Explanation	Reason for cash mismatch.	Text (Max 500 chars)	Required if Variance $\ne 0$	Explains shortage or overage before posting.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Reconciling an Orphan M-Pesa Payment to an Order

1. A customer calls claiming they paid via M-Pesa Paybill, but their web order still shows "Pending Payment".
2. Navigate to /admin/reconciliation.
3. In the Unmatched Payments table, search by the customer's phone number or the M-Pesa transaction code (e.g. QA9912048X).
4. The payment is located: "KES 4,500.00 received at 11:42 AM from 0712345678".
5. In the Candidate Orders pane on the right, the system automatically suggests Order #ORD-10928 based on matching phone number and shilling amount.

6. Click Link Payment to Order.
7. An InlineConfirm prompt appears: "Match transaction QA9912048X (KES 4,500.00) to Order ORD-10928?"
8. Click Confirm Match.
9. The order status updates instantly to Paid, stock is reserved, a confirmation SMS is sent to the customer, and the payment is marked reconciled in the transaction ledger.

Action 2: Executing an End-of-Day Cash Till Count

1. At store closing, the cashier navigates to /admin/finance/cash-counts and clicks New Cash Count.
2. Select Cash Drawer: "Till #1 - Westlands".
3. The system shows that the cashier's till is locked for new sales during counting.
4. Count the physical banknotes and enter the counts:

- KES 1,000 notes: 45 ($= \text{KES } 45,000$)
- KES 500 notes: 12 ($= \text{KES } 6,000$)
- KES 200 notes: 8 ($= \text{KES } 1,600$)
- KES 100 notes: 15 ($= \text{KES } 1,500$)
- Coins: KES 240.00

5. Total Counted calculates: KES 54,340.00.
6. System Book Balance shows: KES 54,350.00 (Variance: - KES 10.00 Shortage).
7. In the explanation box, enter: "Small coin shortage due to change deficit".
8. Select the Supervisor Witness and click Submit & Close Till.
9. The count is saved, the till is closed, and an adjustment journal for KES 10.00 is posted to the Cash Shortage expense account.

---

7. Conditions, Business Rules & Edge Cases

Payment Provider Webhook Signature Verification

- Inbound webhooks from payment gateways (Safaricom Daraja, Pesapal) are cryptographically validated against sealed company secrets.
- Any request lacking a valid HMAC signature or originating from an unauthorized IP address is rejected immediately with HTTP 401 and logged in the Integration Audit Log as a security incident.

Reconciling Multi-Currency Bank Accounts

- A bank account denominated in USD can only be reconciled against statement CSVs in USD.
- When customer payments in USD are allocated to KES invoices, the system automatically uses the active spot exchange rate and writes any currency conversion discrepancy to the Foreign Exchange Gain/Loss G/L Account.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create Bank Account and Cash Drawer master records.	Accounts render in Bank Accounts directory with correct GL bindings.	[ ]

2	Configure M-Pesa Daraja STK Push credentials.	Test KES 10.00 STK push successfully prompts test phone.	[ ]
3	Verify Payment Activity Webhook Stream.	Gateway callback response renders in real time with status "Confirmed".	[ ]
4	Upload a sample Bank Statement CSV.	Auto-Match button successfully pairs statement lines to system receipts.	[ ]
5	Perform an End-of-Day Till Count with a KES 50 variance.	Variance calculation accurate; adjustment journal posted upon supervisor sign-off.	[ ]

Interactive version with live navigation: /admin/docs/11-banking-cash-and-payment-gateways

CHAPTER 12

Guide 12: Approvals & Internal Controls (`/admin/approvals`, `/admin/settings/approvals`)

Category: Governance & Security | Reading Time: ~10 min | Words: 1,934

*The **Approvals & Governance** module delivers a zero-code, runtime corporate approval engine inspired by Microsoft Dynamics 365 Business Central. It allows administrators to define, version, simulate, and enforce approval hierarchies acros*

The Approvals & Governance module delivers a zero-code, runtime corporate approval engine inspired by Microsoft Dynamics 365 Business Central. It allows administrators to define, version, simulate, and enforce approval hierarchies across purchase orders, price discounts, sales credit limits, inventory adjustments, and customer refunds—with tiered signing limits, automatic substitute rerouting, and real-time bottleneck monitoring.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- Rigid, Hard-Coded Approval Rules: In most software, if a board of directors changes the purchase order signing limit from KES 100,000 to KES 250,000, software developers must be paid to rewrite and redeploy code. QSell's approval rules are 100% Runtime Configuration: administrators change thresholds, add approvers, or toggle approval rules on or off in seconds with zero downtime.
- Operational Gridlock When Managers Travel: In traditional organizations, business grinds to a halt when the Managing Director goes on leave or travels abroad without connectivity; urgent purchase orders and customer deliveries sit unapproved. QSell features native Substitute Approver Chains: if a designated manager is on leave, requests automatically route to their certified substitute, or escalate to the next qualified manager after a configurable due-time threshold.
- In-Flight Workflow Corruption: In legacy systems, editing an approval rule while orders are currently pending either crashes the system or leaves documents in an orphaned "limbo" state. QSell uses Immutable Versioned Workflows: active in-flight documents complete under the exact workflow version they started with, while new requests seamlessly use the updated version.
- Unexplained Approval Rejections: Managers often reject employee requests without explanation, creating workplace frustration and delayed customer service. QSell enforces Mandatory Reason Codes and Comments: an approver cannot click "Reject" without selecting a standardized reason code and providing an explanatory note.

Target Client Profiles

- Corporate Enterprises & Institutions: Requiring strict financial governance and tiered signing authorities (e.g. Supervisor up to KES 50k, Manager up to KES 250k, Director Unlimited).
- Multi-Branch Retail Chains: Requiring localized approval hierarchies (e.g. Mombasa store manager approves Mombasa write-offs, while Head Office approves capital expenditures).
- Audit-Regulated Organizations: Subject to internal and external financial control audits (ISO 9001, IFRS, Sarbanes-Oxley).

Commercial Sales Pitch & Talk Track

> "With QSell, you enforce total financial discipline without slowing down your business. You can set up multi-step approval workflows for large purchase orders, high discounts, or stock write-offs without writing a single line of code. If your Commercial Director is out of office, the system automatically routes the request to their appointed deputy. And our visual Approval Monitor gives you complete visibility over what is waiting on whom, highlighting bottlenecks before they delay a customer."

Measurable ROI & Safeguards

- 100% Zero-Code Workflow Maintenance: Modify signing limits and approvers in seconds without IT developer costs.
- Zero Document Deadlock: Automated timeout escalations prevent documents from stalling during executive absence.
- Complete Governance Audit Trail: Every approval, rejection, delegation, and escalation is timestamped and permanently logged.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    USERS[1. Staff Accounts & Hierarchy Created] --> PROF[2. Approver Profiles & Amount Limits Assigned]
    PROF --> GRP[3. Approver User Groups Configured]
    GRP --> WORK[4. Workflows Designed with Triggers & Steps]
    WORK --> SWITCH[5. Module & Scenario Switches Activated]
    SWITCH --> SIM[6. Workflow Simulation Verification]
```

1. Staff Directory: Users must exist under /admin/users.
2. Approver Profiles & Limits: Configured under /admin/settings/approvals/approvers (Assigning signing limits and substitutes).
3. Approver Groups: Configured for multi-person approval committees (e.g. Tender Committee).
4. Workflow Designer: Built under /admin/settings/approvals/workflows.
5. Switches: Toggled active under /admin/settings/approvals/switches.
6. Required Permissions: approvals:act to approve/reject; approvals:request to submit; approvals:manage to design workflows.

---

3. Module & Sub-Module Architecture

The Approvals engine separates operational tasks from administrative configuration:

```
graph TD
    subgraph Operations["Front-Office Operations (/admin/approvals)"]
        INBOX["Waiting on Me (Actionable Inbox)"]
        MYREQ["My Requests (Submissions Tracker)"]
    end

    subgraph Administration["Governance Configuration (/admin/settings/approvals)"]
        WORKFLOWS["Workflow Designer & Versioning"]
        LIMITS["Approvers, Limits & Substitutes"]
        SWITCHES["Module & Scenario On/Off Switches"]
        MONITOR["Real-Time Bottleneck Monitor"]
    end

    Administration --> Operations
```

The Four Approver Resolution Types

When a document enters a workflow step, the engine resolves who must act based on one of four rules:

1. Specific Approver / Group: Sent directly to a named individual or an ordered user committee (e.g. "Finance Director" or "Tender Committee").
2. Requester's Line Manager: Sent to the direct supervisor of the person who initiated the document.
3. Approver Chain by Limit: Evaluates the requester's supervisor; if the document amount exceeds their signing limit, it automatically climbs to their supervisor, and so on until a qualified approver is reached.
4. First Qualified Approver: Skips intermediate managers and jumps directly to the first person up the chain whose signing limit covers the document amount.

---

4. Interface-by-Interface Walkthrough

Part A: Front-Office Operations (/admin/approvals)

1. Waiting on Me (/admin/approvals/inbox)

- The actionable executive queue: Displays all documents currently requiring the logged-in user's approval.
- Summary columns: Document Type (Purchase Order, Sales Order, Credit Memo, Stock Write-Off, Customer Refund), Document No, Requester Name, Total Amount, Due In / Overdue indicator.
- Action Buttons:
- Approve: Instantly releases the document to the next step or final approved state.
- Reject: Opens mandatory rejection modal requiring a reason code and explanatory note.
- Delegate: Forwards the request to another qualified user or designated substitute.

2. My Requests (/admin/approvals/requests)

- Personal submissions tracking: Shows documents submitted by the user that are currently in-flight.
- Shows current step, assigned approver, submission date, and time elapsed.
- Action: Cancel Request (withdraws the document back to Draft state before an approver acts).

Part B: Governance Configuration (/admin/settings/approvals)

3. Workflows (/admin/settings/approvals/workflows)

- Visual workflow designer:
- Trigger Catalogue: Select event (e.g. Purchase Order Released, Order Exceeds Credit Limit, Stock Count Variance > Threshold).
- Condition Tree Builder: Build logical rules using AND / OR blocks (e.g. Order Amount > KES 100,000 AND Branch = Nairobi).
- Step Sequence Builder: Add ordered approval steps, assign resolution rules, set due hours, and configure overdue behaviors (Remind, Escalate to Supervisor, Auto-Approve, Auto-Reject).
- Dry-Run Simulator: Select a real past document and simulate the workflow in sandbox mode to preview who would be assigned.

4. Approvers & Limits (/admin/settings/approvals/approvers)

- Matrix of staff signing authorities:
- Assign maximum approval limits per document kind (e.g. John Kamau: PO Limit KES 200,000; Refund Limit KES 20,000).
- Unlimited Checkbox: For executive directors.
- Substitute Approver Picker: Assign an alternate staff member and activate substitute mode when traveling.

5. Switches (/admin/settings/approvals/switches)

- Three-tier on/off control switches:

1. Module Switches: Turn approvals on/off globally for an entire module (e.g. disable all Purchase Order approvals).
2. Scenario Switches: Toggle specific triggers (e.g. enable Order Credit Limit Approvals, but disable Margin Floor Approvals).
3. Branch Overrides: Apply switches to specific branches.

6. Monitor (/admin/settings/approvals/monitor)

- Executive bottleneck control room:
- Lists all approval requests currently in-flight across the entire company.
- Highlights requests that are overdue against target SLAs.
- Reassign Action: Super Admins can reassign stuck requests to an alternate manager to prevent operational delays.

---

5. Granular Field-by-Field Reference

Workflow Step Fields (/admin/settings/approvals/workflows/[id])

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Step Sequence	Execution order.	Integer	System	Re-orderable (1, 2, 3...).
Approver Rule	How approver is chosen.	Dropdown	Required	Specific User, User Group, Line Manager, Chain by Limit, First Qualified.
Target User / Group	Selected actor or committee.	Lookup Autocomplete	Required if Specific	Identifies user or group.
Requirement Rule	Quorum for group steps.	Dropdown	Required if Group	Any One Member, All Members Must Approve, N of Group.
Due After (Hours)	SLA target window.	Integer (Hours)	Required	Number of hours before marked overdue (e.g. 24).
Overdue Action	What happens on SLA breach.	Dropdown	Required	Remind Only, Escalate to Next in Chain, Auto-Approve, Auto-Reject.
Self-Approval	Can requester approve own request?	Toggle (On/Off)	Required	Default Off. Enforces strict separation of duties.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Reviewing and Approving a Purchase Order in the Inbox

1. The Finance Manager logs in and navigates to /admin/approvals/inbox.
2. A badge shows: "Waiting on Me (1)".
3. Click the request: "Purchase Order #PO-2026-0042 from Acme Supplies (KES 380,000.00)".
4. A slide-over drawer opens displaying the full purchase order document, vendor details, and the approval chain history.
5. Inspect the line items and budget availability.
6. Click the green Approve button in the header.
7. A confirmation banner displays: "Purchase Order #PO-2026-0042 approved successfully."
8. The document transitions to Approved, releases for procurement, and vanishes from the manager's pending inbox.

Action 2: Rejecting a Discount Request with a Mandatory Reason

1. In the inbox, select a Sales Order discount request: "Order #ORD-1092 - 18% Discretionary Discount".
2. The manager determines that 18% wipes out product margin.
3. Click the red Reject button.

4. The Rejection Modal appears (the action cannot proceed without it):
   - Select Reason Code: Unacceptable Margin Loss.
   - In the Mandatory Comment box, type: "Maximum allowable discount on this inverter model is 8%. Please revise quote with customer."
5. Click Confirm Rejection.
6. The request status updates to Rejected, the sales order reverts to Draft state, and an instant alert is sent to the sales representative with the manager's explanation.

Action 3: Reassigning a Stuck Document via the Approval Monitor

1. A sales order for a priority customer is stuck because the designated Branch Manager is hospitalized.
2. The Operations Director navigates to /admin/settings/approvals/monitor.
3. Locate the stuck document, flagged in red: "Overdue by 42 hours (Assigned to: Peter Mwangi)".
4. Click Reassign Request.
5. Select New Approver: "Susan Kiprop (Regional General Manager)".
6. Enter Reassignment Reason: "Emergency reassignment due to medical leave of primary approver".
7. Click Confirm Reassignment.
8. The request instantly transfers to Susan Kiprop's inbox, and the audit log records the emergency reassignment.

---

7. Conditions, Business Rules & Edge Cases

The Self-Approval Block

- By default, allow_self_approval = false.
- If a Managing Director creates a Purchase Order, the system detects that Requester ID == Approver ID and automatically escalates the document to the Board Chairman or appointed second signatory.

In-Flight Workflow Immutability

- When a workflow definition is edited and saved, the system creates a new version (e.g. Version 3).
- Any document that was submitted under Version 2 will continue executing through the Version 2 step sequence until completion. New submissions automatically use Version 3.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Assign approval limit of KES 100,000 to Supervisor and Unlimited to Director.	Limits accurately reflect on Approver Profiles screen.	[ ]
2	Build a Workflow for Purchase Orders exceeding KES 100,000.	Workflow compiles cleanly with active trigger and steps.	[ ]
3	Submit a test PO for KES 150,000 as a Buyer.	PO halts in Pending Approval and appears in Director's Inbox.	[ ]
4	Attempt to reject a request without entering a reason.	System blocks rejection until reason code and comment are provided.	[ ]
5	Verify Real-Time Approval Monitor.	In-flight request appears with accurate SLA timer countdown.	[ ]

Interactive version with live navigation: /admin/docs/12-approvals-and-internal-controls

CHAPTER 13

Guide 13: Settings, Branches, Numbering & KRA eTIMS (`/admin/settings`)

Category: Fiscal & Administration | Reading Time: ~10 min | Words: 1,952

*The **Settings & Configuration** module forms the bedrock of the entire enterprise deployment. It governs organizational Dimensions, physical operating Branches, Payment Provider integrations, gapless Number Series engines, Document Number*

The Settings & Configuration module forms the bedrock of the entire enterprise deployment. It governs organizational Dimensions, physical operating Branches, Payment Provider integrations, gapless Number Series engines, Document Number Bindings, Order Stage Target Timers (SLAs), and complete Kenya Revenue Authority (KRA TIMS & eTIMS) fiscal compliance.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- The KRA eTIMS Compliance Trap: Under Kenyan tax law, every B2B and B2C commercial invoice must be validated, digitally signed, and transmitted to the KRA eTIMS system in real time, complete with cryptographic control codes and verification QR codes. Businesses using legacy POS software must buy expensive external hardware "Type C" fiscal boxes (KES 40,000 to KES 80,000 per till) that regularly jam, drop serial connections, and halt store sales. QSell features native Direct Cloud eTIMS Digital Signing: no external boxes required; invoices sign in milliseconds via direct API integration.
- Audit Disasters from Gapped Invoice Numbering: Tax auditors from the KRA routinely penalize businesses if invoice numbers contain missing gaps (e.g. INV-0012 followed by INV-0014 without an explanatory audit trail for 0013). QSell enforces Strict Gapless Number Series Engines: document numbers are reserved transactionally; if a transaction aborts, the gap is tracked or the sequence is preserved cleanly.
- Multi-Branch Tax Inconsistencies: A company with branches in Nairobi, Mombasa, and Kisumu may have different physical fiscal devices or branch-specific KRA middleware. QSell supports Independent Branch Tax Modes: branches can either inherit the company-wide eTIMS configuration or operate under independent fiscal credentials.
- Unnoticed Operational Sluggishness: Orders sit for hours between "Order Placed" and "Picking Started" without anyone noticing. QSell's Order Target Timers alert management when any stage exceeds its SLA threshold.

Target Client Profiles

- Kenyan Retail & Wholesale Enterprises: Subject to mandatory KRA eTIMS invoice transmission and VAT compliance.
- Multi-Branch Retail Chains: Requiring standardized document numbering (e.g. INV-NBI-0001 vs INV-MBA-0001) and localized branch reporting.
- SLA-Sensitive Delivery & E-Commerce Merchants: Guaranteeing customers fast fulfillment and delivery windows.

Commercial Sales Pitch & Talk Track

> "With QSell, regulatory tax compliance and enterprise setup are solved out of the box. You do not need to buy expensive, unreliable fiscal gadget boxes for each of your cashiers. Our native KRA eTIMS integration signs and transmits your invoices in the cloud in real time, generating official QR codes on every printed receipt. Furthermore, our gapless number series protect you from tax audit penalties, and our multi-branch architecture lets you scale to 50 stores with complete central visibility."

Measurable ROI & Safeguards

- 100% KRA eTIMS Compliance: Zero risk of non-compliance fines (saving up to KES 1,000,000 per violation).

- Elimination of Fiscal Box Hardware Costs: Saves KES 50,000+ per counter till by utilizing software-based cloud signing.
- Zero Gapless Audit Errors: Guaranteed sequential numbering satisfies statutory book requirements.

---

2. Prerequisites & Configuration Sequence

```
graph TD
    DIM[1. Dimensions Configured: Cost Centers & Regions] --> BRAN[2. Physical Operating Branches Created]
    BRAN --> NUM[3. Number Series & Bindings Established]
    NUM --> SLA[4. Order Targets & SLA Timers Defined]
    SLA --> TAX[5. KRA Tax Classifications & eTIMS Mode Setup]
```

1. Dimensions: Established under /admin/settings/dimensions (Optional financial tags).
2. Branches: Created under /admin/settings/branches.
3. Number Series: Formats defined under /admin/settings/number-series and bound under /admin/settings/number-series/bindings.
4. Order Targets: SLA thresholds set under /admin/settings/order-targets.
5. KRA eTIMS Mode: Configured under /admin/settings/tax-compliance/kra-tims-etims.
6. Required Permissions: settings:read/manage, branches:read/manage, numbering:manage, taxes:manage.

---

3. Module & Sub-Module Architecture

The Settings module is organized into six administrative domains:

```
graph TD
    subgraph Structure["1. Organizational Infrastructure"]
        DIM["Dimensions (Cost Centers, Territories)"]
        BRAN["Operating Branches & Banking Links"]
    end

    subgraph Numbering["2. Sequential Integrity"]
        NUM["Number Series (Prefixes, Resets, Padding)"]
        BIND["Document Bindings (Order, Invoice, PO)"]
    end

    subgraph ServiceLevel["3. Operational SLAs"]
        TARG["Order Stage Targets & Escalations"]
    end

    subgraph Compliance["4. KRA Tax & eTIMS Engine"]
        CLASS["Tax Classification (16%, 0%, Exempt)"]
        ITEMS["eTIMS Item Master Synchronization"]
        MODE["Company vs Independent Branch Mode"]
        LOG["Forensic Signing Log & QR Audit"]
    end

    Structure --> Numbering
    Numbering --> ServiceLevel
    ServiceLevel --> Compliance
```

---

4. Interface-by-Interface Walkthrough

1. Dimensions (/admin/settings/dimensions)

- Multi-dimensional reporting axes: Create custom tags (e.g. Cost Center, Department, Project, Sales Territory).
- Allows slicing Trial Balance, Sales Reports, and Expense Ledgers by dimension values without cluttering the Chart of Accounts.

2. Branches (/admin/settings/branches)

- Physical business units:
- Branch Code (e.g. BR-NBI-01), Branch Name (e.g. "Nairobi Central Depot").
- Physical Street Address, City, County, GPS Coordinates.
- Linked Default Cash Drawer and linked Bank Account.
- Public Base URL configuration for localized payment gateway webhooks.

3. Payments (/admin/settings/payments)

- Payment provider configuration:
- Connect Safaricom M-Pesa Daraja APIs (Shortcode, Consumer Key, Consumer Secret, Passkey, B2C initiator credentials).
- Connect Card Processors (Pesapal, Bank PGWs).
- All secrets encrypted with AES-256 GCM sealed storage.

4. Number Series (/admin/settings/number-series)

- Document sequential engine:
- Define formats (e.g. Prefix INV-, Starting No 00001, Ending No 99999, Increment 1).
- Configure Yearly Restarts (e.g. resetting sequence to 0001 on January 1st).
- Enforce Gap-Less Rules for fiscal documents.

5. Document Numbering Bindings (/admin/settings/number-series/bindings)

- Connects document types to specific number series per branch:
- Branch Nairobi: Invoices bind to series INV-NBI-, Quotes bind to QT-NBI-.
- Branch Mombasa: Invoices bind to series INV-MBA-, Quotes bind to QT-MBA-.

6. Order Targets & SLA Timers (/admin/settings/order-targets)

- SLA management:
- Define target duration per order stage (e.g. Picking: Max 45 mins, Packing: Max 30 mins, Delivery: Max 180 mins).
- Configure notification triggers and escalation email alerts when orders languish in a stage.

7. Tax Compliance & KRA eTIMS (/admin/settings/tax-compliance)

The comprehensive regulatory suite divided into five sub-screens:

1. Tax Classification (.../classification): Maps company tax categories to official KRA eTIMS tax codes:
- Code A: Standard Rate (16% VAT).
- Code B: Zero-Rated (0% VAT - exports, medicines).
- Code C: Exempt (financial services, unmanufactured agricultural produce).
- Code D: Special / Petroleum (8% Fuel VAT).
2. eTIMS Items Master (.../items):
- Synchronizes product catalog items with KRA's central item registry.
- Shows registration status, KRA Item Code, packaging unit code, and stock sync status.
- Action: Register Item with KRA, Update Item, Transmit Restock Notification.
3. Signing Log (.../signing-log):

- Forensic audit trail of every invoice signing attempt.
- Displays Document No, Timestamp, Result (Signed, Skipped, Failed), KRA Invoice Code, Cryptographic Middleware Signature, and QR URL.
- For skipped or failed documents, displays exact reason trace (e.g. "Skipped: Proforma quote is not a tax document" or "Failed: KRA VSCU endpoint timeout, retrying via outbox").

4. Company Mode (.../company): Sets the global default mode: Disabled, TIMS (Hardware Control Unit), or eTIMS (Cloud Digital Signing).
5. Branch Modes (.../branches): Overrides company mode for specific branches operating independently.

---

5. Granular Field-by-Field Reference

Number Series Form Fields (/admin/settings/number-series/new)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Series Code	Identifier for this sequence.	Text (Max 20 chars, uppercase)	Required	e.g. SERIES-INV-2026.
Description	Human-readable explanation.	Text (Max 120 chars)	Required	e.g. "2026 Sales Invoices Series".
Prefix	Characters leading the number.	Text (Max 10 chars, uppercase)	Required	e.g. INV- or PO-.
Starting Number	First sequential integer.	Integer	Required	Typically 1 or 1001.
Ending Number	Ceiling for this series.	Integer	Required	Must be $>$ Starting Number. Generates warning at 95% capacity.
Padding Length	Total digits with leading zeros.	Integer (1-10)	Required	e.g. 5 yields 00001, 6 yields 000001.
Allow Gaps	Can sequence skip cancelled numbers?	Toggle (On/Off)	Required	Must be Off for KRA tax invoices and credit notes.

KRA Tax Classification Fields (/admin/settings/tax-compliance/.../classification)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Tax Label	Display name in app.	Text (Max 60 chars)	Required	e.g. "General VAT (16%)".
Tax Rate %	Percentage applied to line.	Percentage	Required	e.g. 16.0 or 0.0.
KRA TIMS Code	Legacy hardware tax code.	Single Character (A-E)	Required	A=16%, B=0%, C=Exempt, D=Other, E=8%.
KRA eTIMS Code	Cloud digital signing code.	Text (Max 10 chars)	Required	Standard KRA HS item tax code.
Applies to Delivery	Does delivery charge take this tax?	Toggle (On/Off)	Required	Delivery fee is a service taxed at 16% VAT by default.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Creating a Branch-Specific Invoice Number Series

1. Navigate to /admin/settings/number-series and click New Series.
2. Enter Series Code: INV-MOMBASA-2026.

3. Enter Prefix: INV-MSA-.
4. Set Starting No: 1, Ending No: 99999, Padding Length: 5.
5. Set Allow Gaps: Off (enforcing strict audit compliance).
6. Click Save Number Series.
7. Navigate to /admin/settings/number-series/bindings.
8. Locate Document Kind: Tax Invoice, and find Branch: "Mombasa Retail Store".
9. In the dropdown, change the binding from Global Series to INV-MOMBASA-2026.
10. Click Save Bindings. All future invoices generated in Mombasa will print numbers formatted as INV-MSA-00001.

Action 2: Inspecting and Verifying the KRA Invoice Signing Log

1. Navigate to /admin/settings/tax-compliance/kra-tims-etims/signing-log.
2. The log renders in chronological order.
3. Locate Invoice #INV-2026-00412.
4. Inspect the columns:

- Result: Signed (Green badge).
- KRA Middleware Signature: eTIMS-KRA-01928374-AF892B.
- Control Code: 0192-8374-AF89.

5. Click View Signing Trace:

- Gate 1 (Document): PASSED (Tax Invoice).
- Gate 2 (Scope): PASSED (Company eTIMS Active).
- Gate 3 (Customer): PASSED (Customer PIN Validated).
- Gate 4 (Readiness): PASSED (Keys valid, response received in 240ms).

6. Click the QR Code Preview link to test that scanning the QR code opens the official KRA tax invoice verification portal page.

---

7. Conditions, Business Rules & Edge Cases

The 7-Gate KRA Invoice Signing Policy

Every invoice generated by the system passes through the 7-Gate Evaluation Policy in strict order:

1. Document Gate: Proforma quotes and draft orders are never signed.
2. Scope Gate: Evaluates whether the branch operates under independent mode or company mode.
3. Mode Gate: eTIMS or TIMS must be configured for that scope.
4. Mode Switch Gate: Signing must be enabled for that mode.
5. Module Switch Gate: Invoicing module switch must be active.
6. Customer Gate: Checks if customer explicitly opted out (diplomatic missions).
7. Readiness Gate: Digital certificates must be valid and unexpired.

If any gate halts, the document records a Skipped or Failed event with an explicit reason string. Crucially, a temporary KRA network outage never halts customer sales: the invoice is issued locally, and the signing task is queued in the transactional outbox for automated background retry.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Create Mombasa and Nairobi branches with unique codes.	Branches appear in branch selection dropdowns across the system.	[ ]
2	Create sequential gapless Number Series for Invoices and POs.	Numbers increment sequentially without skips.	[ ]
3	Configure KRA eTIMS Tax Classifications (16% VAT, 0%, Exempt).	Product catalog reflects mapped tax codes accurately.	[ ]
4	Post a test cash sale invoice and inspect KRA Signing Log.	Document signs successfully; QR code and KRA signature code generated.	[ ]
5	Verify Order Stage SLA target timers.	Lapsed orders in Packing stage trigger late warning alerts on dashboard.	[ ]

Interactive version with live navigation: /admin/docs/13-settings-branches-numbering-and-kra-etims

CHAPTER 14

Guide 14: External & Field Portals (`/(store)`, `/rider`, `/vendor`)

Category: External Ecosystem | Reading Time: ~12 min | Words: 2,391

*While administrative staff manage operations via `/admin`, QSell delivers three purpose-built external interfaces: the **Customer Web Storefront** (`/(store)`), the **Rider Mobile App** (`/rider`), and the **Vendor Self-Service Portal** (`/*

While administrative staff manage operations via /admin, QSell delivers three purpose-built external interfaces: the Customer Web Storefront (/(store)), the Rider Mobile App (/rider), and the Vendor Self-Service Portal (/vendor). These portals extend the system's core capabilities directly to shoppers, field delivery crews, and supply chain partners.

---

1. Executive Overview & Commercial Value Proposition

Operational Problems Solved

- Customer Phone Ordering Friction: Customers have to call or WhatsApp sales reps to inquire about stock availability, ask for prices, and track their delivery status. The Customer Storefront empowers shoppers to browse live stock, calculate exact delivery charges via map pin drops, checkout via M-Pesa, and track their order's timestamped progress in real time.
- Rider Paper Run Sheets & Misplaced Delivery Notes: Delivery riders traditionally carry loose paper receipts and invoices on their motorbikes, which get stained by rain, torn, or lost. The Rider Mobile App delivers an interactive, mobile-optimized daily run sheet directly to the driver's smartphone, complete with one-tap Google Maps directions, customer phone dialing, and digital proof-of-delivery PIN verification.
- Supplier Invoice Phone Tag & Lost Invoices: Suppliers regularly email invoices that get trapped in spam filters or sit forgotten on buyers' desks, resulting in delayed orders and supplier relationship friction. The Vendor Portal enables prequalified suppliers to log in, view open RFQs, submit competitive price bids, acknowledge purchase orders, and generate electronic supplier invoices matched directly to Goods Received Notes.

Target Client Profiles

- Direct-to-Consumer (D2C) & B2B Web Merchants: Selling to both retail end-consumers and trade account holders on one unified website.
- Last-Mile Logistics Fleets: In-house motorcycle dispatchers, delivery vans, and third-party contract drivers.
- Multi-Vendor Procurement Networks: Sourcing materials from tens or hundreds of trade vendors and service contractors.

Commercial Sales Pitch & Talk Track

> "With QSell, your business operations connect seamlessly to the outside world. Your customers shop on a fast, modern web store customized with your brand and colors. Your delivery riders carry our dedicated mobile app that guides them stop by stop and verifies deliveries using customer SMS PINs. And your suppliers manage their bids, orders, and invoices in their own private portal—giving you an enterprise ecosystem without paying for external third-party software licenses."

Measurable ROI & Safeguards

- Zero Third-Party Portal Subscription Costs: Built-in customer storefront, rider dispatch, and vendor portals eliminate separate SaaS subscriptions.
- 100% Secure Tenant Isolation: External users are cryptographically restricted to their own session records; vendors can never see competing bids or internal staff notes.

- Paperless Field Operations: Digital run sheets, doorstep PINs, and electronic delivery notes eliminate paper loss.

---

2. Prerequisites & Portal Access Setup

```
graph TD
    COMP[1. Company Profile, Logo & Brand Colors Configured] --> STORE[2. Storefront Catalog & Payment
Providers Live]
    STORE --> RIDER[3. Riders Added; Riders with Their Own Account Given a Phone Number and PIN]
    RIDER --> VEND[4. Vendors Invited to Register on Vendor Portal]
```

1. Branding & Gateways: Storefront appearance, payment providers, and delivery zones must be configured in /admin.
2. Sellable Products: Catalog items must have Show on Web Shop = True and active selling prices.
3. Rider Logins: Riders are added under /admin/delivery/riders. A rider with an account of their own needs a Kenyan mobile number, and the office gives them a PIN there (see Guide 06 (06-delivery-logistics-and-field-dispatch.md, Action 4). A member of staff who rides signs in with their own email and password.
4. Vendor Portal Access: Vendors must receive an invitation link or register at /vendor/register.

---

3. Architecture of the Three External Surfaces

```
graph LR
    subgraph Storefront["Customer Storefront (/(store))"]
        CAT[Product Catalog & Search]
        CART[Shopping Cart & Wishlist]
        CHK[Pin-on-Map Checkout & M-Pesa]
        TRK[Real-Time Order Tracking]
    end

    subgraph RiderApp["Rider Mobile App (/rider)"]
        AUTH[Rider Phone Number and PIN Sign-In]
        RUNS[Active Run Sheets & Multi-Drop Stops]
        POD[Customer Delivery Code & Proof of Delivery]
        COD[Door Cash & M-Pesa Collection]
    end

    subgraph VendorPortal["Vendor Portal (/vendor)"]
        PRE[Prequalification Application]
        BID[RFQ Bidding & Quote Submission]
        PO[Purchase Order Acknowledgment]
        INV[Direct Invoice Generation against RGN]
    end

    Storefront --> APP["QSell Central Platform"]
    RiderApp --> APP
    VendorPortal --> APP
```

---

4. Interface-by-Interface Walkthrough

Part A: Customer Web Storefront (/(store))

1. Home & Catalog (/)

- Brand header, promotional announcement banners, search bar with live autocomplete, and mega-menu categories.
- Responsive product cards displaying Title, Brand, Star Rating, Price (KES), Unit of Measure, Stock Availability Badge, and Add to Cart button.

2. Shopping Cart & Wishlist (/cart, /wishlist)

- Line item quantity adjusters, subtotal calculation, coupon code entry box, and shipping fee estimator.

3. Checkout (/checkout)

- The frictionless, single-page checkout flow:

1. Customer Information: Name, email, mobile phone number.
2. Delivery Method & Location:

- Choose Delivery or In-Store Pickup / Click & Collect.
- Drop a GPS pin on the map or select County $\rightarrow$ Town $\rightarrow$ Estate.
- System calculates exact delivery fee in real time.

3. Payment Selection: Safaricom M-Pesa STK Push, Credit/Debit Card, or Cash on Delivery.
4. Order Submission: Dispatches M-Pesa prompt directly to customer's phone and transitions to order tracking upon PIN entry.

4. Order Tracking (/track/[id])

- Customer self-service portal: Displays real-time progress bar through the order stage timeline (Order Confirmed $\rightarrow$ Preparing in Warehouse $\rightarrow$ Out with Rider $\rightarrow$ Delivered).
- Displays the delivery code (6 digits) to give to the rider upon arrival, the promised delivery window until it is delivered, and a link to the courier's own page when a courier handles the parcel.

Part B: Rider Mobile Web App (/rider)

1. Rider Sign-In (/rider/sign-in, /rider/pin)

- Clean, high-contrast mobile interface designed for one-handed operation on low-end smartphones.
- Sign in (/rider/sign-in) with the rider's Kenyan mobile number and a 6-digit PIN (no password to type with gloves on). Opening /rider while signed out leads here. A link underneath offers email and password sign-in for staff who ride.
- Choose your own PIN (/rider/pin): the PIN the office gives is temporary, and the rider must replace it with their own before anything else in the app opens. The same screen (Change my PIN) is used later.
- Five wrong tries in a row shut the rider out for a while (15 minutes, growing to 4 hours); the office can let them try again at once. A PIN opens the rider app only, and only for riders with an account of their own.
- Sign out is at the top of every screen.

2. Active Run Sheet (/rider and a run's own page)

- Displays the assigned multi-drop run for the day:
- Header: Run Number, Total Stops (e.g. "Stop 2 of 5"), Total Cash Float, Total Run Weight.
- Ordered stop cards: Customer Name, Phone, Delivery Address, Estimated Arrival Time, Payment Method (Paid Online vs Collect KES 3,500 at Door).
- Action: Open Google Maps / Waze Directions, Call Customer.

3. Delivery Confirmation & Door Payment (on a stop of the run)

- Execution of the handover:
- Input field for the customer's 6-digit delivery code (when the company's rule at the door asks for it).
- Door Payment collection confirmation (Cash received or M-Pesa Paybill confirmation code).
- Digital signature capture or photo upload of delivered parcel.
- Action: Confirm Delivery or Report Failed Attempt (with mandatory reason selection).

4. My Pay (/rider/pay)

- The rider's own pay: Owed to me now, Earned this month, the latest 30 pay entries (each Paid or Not paid yet; pay the office took back shows as Taken back by the office) and the latest 10 payouts. A rider sees only their own pay. See Guide 06 (06-delivery-logistics-and-field-dispatch.md.

Part C: Vendor Self-Service Portal (/vendor)

1. Registration & Prequalification (/vendor/register, /vendor/prequalification)

- Supplier onboarding:
- Complete legal business profile, KRA PIN, physical address, and bank wire details.
- Select procurement categories applied for (e.g. Electrical Supplies, Solar Inverters).
- Upload mandatory compliance documentation (Certificate of Incorporation, KRA Tax Compliance Certificate, CR12).

2. RFQ Bidding Workspace (/vendor/(portal)/rfqs)

- View active Requests for Quotation invited to bid on.
- Enter unit prices per line item, lead time in days, payment terms, and validity dates.
- Submit sealed bid before the published round deadline.

3. Purchase Orders & Acknowledgment (/vendor/(portal)/orders)

- View official purchase orders issued by the company.
- Download official PDF Purchase Orders.
- Action: Acknowledge PO (confirms commitment to deliver by promised date).

4. Supplier Invoicing (/vendor/(portal)/invoices)

- Direct electronic invoice submission:
- Select the matching Goods Received Note (RGN) issued by the warehouse.
- Type vendor invoice number and attach invoice PDF.
- Submit directly into the back-office 3-Way Match queue.

5. Withholding Tax Certificates (/vendor/(portal)/certificates)

- Self-service tax portal: Suppliers can search by month and download official KRA Withholding VAT and Withholding Tax certificates without calling the accounts department.

---

5. Granular Field-by-Field Reference

Storefront Checkout Fields (/(store)/checkout)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Full Name	Recipient identity.	Text (Max 120 chars)	Required	Recipient of the package.
Phone Number	Contact & payment phone.	Phone (Max 25 chars)	Required	M-Pesa phone number e.g. 0712345678.
Delivery Address	Physical location description.	Text (Max 160 chars)	Required if Delivery	Landmark, estate, apartment/house number.
Map Pin Location	Exact GPS coordinates.	Map Pin Dropper	Optional	Used for precise distance rate calculation.

Payment Method	Ingress payment choice.	Radio Selector	Required	M-Pesa STK Push, Credit Card, Cash on Delivery.
Customer Order Notes	Special delivery instructions.	Text (Max 300 chars)	Optional	e.g. "Call upon arrival at the security gate".

Rider Proof of Delivery Fields (on a stop of a run)

Field Label	Purpose	Type & Format	Required?	Validation Rules & Hints
Delivery code	6-digit customer security code.	Numeric (6 digits)	Required when the company's rule at the door asks for the code	Matches the code sent to the customer when the delivery left.
Cash Collected	Monetary sum taken at door.	Currency (KES)	Required if COD	Verified against order balance. Added to rider custody.
M-Pesa Trans ID	Doorstep M-Pesa receipt code.	Text (10 chars, uppercase)	Required if Door M-Pesa	Valid Safaricom receipt e.g. QA81273918.
Recipient Name	Person accepting parcel.	Text (Max 80 chars)	Required	Customer or authorized agent at premises.

---

6. Step-by-Step User Actions (Down to the Last Click)

Action 1: Shopper Checkout with M-Pesa STK Push

1. A customer browses the storefront, adds a "Solar Charge Controller" to the cart, and clicks Checkout.
2. Fills in Name and Mobile Phone: 0712 345 678.
3. Selects Delivery: Picks County: Nairobi, Estate: Westlands. The system quotes delivery fee: KES 250.00.
4. Selects Payment Method: M-Pesa STK Push.
5. Clicks the Place Order & Pay button.
6. The screen displays a loading spinner: "Prompt sent to your phone. Please enter your M-Pesa PIN."
7. Within 3 seconds, the customer's phone vibrates with the Safaricom STK prompt: "Do you want to pay KES 4,750.00 to [Company Trading Name]?"
8. The customer enters their PIN.
9. The checkout screen automatically detects payment confirmation, displays a green success checkmark, and redirects to /track/[orderId] displaying the 6-digit delivery code.

Action 2: Rider Executing a Delivery Stop on Mobile

1. The delivery rider opens /rider on their mobile phone and signs in with their phone number and 6-digit PIN (on the first time after the office gives a PIN, they choose their own).
2. The active run sheet displays: "Stop 1: Jane Wanjiku - Westlands".
3. Clicks Navigate to open Google Maps turn-by-turn directions.
4. Upon arrival, clicks I Have Arrived.
5. Hands the parcel to Jane Wanjiku and asks for her 6-digit delivery code.
6. Types the code (for example 048213) into the app.
7. Clicks Complete Delivery.
8. The app vibrates with a success chime, marks Stop 1 completed, and automatically queues Stop 2.

Action 3: Vendor Submitting an Electronic Invoice via the Vendor Portal

1. The supplier logs into /vendor.
2. Navigates to Orders & Receipts.
3. Locates Purchase Order #PO-2026-0038 and clicks Create Supplier Invoice.
4. The system pre-fills the screen with the quantities and prices from Goods Received Note #RGN-2026-0024.
5. Types their company's internal invoice number: INV-SOLAR-8819.
6. Uploads their signed invoice PDF.
7. Clicks Submit Invoice to Accounts.
8. The invoice immediately appears in the back-office queue at /admin/vendors/invoices ready for 3-way matching and payment.

---

7. Conditions, Business Rules & Security Boundaries

Strict Vendor Data Isolation

- Vendor Portal users are cryptographically restricted to their own corporate entity:
- Every database query automatically applies WHERE vendor_id = session.vendor_id.
- Vendors cannot see other vendors, competing bid prices, internal buyer notes, or uninvited RFQs.
- Passing an arbitrary vendorId in the URL or payload is rejected with an instant security exception.

Rider Cash Custody Block

- If a delivery rider has unremitted cash custody exceeding their maximum allowed cash ceiling (configured per rider under /admin/delivery/riders), the system blocks the dispatcher from assigning the rider to new delivery runs until previous cash is reconciled at the cashier desk.

---

8. Implementor Checklist & Go-Live Verification

Step	Verification Task	Expected Result	Pass/Fail
1	Complete a storefront checkout using M-Pesa STK Push.	Order created; payment confirmed; customer receives SMS with tracking PIN.	[ ]
2	Open /track/[id] on a mobile browser.	Order stage timeline renders with live milestone indicator and verification PIN.	[ ]
3	Log into /rider on a mobile device using a test rider PIN.	Assigned delivery run sheet renders cleanly with stop sequence.	[ ]
4	Complete a test delivery stop using verification PIN.	Delivery confirmed; status updates to Delivered across admin and customer tracking.	[ ]
5	Invite a test supplier to the Vendor Portal.	Supplier registers, accesses open PO, and submits an invoice successfully.	[ ]

Interactive version with live navigation: /admin/docs/14-external-portals-customer-rider-vendor